

Toyota Safety Sense (TSS) testing and research of VCH, PCS-O and PCS-I

Tim Reust and Jim Morgan

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Terminology

TSS, Toyota Safety Sense

LSS, Lexus Safety System

PCS, Pre-Collision System

TIS, Technical Information System

VCH, Vehicle Control History

PCS-O, Pre-Collision System Operational data

PCS-I, Pre-Collision System Image data

Helping to make the road a safer place.

Camry's standard Toyota Safety Sense™ P (TSS-P)[11] is an active safety package that combines Pre-Collision System with Pedestrian Detection (PCS w/PD),[13] Lane Departure Alert with Steering Assist (LDA w/SA),[23] Automatic High Beams (AHB)[18] and Dynamic Radar Cruise Control (DRCC)[12] to assist you. TSS-P's comprehensive features create in-the-moment safety designed to help keep you safe under certain circumstances.



LANE DEPARTURE ALERT WITH STEERING ASSIST (LDA w/SA)[23]

Under certain circumstances, Lane Departure Alert with Steering Assist (LDA w/SA) is designed to help detect lane departure on roads with visible lane markings. LDA only activates when you're traveling at a speed of 32 mph or faster. When you hear and see the alert, after carefully checking the road around you, safely direct your vehicle back to the center of your lane. In addition, if the system senses that you're unintentionally drifting, it may automatically make small corrective steering inputs to help keep your vehicle in its lane.



AUTOMATIC HIGH BEAMS (AHB)[18]

The Automatic High Beams (AHB) feature is a safety system designed to help you see more clearly at night — without distracting other drivers. Designed to activate at speeds above 25 mph, AHB relies on an in-vehicle camera to help detect the headlights of oncoming vehicles and taillights of preceding vehicles, then automatically toggles between high and low beams accordingly to provide the appropriate amount of light. By using high beams more frequently, the system may allow for earlier detection of pedestrians and obstacles.



PRE-COLLISION SYSTEM WITH PEDESTRIAN DETECTION (PCS w/PD)[13]

Pre-Collision System with Pedestrian Detection (PCS w/PD) is designed to help keep the road safe by detecting a vehicle or a pedestrian in certain situations. By combining millimeter-wave radar with a camera capable of certain shape recognition, the system provides an audio/visual alert, warning you of a possible collision under certain circumstances. If you don't react, the system is designed for automatic braking support to help mitigate the potential for a collision.



DYNAMIC RADAR CRUISE CONTROL (DRCC)[12]

Intended for highway use and similar to "constant speed" cruise control, available Full-Speed Range Dynamic Radar Cruise Control (DRCC) lets you drive at a preset speed. DRCC uses vehicle-to-vehicle distance control, which is designed to adjust your speed, to help you maintain a preset distance from the vehicle ahead of you that is driving at a slower speed. DRCC uses a front-grille-mounted radar and an in-vehicle camera designed to detect vehicles and their distance. If a driver is traveling slower than you, or within your preset range, DRCC is designed to automatically slow your vehicle down without deactivating cruise control. If DRCC determines you need to slow down more, an audio and visual alert notifies you and brakes may be applied. When there's no longer a vehicle driving slower than your preset speed in front of you, DRCC will then accelerate to the preset speed and regular cruise control will resume.

See numbered footnotes in Disclosures section.



TOYOTA SAFETY SENSE™ 2.0 (TSS 2.0)

Peace of mind comes standard.

Toyota Safety Sense™ 2.0 (TSS 2.0) is a bundle of active safety features included on many new Toyota vehicles at no additional cost. These innovative features were designed to help protect you and your passengers from harm.

- Pre-Collision System with Pedestrian Detection[1]
- Full-Speed Range Dynamic Radar Cruise Control[1,2,3]
- Lane Departure Alert with Steering Assist[1]
- Automatic High Beams[1]
- Lane Tracing Assist[1]
- Road Sign Assist[1]

[1] For more information, visit toyota.com/safety-sense. Feature availability may vary by model grade. See Limited Warranty in Owner's Manual.



TOYOTA SAFETY SENSE™ 2.5+ (TSS 2.5+)

Peace of mind comes standard.

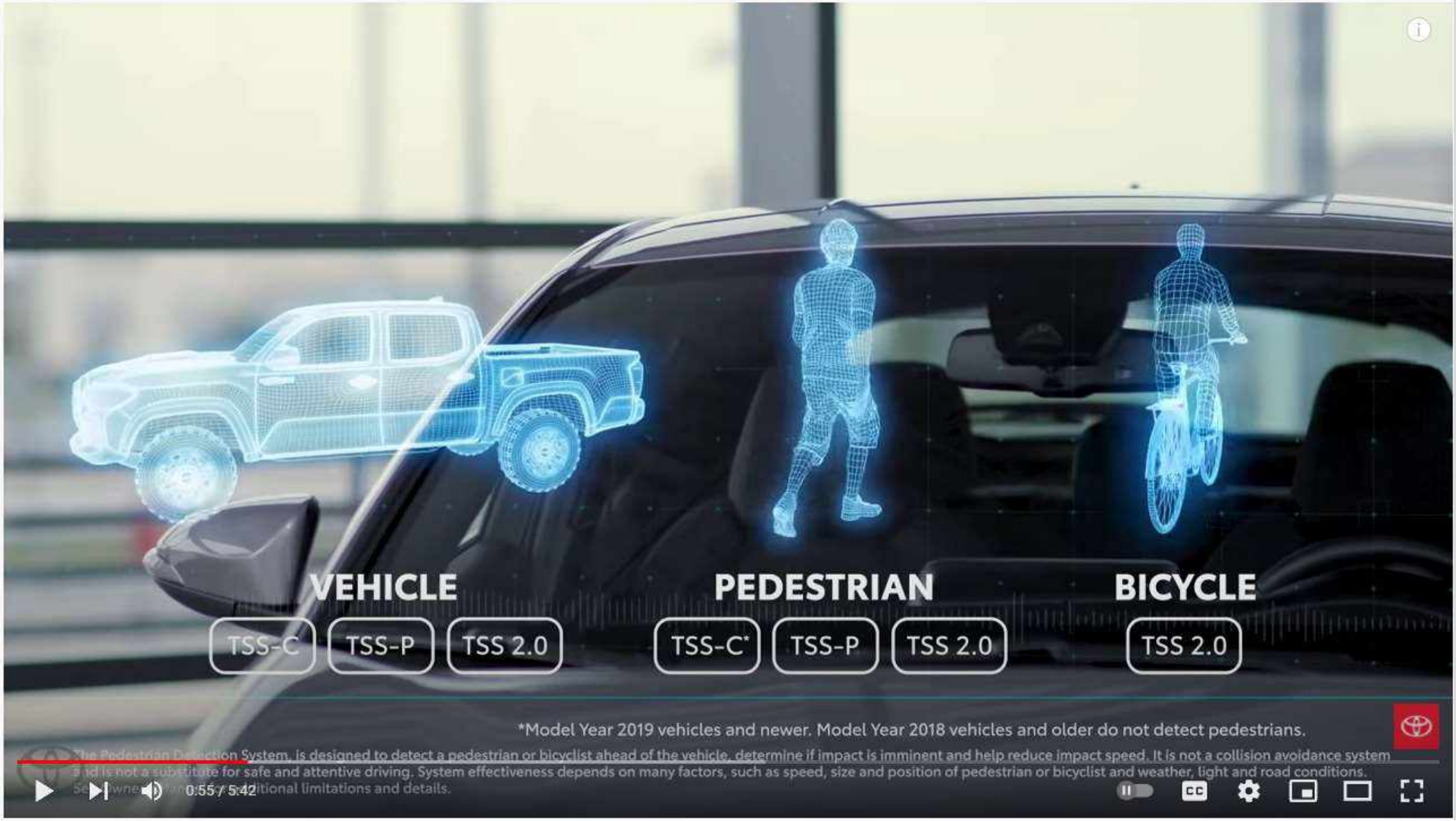
Toyota Safety Sense™ 2.5+ (TSS 2.5+)[1] is the latest bundle of active safety features included on many new Toyota vehicles at no additional cost. These innovative features were designed to help protect you and your passengers from harm.

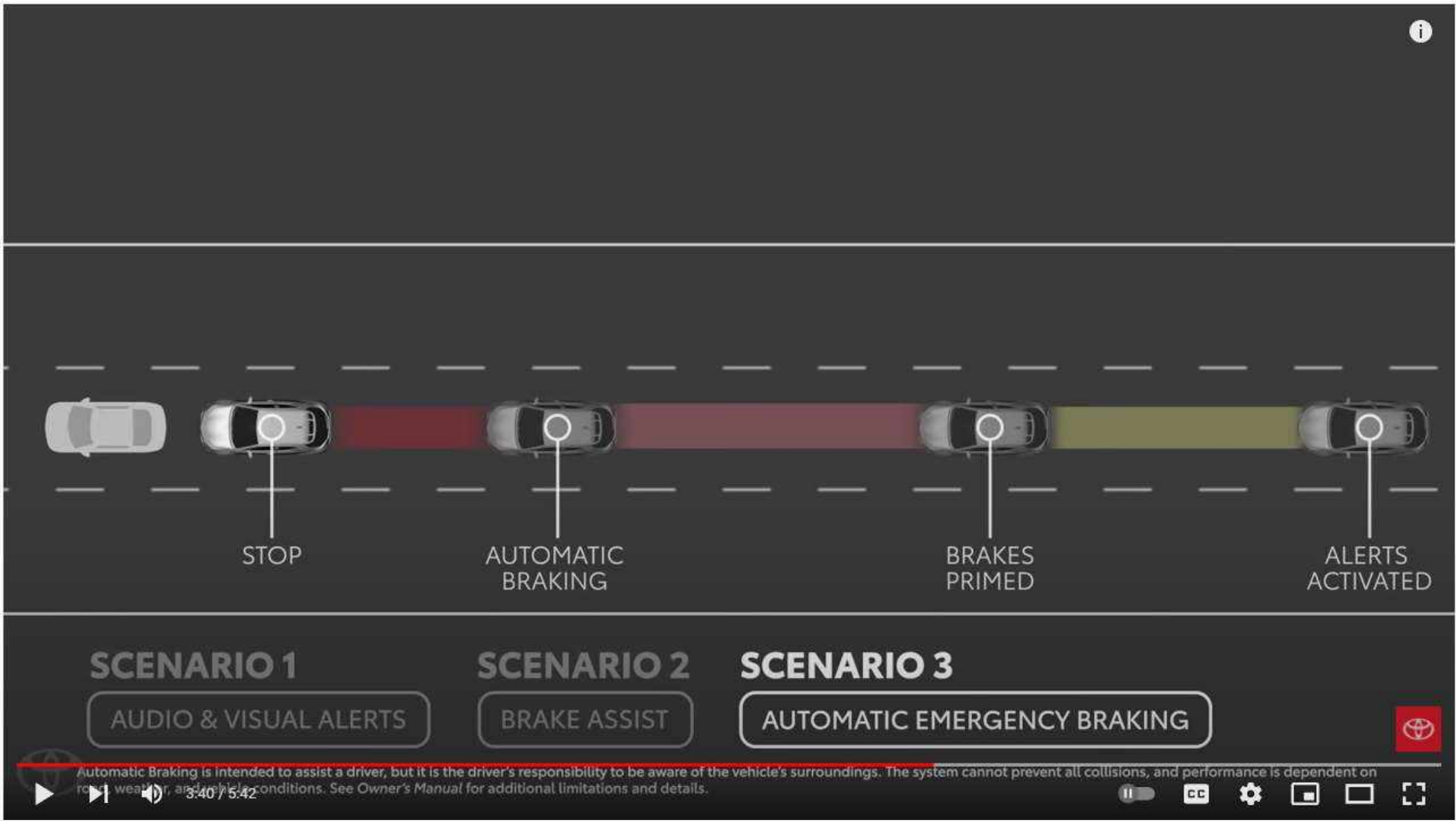
- Pre-Collision System with Pedestrian Detection[2]
- Full-Speed Range Dynamic Radar Cruise Control[3]
- Lane Departure Alert with Steering Assist[4]
- Automatic High Beams[5]
- Lane Tracing Assist[6]
- Road Sign Assist[6]

For more information, visit toyota.com/safety-sense. Features available by trim; see for model years. See numbered footnotes in Disclosures section.


Toyota
Safety
Sense
and millimeter-wave radar.
0:39 / 4:54 • Overview: Toyota Safety Sense 2.5/2.5 >


Pre-Collision System (PCS)
Pre-Collision System, or PCS,
is designed to help mitigate
mpact is imminent and help reduce
bicyclist. PCS w/PD is not a substitute
for safe and attentive driving. System effectiveness depends on many factors, such as speed, size and position of pedestrian or
bicyclist and weather, light and road conditions. See Owner's Manual for additional limitations and details.
Play (k)
0:48 / 4:54 • Pre-Collision System


VEHICLE
TSS-C
TSS-P
TSS 2.0
PEDESTRIAN
TSS-C*
TSS-P
TSS 2.0
BICYCLE
TSS 2.0
*Model Year 2019 vehicles and newer. Model Year 2018 vehicles and older do not detect pedestrians.
The Pedestrian Detection System, is designed to detect a pedestrian or bicyclist ahead of the vehicle, determine if impact is imminent and help reduce impact speed. It is not a collision avoidance system and is not a substitute for safe and attentive driving. System effectiveness depends on many factors, such as speed, size and position of pedestrian or bicyclist and weather, light and road conditions.
0:55 / 5:42


STOP
AUTOMATIC BRAKING
BRAKES PRIMED
ALERTS ACTIVATED
SCENARIO 1
AUDIO & VISUAL ALERTS
SCENARIO 2
BRAKE ASSIST
SCENARIO 3
AUTOMATIC EMERGENCY BRAKING
Automatic Braking is intended to assist a driver, but it is the driver's responsibility to be aware of the vehicle's surroundings. The system cannot prevent all collisions, and performance is dependent on
conditions. See Owner's Manual for additional limitations and details.
3:40 / 5:42

TOYOTA | **TOYOTA SAFETY SENSE™ P (TSS-P)**
Features and Operation Overview
For model year 2019 and newer vehicles equipped with TSS-P

AUG 2018

TOYOTA SAFETY SENSE™ P

AN ADVANCED ACTIVE SAFETY PACKAGE
FOR MID-SIZE AND LARGE VEHICLES



Crash protection starts with crash prevention. Collisions that result in injury may be caused by the delay in a driver's recognition of the situation and their ability to react accordingly. According to NHTSA, there were over 7.2 million reported crashes in 2016 – many of which were avoidable.

Toyota Safety Sense™ (TSS)[1] is designed to help protect drivers, passengers, people in other vehicles on the road, and pedestrians from harm. TSS represents a major milestone in Toyota's long history of creating advancements and innovations in safety that help prevent crashes and protect people.

TSS ADDRESSES THE THREE MOST COMMON ACCIDENT TYPES


Frontal Collisions


Unintended Lane Departures


Nighttime Accidents

Under certain conditions, TSS[1] is designed to support driver awareness, decision making and vehicle operation over a wide range of speeds. Packaged together in an integrated system, TSS features help address three key areas of accident protection: preventing or mitigating **frontal collisions**[2], keeping drivers **within their lane**[3], and enhancing road safety during **nighttime driving**[4]. However, drivers are responsible for their own safety, and must always drive safely, obey traffic speed limits and laws and focus on the road while driving.

TSS-P[5] consists of four active safety and driver assistance systems:

- **Pre-Collision System (PCS)**[2]
  with Vehicle and Pedestrian Detection[3]
- **Dynamic Radar Cruise Control (DRCC)**[3]
- **Lane Departure Alert (LDA)**[3]
- **Automatic High Beams (AHB)**[4]

- Pre-collision braking:
  - Vehicle speed is between approximately 7 and 110 mph (10 and 180 km/h). (For detecting a pedestrian, vehicle speed is between approximately 7 and 50 mph [10 and 80 km/h].)
  - The relative speed between your vehicle and the vehicle or pedestrian ahead is approximately 7 mph (10 km/h) or more.

TOYOTA SAFETY SENSE™
2.5 & 2.5+

PLAY VIDEO

What Is It and How Does It Work?

Toyota Safety Sense™, or TSS, is **a suite of active safety technologies and advanced driver assistance systems** designed to help address three key areas of accident protection:

- **Preventing or mitigating frontal collisions**
- **Keeping vehicles within their lane**
- **Enhancing road safety during nighttime driving**

TSS 2.5 and 2.5+ build on the previous TSS 2.0 suite by enhancing the functionalities of its six features, further utilizing a high-resolution camera and millimeter-wave radar to help keep occupants safe.



PRE-COLLISION SYSTEM (PCS)

PCS[2] is designed to detect a **vehicle or a pedestrian in either daytime or low-light situations**, as well as **daytime detection of a bicyclist**. When a potential hazard is detected, it may provide **audible and visual alerts**, and **automatic braking** in certain conditions.

TSS 2.5 and 2.5+ take PCS a step further with two additional functionalities: **intersection support** and **emergency steering assist**.

At intersections, the system may detect an oncoming **vehicle or pedestrian when performing a left-hand turn**, as well as an oncoming **pedestrian when performing a right-hand turn**.

Meanwhile, emergency steering assist is designed to help stabilize the vehicle when the driver initiates an emergency steering maneuver around an obstacle within their lane, reacting to a preceding **pedestrian for TSS 2.5**, as well as a preceding **vehicle or bicyclist for 2.5+**.

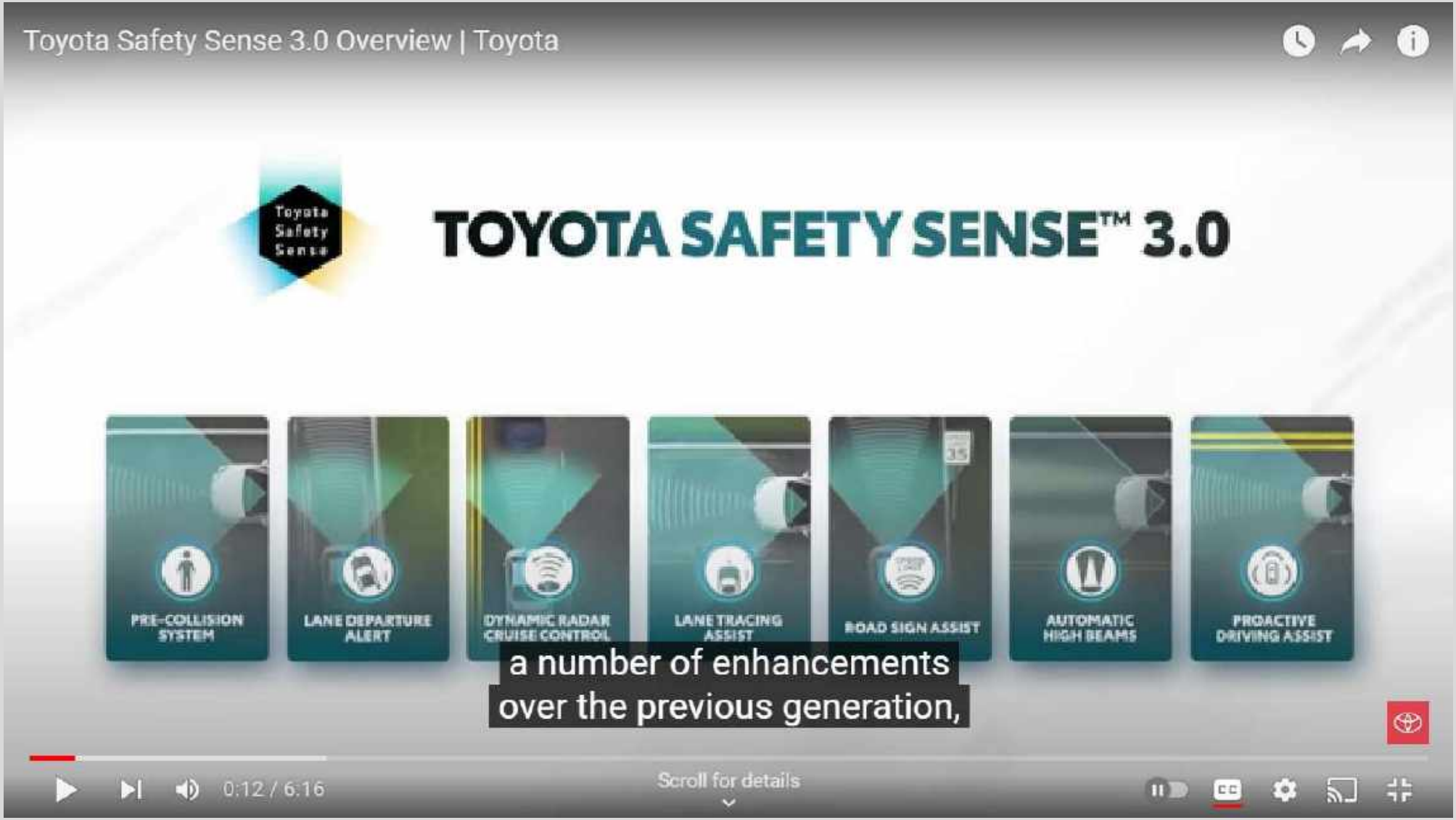


LANE DEPARTURE ALERT WITH STEERING ASSIST (LDA W/SA)

By detecting visible white/yellow lane markings or the road's edge at speeds above 32 mph, LDA w/SA[3] is designed to issue an **audio/visual alert**, as well as **steering wheel vibrations** on select vehicles, if an inadvertent lane departure is detected. If the driver does not take corrective action, the Steering Assist function is designed to provide **gentle corrective steering**.

● Pre-collision braking

Detectable objects	Vehicle speed	Relative speed between your vehicle and object
Preceding and stopped vehicles	Approx. 7 to 110 mph (10 to 180 km/h)	Approx. 7 to 110 mph (10 to 180 km/h)
Oncoming vehicles	Approx. 7 to 110 mph (10 to 180 km/h)	Approx. 13 to 110 mph (20 to 180 km/h)
Bicyclists and pedestrians	Approx. 7 to 50 mph (10 to 80 km/h)	Approx. 7 to 50 mph (10 to 80 km/h)


Toyota Safety Sense 3.0 Overview | Toyota
TOYOTA SAFETY SENSE™ 3.0
PRE-COLLISION SYSTEM
LANE DEPARTURE ALERT
DYNAMIC RADAR CRUISE CONTROL
LANE TRACING ASSIST
ROAD SIGN ASSIST
AUTOMATIC HIGH BEAMS
PROACTIVE DRIVING ASSIST
a number of enhancements
over the previous generation,
0:12 / 6:16
Scroll for details

The 3.0 system is currently only found on a few 2023 models and the 2022 Lexus NX

The system has a wider range of coverage and can detect Motorcycles in some instances

TOYOTA	LEXUS
bZ4X	NX
Corolla	RX

May not operate if ……..

Pre-collision braking

- When the pre-collision braking function is operating, a large amount of braking force will be applied.
- If the vehicle is stopped by the operation of the pre-collision braking function, the pre-collision braking function operation will be canceled after approximately 2 seconds. Depress the brake pedal as necessary.
- The pre-collision braking function may not operate if certain operations are performed by the driver. If the accelerator pedal is being depressed strongly or the steering wheel is being turned, the system may determine that the driver is taking evasive action and possibly prevent the pre-collision braking function from operating.
- In some situations, while the pre-collision braking function is operating, operation of the function may be canceled if the accelerator pedal is depressed strongly or the steering wheel is turned and the system determines that the driver is taking evasive action.
- If the brake pedal is being depressed, the system may determine that the driver is taking evasive action and possibly delay the operation timing of the pre-collision braking function.

Accelerator Pedal

Steering

Brake depressed

- Do not attempt to test the operation of the pre-collision system yourself. Depending on the objects used for testing (dummies, cardboard objects imitating detectable objects, etc.), the system may not operate properly, possibly leading to an accident.

How can you determine what version of the system is on a vehicle

TOYOTA

TSS-C

TSS-P

TSS 2.0

TSS 2.5

TSS 2.5+

TSS 3.0

Owners Manual?

Window Sticker?

Car Brochure?

Lexus has a Similar system
LSS Lexus Safety System

LEXUS

LSS

LSS+

LSS 2.0

LSS 2.5

LSS 3.0



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TOYOTA

Toyota PDF Brochures

4Runner, 86, Avalon, Camry, Celica, CH-R, Corolla, Corolla Cross, Corolla iM, Corona, Echo, FJ Cruiser, Highlander, Land Cruiser, Mark II, Matrix, Mirai, MR2, Prius, RAV4, Sequoia, Sienna, Solara, Supra, Tacoma, Tundra, Venza & Yaris

Toyota Model Range

2022 Toyota Full Line PDF Brochure (3.4 MB)
2021 Toyota Full Line PDF Brochure (3.9 MB)
2020 Toyota Full Line PDF Brochure (7.2 MB)
2019 Toyota Full Line v2 PDF Brochure (9.4 MB)
2019 Toyota Full Line v1 PDF Brochure (5.2 MB)
1986 Toyota Full Line PDF Brochure (7.1 MB)
1984 Toyota Full Line PDF Brochure (5.1 MB)
1975 Toyota Full Line PDF Brochure (38.7 MB)
1970 Toyota Full Line PDF Brochure (3.0 MB)
1962 Toyota Full Line PDF Brochure (27.4 MB)



page 3 - 1975 Toyota Model Range catalogue - Toyota Celica

Toyota Prius PDF Sales Brochures

2022 Toyota Prius PDF Brochure (1.5 MB)
2021 Toyota Prius PDF Brochure (1.5 MB)
2020 Toyota Prius PDF Brochure (2.6 MB)
2019 Toyota Prius PDF Brochure (4.5 MB)
2018 Toyota Prius v2 PDF Brochure (2.1 MB)
2018 Toyota Prius v1 PDF Brochure (1.3 MB)
2017 Toyota Prius PDF Brochure (3.1 MB)
2016 Toyota Prius PDF Brochure (6.5 MB)
2015 Toyota Prius PDF Brochure (1.9 MB)
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2007 Toyota Prius PDF Brochure (0.6 MB)
2006 Toyota Prius v2 PDF Brochure (0.6 MB)
2006 Toyota Prius v1 PDF Brochure (1.0 MB)
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2004 Toyota Prius PDF Brochure (0.2 MB)
2003 Toyota Prius PDF Brochure (0.2 MB)
2002 Toyota Prius PDF Brochure (0.2 MB)
2001 Toyota Prius PDF Brochure (5.7 MB)



Toyota Camry

2022 Toyota Camry PDF Brochure (1.7 MB)
2021 Toyota Camry PDF Brochure (1.6 MB)
2020 Toyota Camry v2 PDF Brochure (5.1 MB)
2020 Toyota Camry v1 PDF Brochure (4.7 MB)
2019 Toyota Camry PDF Brochure (3.9 MB)
2018 Toyota Camry v2 PDF Brochure (3.0 MB)
2018 Toyota Camry v1 PDF Brochure (3.5 MB)
2017 Toyota Camry PDF Brochure (3.2 MB)
2016 Toyota Camry PDF Brochure (3.4 MB)
2015 Toyota Camry PDF Brochure (3.2 MB)
2014.5 Toyota Camry PDF Brochure (3.6 MB)
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2004 Toyota Camry PDF Brochure (0.4 MB)
2003 Toyota Camry PDF Brochure (0.3 MB)
2002 Toyota Camry v2 PDF Brochure (1.6 MB)
2002 Toyota Camry v1 PDF Brochure (0.3 MB)
2001 Toyota Camry v2 PDF Brochure (12.0 MB)
2001 Toyota Camry v1 PDF Brochure (0.2 MB)
1996 Toyota Camry PDF Brochure (2.3 MB)



TOYOTA

2022 Camry

CAMRY MODELS



SE

Includes these key features

Mechanical/Performance

- ✓ Available All-Wheel Drive (AWD)
- ✓ 28 city/39 highway/32 combined EPA-est. mpg[1]

Exterior Features

- ✓ Bi-LED combination headlights with black accents and auto on/off feature
- ✓ Black front grille with sport mesh insert
- ✓ Color-keyed sport side rocker panels
- ✓ 18-in. black machined-finish alloy wheels
- ✓ Color-keyed rear spoiler
- ✓ Single exhaust with dual chrome tips

Interior Features

- ✓ Single-zone automatic climate control
- ✓ Sport SofTex®-trimmed[29] front seats
- ✓ Leather-trimmed tilt/telescopic steering wheel with paddle shifters

Audio Multimedia

- ✓ Audio includes:
  - 7-in. touchscreen and six speakers
  - Android Auto™[30] & Apple CarPlay®[31] compatible
  - SiriusXM®[32] with 3-month All Access trial subscription[5]
- ✓ Connected Services[6] with Safety Connect®[7]

Safety Features

- ✓ Toyota Safety Sense 2.5+ (TSS 2.5+)[8]
- ✓ Dynamic Radar Cruise Control (DRCC)[26]
- ✓ Star Safety System™
- ✓ Ten airbags[27]
- ✓ Tire Pressure Monitor System (TPMS)[28] with direct pressure readout

Safety Features

- ✓ Toyota Safety Sense 2.5+ (TSS 2.5+)[18]
- ✓ Full-Speed Dynamic Radar Cruise Contol (DRCC)[20]
- ✓ Blind Spot Monitor (BSM)[32] with Rear Cross-Traffic Alert (RCTA)[33]
- ✓ Star Safety System™
- ✓ Ten airbags[27]
- ✓ Tire Pressure Monitor System (TPMS)[28] with direct pressure readout

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Activate Windows
Go to Settings to activate Windows.

Year: 2021
Make: Toyota
Model: Camry SE SEDAN
VIN: 4T1G11AK8MU470837

Engine: 2.5L 4-Cyl. Dynamic Force Engine
Transmission: 8AT
Exterior: Supersonic Red (3U5)
Interior: Black Sof Tex®

Safety And Convenience

- Toyota Safety Sense™ 2.5+ — Pre-Collision System with Pedestrian Detection , Lane Departure Alert with Steering Assist , Automatic High Beams , Lane Tracing Assist , Road Sign Assist and Dynamic Radar Cruise Control
- Dynamic Radar Cruise Control
- Tire Pressure Monitor System with direct pressure readout
- Hill Start Assist Control
- Child-protector rear door locks and power window lockout control
- Star Safety System™ — includes Enhanced Vehicle Stability Control , Traction Control , 4-wheel Anti-lock Brake System , Electronic Brake-force Distribution , Brake Assist and Smart Stop Technology®
- Energy-absorbing collapsible steering column
- Automatic engine shut off
- Side-impact door beams
- Rear-Seat Reminder
- Front and rear energy-absorbing crumple zones
- 3-point seatbelts for all seating positions; driver-side Emergency Locking Retractor and Automatic/Emergency Locking Retractor on all passenger seatbelts
- Anti-theft system with alarm
- LATCH includes lower anchors on outboard rear seats
- Driver and front passenger Whiplash-Injury-Lessening seats
- Ten airbags — includes driver and front passenger Advanced Airbag System, driver and front passenger seat-mounted side airbags, driver and front passenger knee airbags, rear seat-mounted side airbags, and front and rear side curtain airbags

Interior

- Covered center console, armrest and storage
- One 12V auxiliary power outlet
- One USB charge and media port
- Interior silver door handles
- Shoulder room
- Leg room
- Soft-material upper door trim
- One USB-C charge port
- Power windows with 4-window auto up/down, jam protection and retained-power features
- LED-illuminated glove compartment
- Audio — 7-in. touchscreen, six speakers, Android Auto™ & Apple CarPlay® & Amazon Alexa compatible, USB media port, two USB charge ports, hands-free phone capability and music streaming via Bluetooth® wireless technology. SiriusXM® with 3-month All Access trial subscription. See toyota.com/audio-multimedia for details. Connected Services — Safety Connect® with 1-year trial, , Wi-Fi Connect with up to 2 GB within 3-month trial. See toyota.com/connected-services for details.
- Toyota + Alexa App

Exterior

- Color-keyed outside door handles
- 18-in. black machined-finish alloy wheels with P235/45R18 tires
- Black front grille with sport mesh insert
- Color-keyed power outside mirrors
- Bi-LED combination headlights with black accents and auto on/off feature
- Color-keyed rear spoiler
- Single exhaust with dual chrome tips
- Track
- Color-keyed sport side rocker panels
- Overall length
- LED combination taillights with black accents
- Shark-fin antenna
- Overall width
- Overall height, unloaded
- Ground clearance
- Wheelbase

CITY MPG	HIGHWAY MPG
28	39

Fuel Economy Information

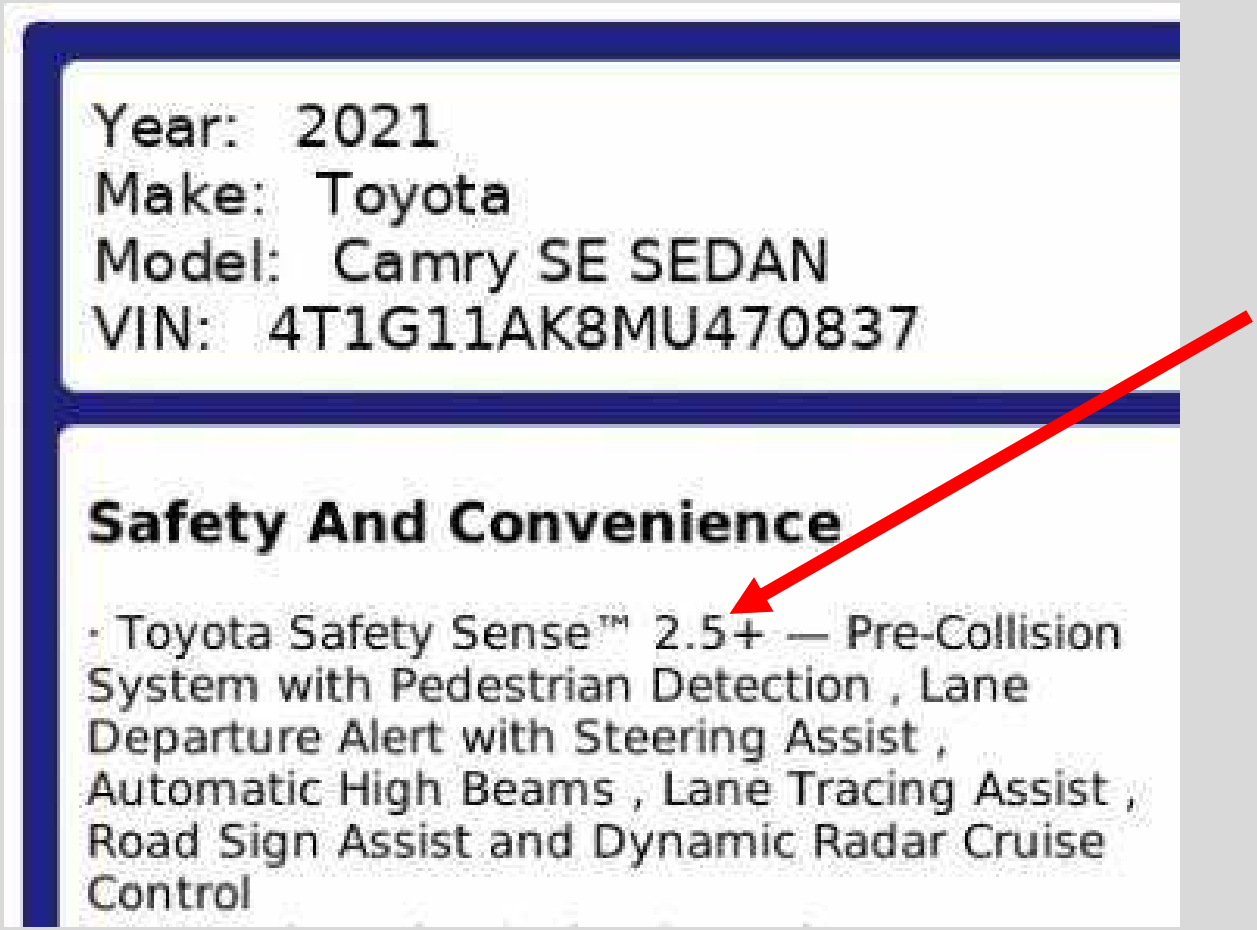
Actual mileage will vary with options, driving conditions, driving habits and vehicle's condition

New

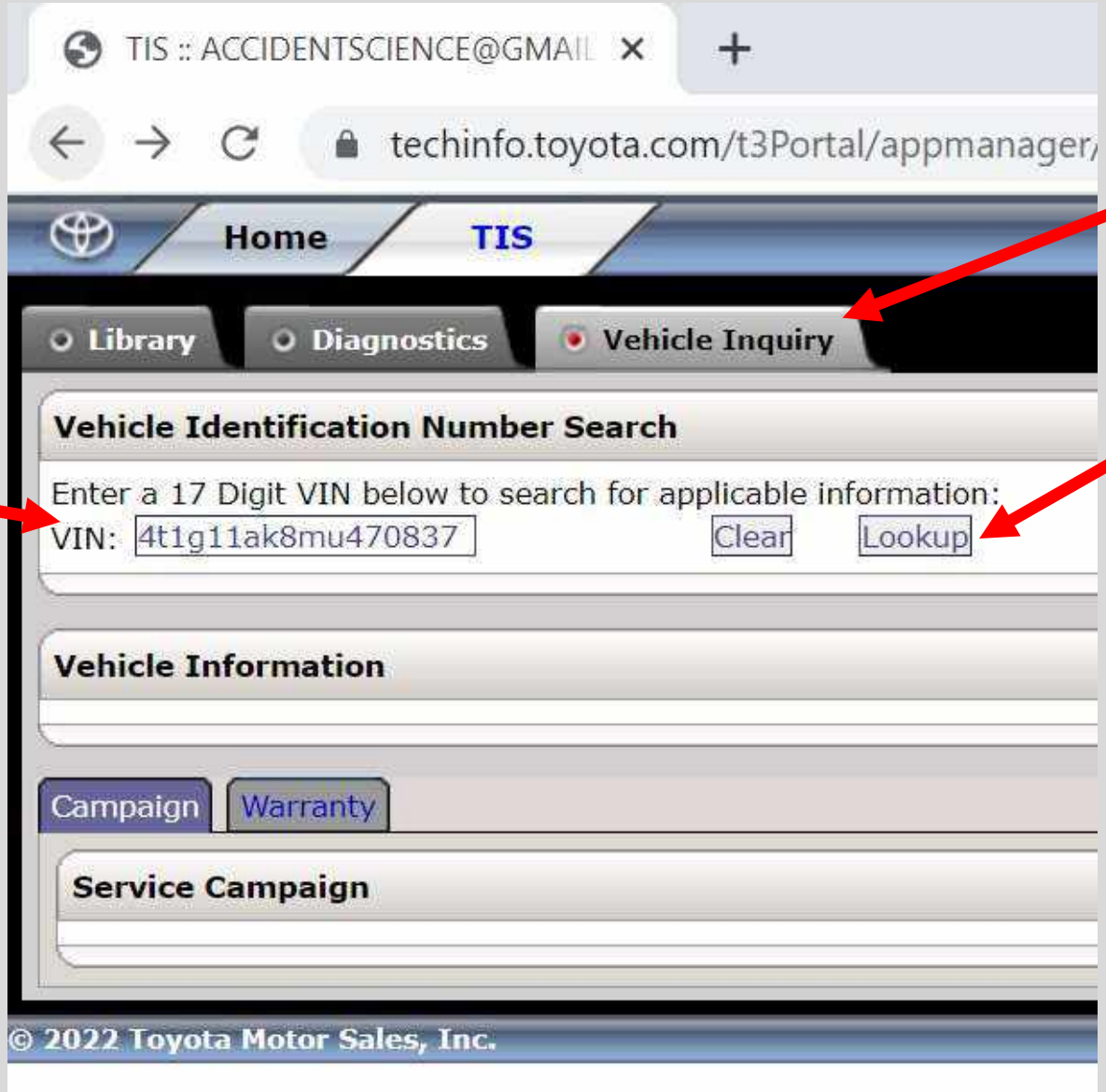
MSRP	**$26,560.00**
INSTALLED OPTIONS	
[FE AT] 50 State Emissions	$0
[H6 CT] Rent A Car Program	$0
[H8 CT] Fleet Credit	$0
[PC AT] Special Color	$425
Original Shipping Charge	$1,025
RETAIL PRICE (ORIGINALLY NEW)	$28,010.00

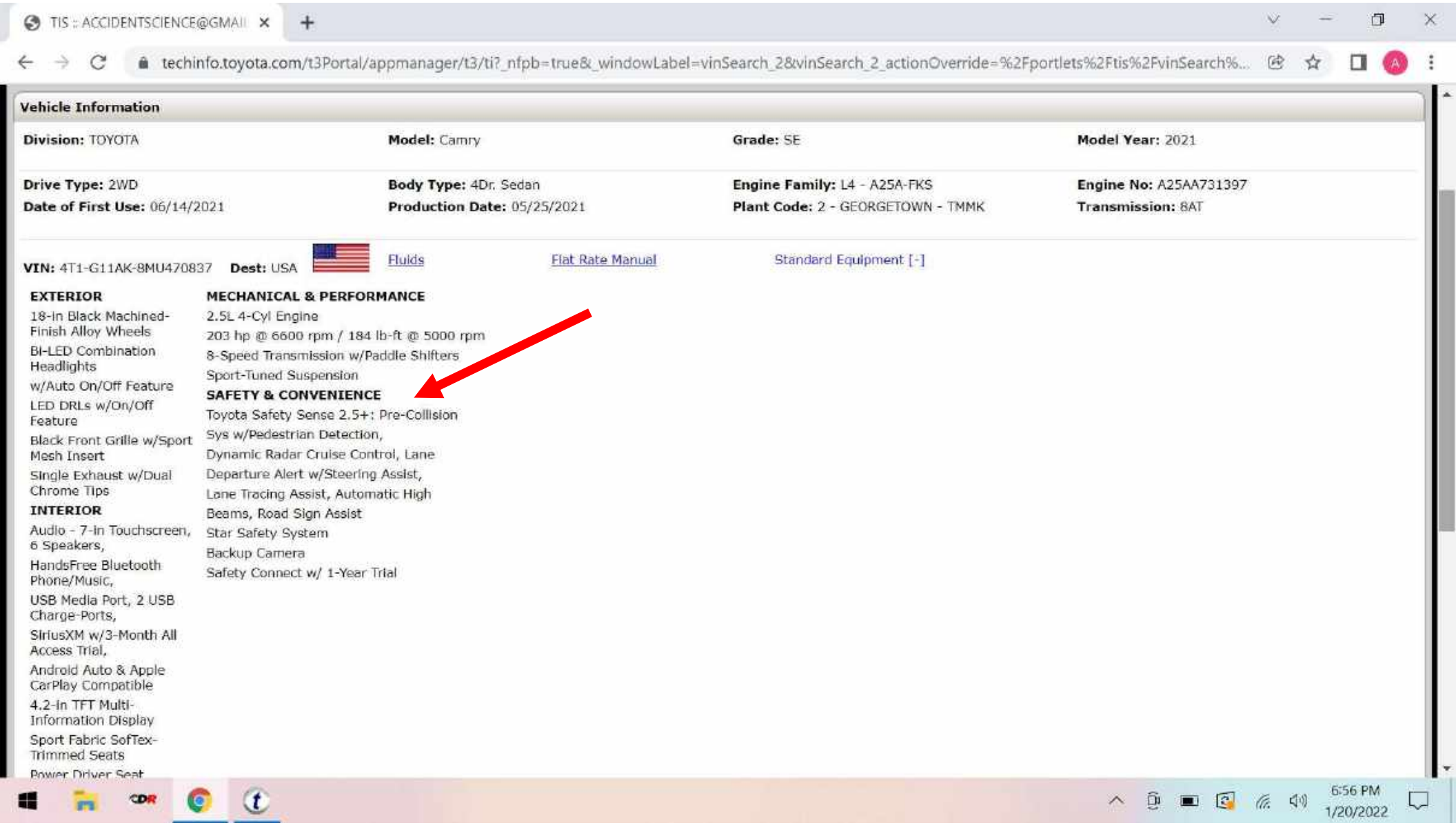
Get more information on your smartphone:

ate Windows
Settings to activate Windows.


Year: 2021
Make: Toyota
Model: Camry SE SEDAN
VIN: 4T1G11AK8MU470837
Safety And Convenience
· Toyota Safety Sense™ 2.5+ — Pre-Collision System with Pedestrian Detection , Lane Departure Alert with Steering Assist , Automatic High Beams , Lane Tracing Assist , Road Sign Assist and Dynamic Radar Cruise Control

Enter the VIN




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Vehicle Information
Division: TOYOTA
Model: Camry
Grade: SE
Model Year: 2021
Drive Type: 2WD
Body Type: 4Dr. Sedan
Engine Family: L4 - A25A-FKS
Engine No: A25AA731397
Date of First Use: 06/14/2021
Production Date: 05/25/2021
Plant Code: 2 - GEORGETOWN - TMMK
Transmission: 8AT
VIN: 4T1-G11AK-8MU470837
Dest: USA
Fluids
Flat Rate Manual
Standard Equipment [-]
EXTERIOR
18-In Black Machined-Finish Alloy Wheels
Bi-LED Combination Headlights
w/Auto On/Off Feature
LED DRLs w/On/Off Feature
Black Front Grille w/Sport Mesh Insert
Single Exhaust w/Dual Chrome Tips
INTERIOR
Audio - 7-in Touchscreen, 6 Speakers,
HandsFree Bluetooth Phone/Music,
USB Media Port, 2 USB Charge-Ports,
SiriusXM w/3-Month All Access Trial,
Android Auto & Apple CarPlay Compatible
4.2-in TFT Multi-Information Display
Sport Fabric SofTex-Trimmed Seats
Power Driver Seat
MECHANICAL & PERFORMANCE
2.5L 4-Cyl Engine
203 hp @ 6600 rpm / 184 lb-ft @ 5000 rpm
8-Speed Transmission w/Paddle Shifters
Sport-Tuned Suspension
SAFETY & CONVENIENCE
Toyota Safety Sense 2.5+: Pre-Collision Sys w/Pedestrian Detection, Dynamic Radar Cruise Control, Lane Departure Alert w/Steering Assist, Lane Tracing Assist, Automatic High Beams, Road Sign Assist
Star Safety System
Backup Camera
Safety Connect w/ 1-Year Trial
6:56 PM
1/20/2022

SAFETY & CONVENIENCE

Toyota Safety Sense 2.5+: Pre-Collision Sys w/Pedestrian Detection, Dynamic Radar Cruise Control, Lane Departure Alert w/Steering Assist, Lane Tracing Assist, Automatic High Beams, Road Sign Assist

Star Safety System

Backup Camera

Safety Connect w/ 1-Year Trial

TOYOTA/LEXUS Vehicles US Market

Gen 1 TSS/LSS pictures only with auto braking, Gen 2 pictures in more circumstances

	#	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
LEXUS	CT200h						discontinued					
	ES				LSS	LSS	LSS	?	?	?	2.5	
	GS				LSS	LSS	LSS	?	?	discontinued		
	GX								LSS	LSS	LSS	
	IS					LSS	LSS	?	?	?	2.5	
	LC						LSS	LSS	LSS	LSS	LSS	
	LS						LSS+	2.0	2.0	2.0	2.5	
	LX				LSS	LSS	LSS	?	?	?	2.5	
	NX						LSS	?	?	?	3.0	
	RC-F						LSS	?	?	?	LSS	
	RX				LSS	LSS	LSS	?	?	?	2.0	
	UX										2.0	
TOYOTA	4Runner								P	P	P	
	86										Subaru	
	Avalon				P	P	P	P	P	P	2.5+	
	Camry						P	P	P	2.5+	2.5+	
	C-HR						P	P	P	2.5	2.5	
	Corolla					P/C	P	2.0	2.0	2.0	2.0	
	Echo											
	FJ											
	Highlander					P	P	P	2.0	2.5+	2.5+	
	Land Cruiser				P	P	P	P	P	P		
	Mirai							2.0	2.0	2.5+	2.5+	
	Prius				P/C	P/C	P	P/C	P	2.0	2.0	
	RAV4				P/C	P/C	P	2.0	2.0	2.0	2.0	
	Sequoia						P	P/C	P	P	P	
	Sienna						P	P/C	P	2.0	2.0	
	Supra (BMW)								(made by BMW)			
	Tacoma						P	P	P	P	P	
	Tundra						P	P	P	P	2.5	
	Venza			out of production						2.0	2.0	
	Yaris					C	C					

Chart is only inteded to cover US models. Content MAY depend on trim level or model.

LSS = Lexus Safety System , Gen 1.0 or higher

P, C = Toyota Safety System, Gen 1.0

2.0 Toyota or Lexus Safety System Gen 2.0

2.5 Toyota or Lexus Safety System Gen 2.5

3.0 Toyota or Lexus Safety System, Gen 3.0

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Accessing the data from the Toyota and Lexus vehicle requires the TIS techstream software,
Professional Diagnostic version. Also using the CDR 900 or Mongoose Cable

Software cost
2 Day $70
Yearly $1,360

Information on the software and downloads

Collison Magazine

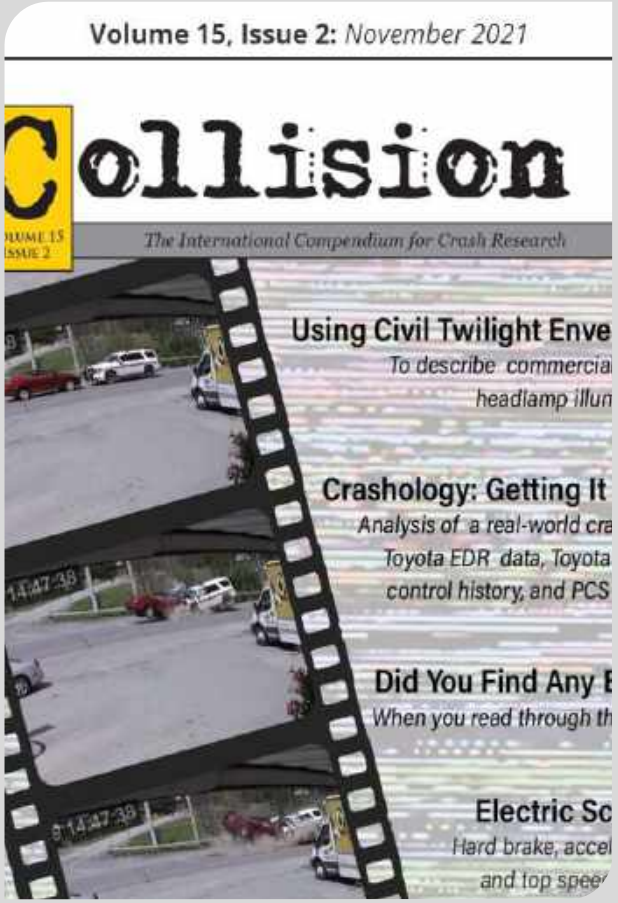
Volume 15, Issue 2

Crashology, Bob Anderson

Collision Magazine

Volume 14, Issue 1

How-To Guide, Alan Moore



2017 Toyota Corolla TSS-P





2021 Toyota Rav4 Prime TSS 2.0





2021 Toyota Camry TSS 2.5+





2021 Toyota Camry TSS 2.5+

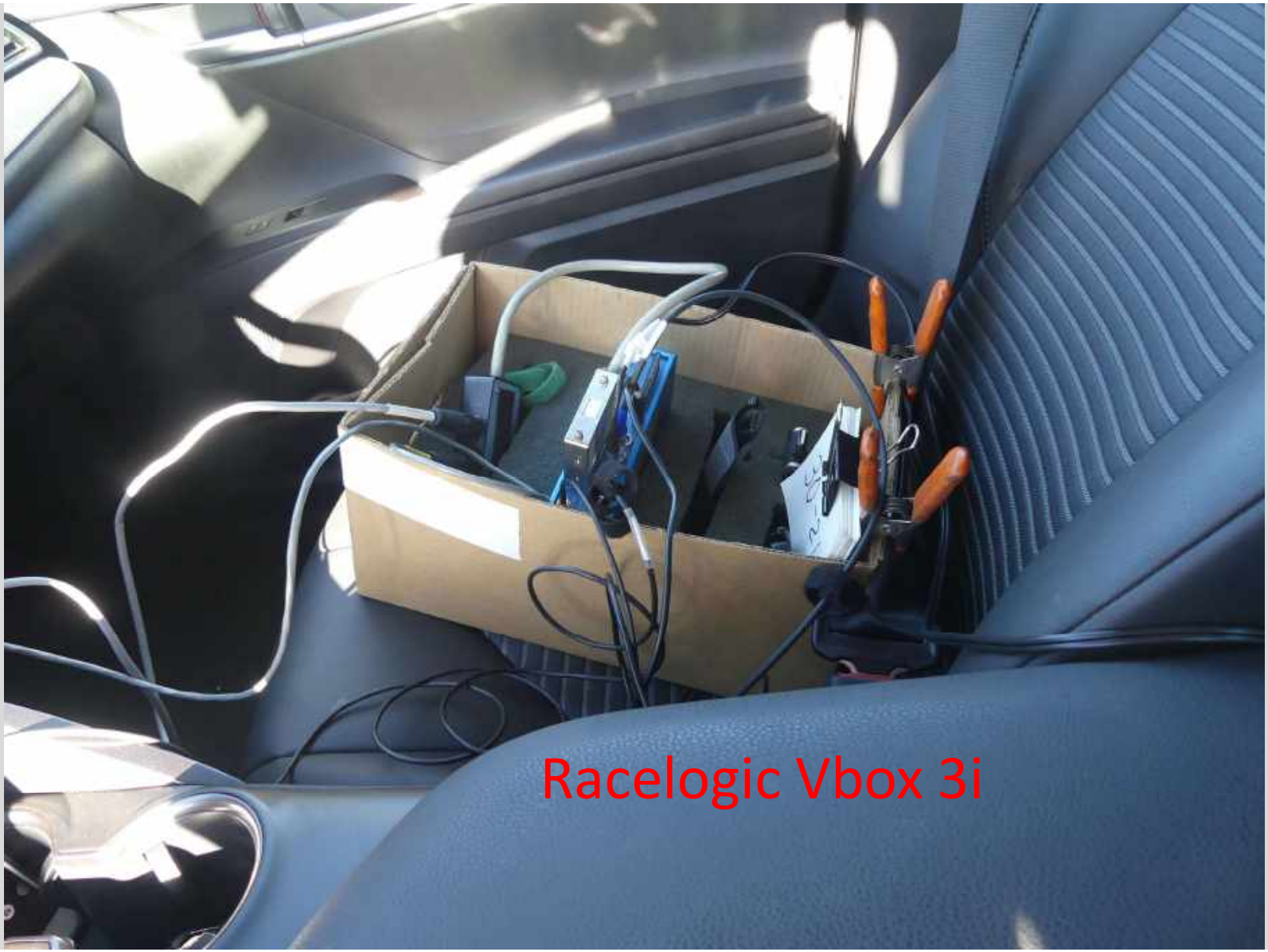




Instrumentation

- Vbox 3i, 100 hz
- IMU
- Load Cell on the brake pedal
- Tape Switch on the front bumper
- Tap into the accelerator pedal sensor
- Tap into the brake light wires

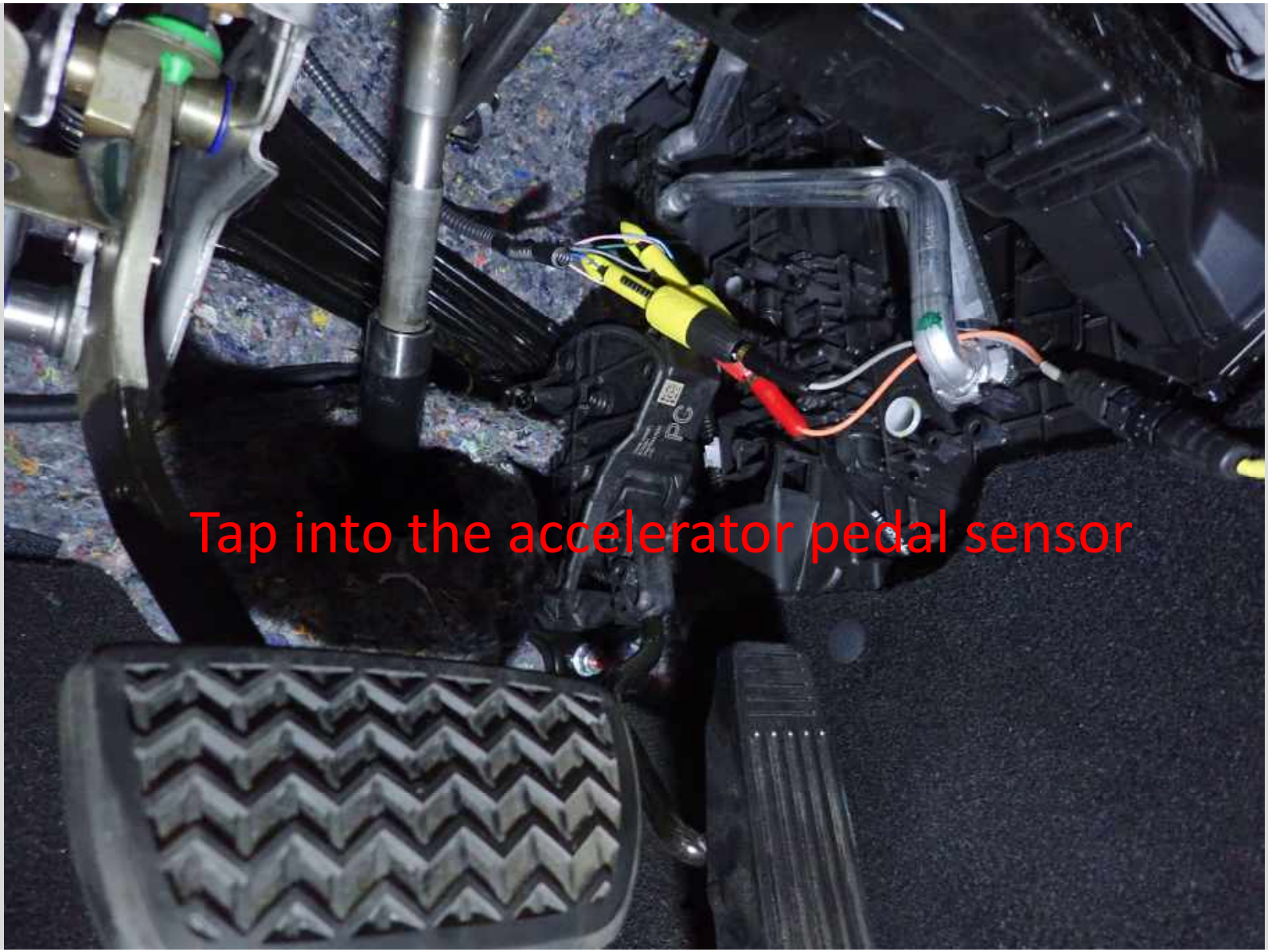



Racelogic Vbox 3i


IMU mounted in the center console
RACELOGIC
VBOX
IMU04


Load Cell on the brake pedal


Tape-switch on the front bumper


Tap into the accelerator pedal sensor

Tap into the brake light wires





Pomona taps are awesome






GoPro camera
Bubble wrap on the hood just in case

For the tests, the PCS event was triggered by a Pedestrian target

<u>Design criteria</u>

Life size

Light weight

“Free-standing” and does not blow over

Does not damage the car

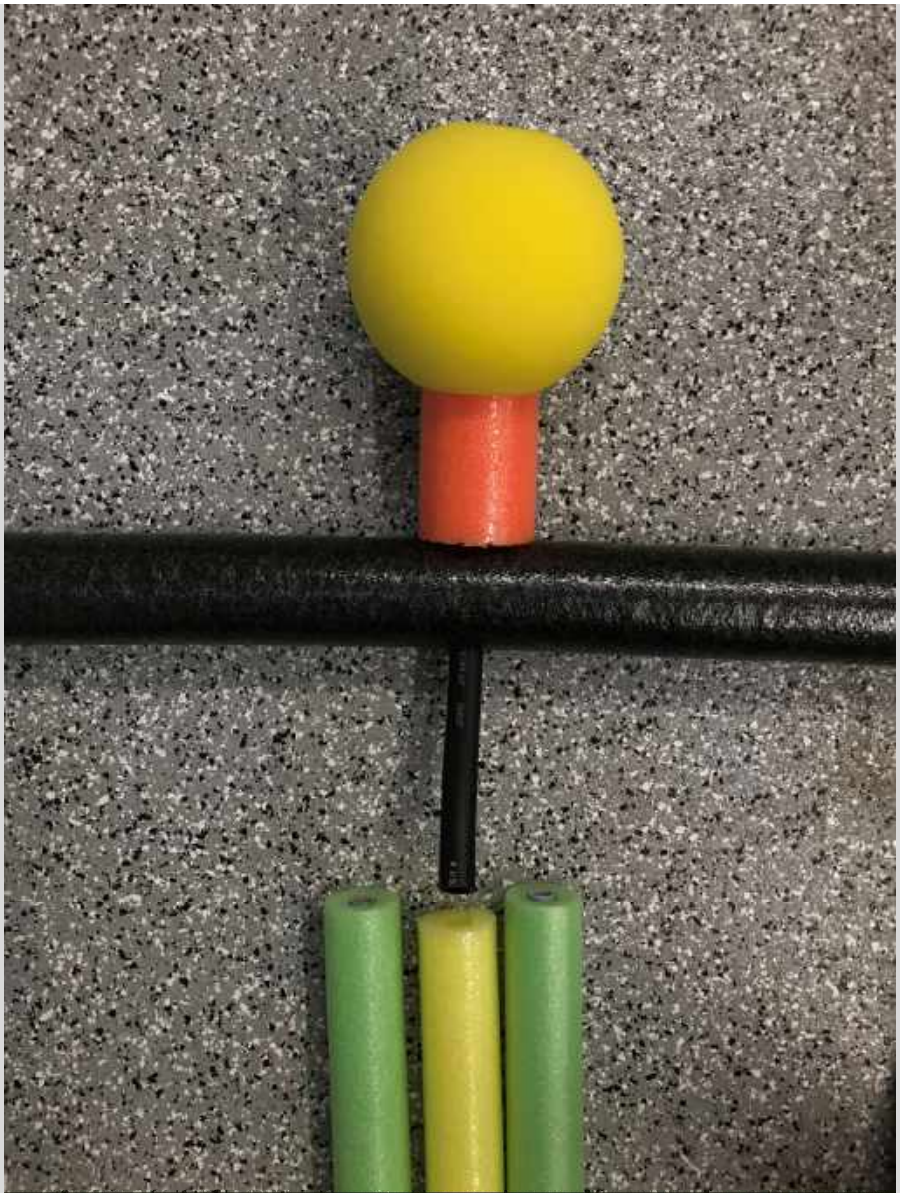
Easy to set up

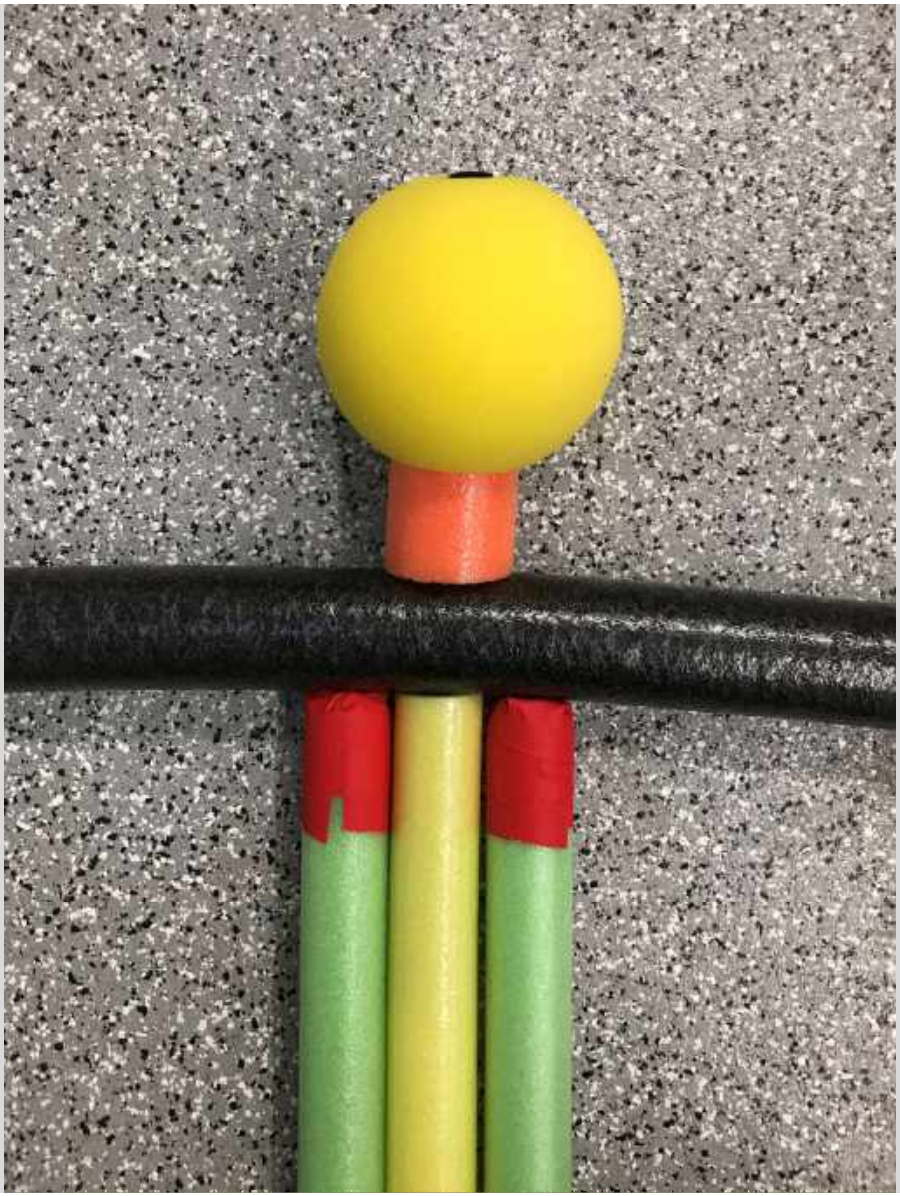
















Test Runs

- We have conducted approximately 60 tests with instrumented Toyota vehicles
- The primary focus was on the triggered PCS event
- Also studied was the Sudden Braking History event and ABS Operation event that are triggered during the PCS event


Test 45-01
Test numbering
45 mph
01 the first test at that speed


Test 35-23 some of you older guys might recognize the crash dummy


Test 35-23


Test 45-01

What happens when approaching a pedestrian with the cruise control activated?
TSS-P System

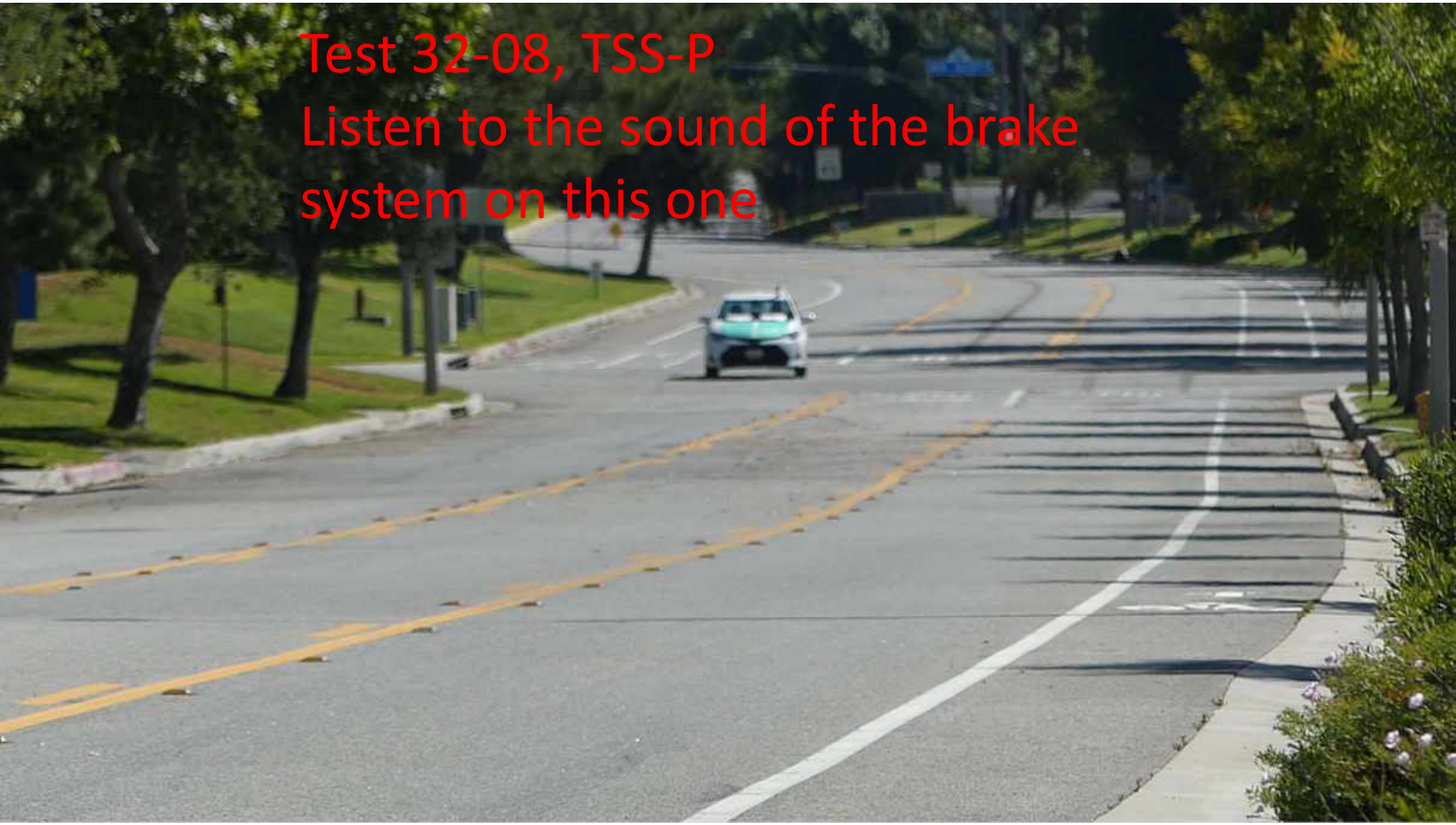

Test 32-04, TSS-P


Test 32-05, TSS-P


Test 32-06, TSS-P

Additional testing with the TSS-P System


Test 45-01, TSS-P


Test 32-08, TSS-P
Listen to the sound of the brake system on this one

Testing with the TSS-2.5+ System


Test 35-21, TSS 2.5+


Test 35-23, TSS 2.5+


Test 35-22, TSS 2.5+

Will the brake lights come on when the automatic braking is activated?

At what point will the brake lights come on?


Test 30-25
The system will continue to apply the brakes
for 2 seconds after the vehicle comes to a stop

PCS Data Viewer - 4T1G11AK8MU470837_20220123111621.vdas

File Help

File Information when GTS read out PCS Data

VIN	4T1G11AK8MU470837		
Key Cycle	1377	Dealer Code	123
ODO	17587mile	Retrieval Date	20220123

Airbag	
Parts No.	06A40
Software	8917F06470

VCH or FFD Data Set in PCS Data

Will store the images for the 3 most recent PCS events

VCH FFD

Key Cycle	Elapsed Time	Clock Type	Time and Date	Trigger	ODO	Number of Overwrite of the Recording Area	Image FFD	Image FFD Trigger
1371	00:02:50.3	-	2022-01-23-10:07:08.200	PCS operation history (warning buzzer operation)	17580	3	-	-
1374	00:02:50.7	-	2022-01-23-10:57:01.800	PCS operation history (warning buzzer operation)	17581	3	Front Recognition Camera	PCS operation history (prior brake operation)
1374	00:10:03.7	-	2022-01-23-11:04:14.800	PCS operation history (warning buzzer operation)	17581	3	Front Recognition Camera	PCS operation history (pre-collision brake assist operation)
1374	00:19:12.3	-	2022-01-23-11:13:23.400	PCS operation history (warning buzzer operation)	17582	3	Front Recognition Camera	PCS operation history (pre-collision brake assist operation)
1087	00:24:39.1	-	2021-12-05-19:02:29.500	PCS operation history (warning buzzer operation)	14093	2	-	-
				PCS operation				

PCS Data Viewer - 4T1G11AK8MU470837_20220123111621.vdas

File Help

File Information when GTS read out PCS Data

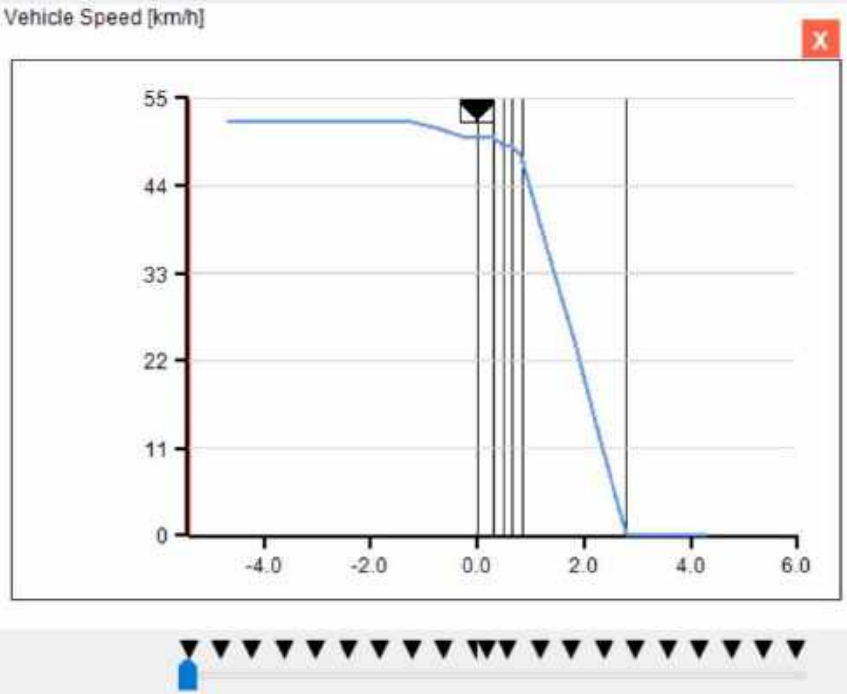
VIN	4T1G11AK8MU470837		
Key Cycle	1377	Dealer Code	123
ODO	17587mile	Retrieval Date	20220123

Airbag	
Parts No.	06A40
Software	8917F06470

Vehicle Control History

Key Cycle 1374
Trigger PCS operation history (warning buzzer operation)
Elapsed Time 00:10:03.7

Parameter	Value
Vehicle Speed [km/h]	-
Accelerator Opening Ratio [%]	-
Engine RPM Data [rpm]	-
Cruise Control	-
Shift Position Signal	-
Brake SW	-
Brake Oil Pressure [MPa]	-
Yaw Rate Sensor Signal [deg/s]	-
Longitudinal G Sensor Signal [m/s^2]	-
Lateral G Sensor Signal [m/s^2]	-
Throttle Opening Ratio [%]	-



Imge FFD(Film Advance)



Image Number(1 /22)

Time Stamp : -5.400s

PCS Data Viewer - 4T1G11AK8MU470837_20220123111621.vdas

File Help

File Information when GTS read out PCS Data

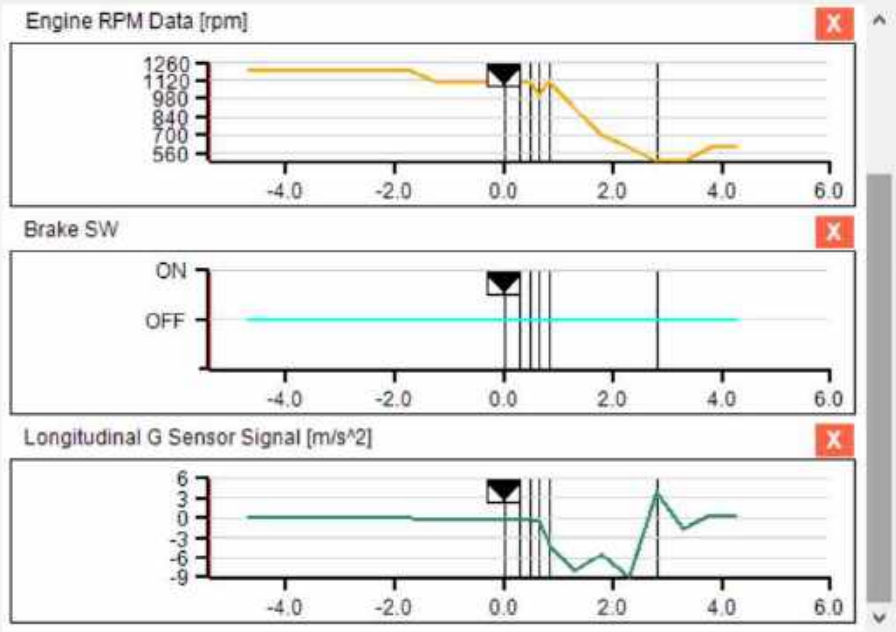
VIN	4T1G11AK8MU470837		
Key Cycle	1377	Dealer Code	123
ODO	17587mile	Retrieval Date	20220123

Airbag	
Parts No.	06A40
Software	8917F06470

Vehicle Control History

Key Cycle 1374
Trigger PCS operation history (warning buzzer operation)
Elapsed Time 00:10:03.7

Parameter	Value
Vehicle Speed [km/h]	-
Engine RPM Data [rpm]	-
Brake SW	-
Longitudinal G Sensor Signal [m/s^2]	-
Accelerator Opening Ratio [%]	-
Cruise Control	-
Shift Position Signal	-
Brake Oil Pressure [MPa]	-
Yaw Rate Sensor Signal [deg/s]	-
Lateral G Sensor Signal [m/s^2]	-
Throttle Opening Ratio [%]	-



Imge FFD(Film Advance)



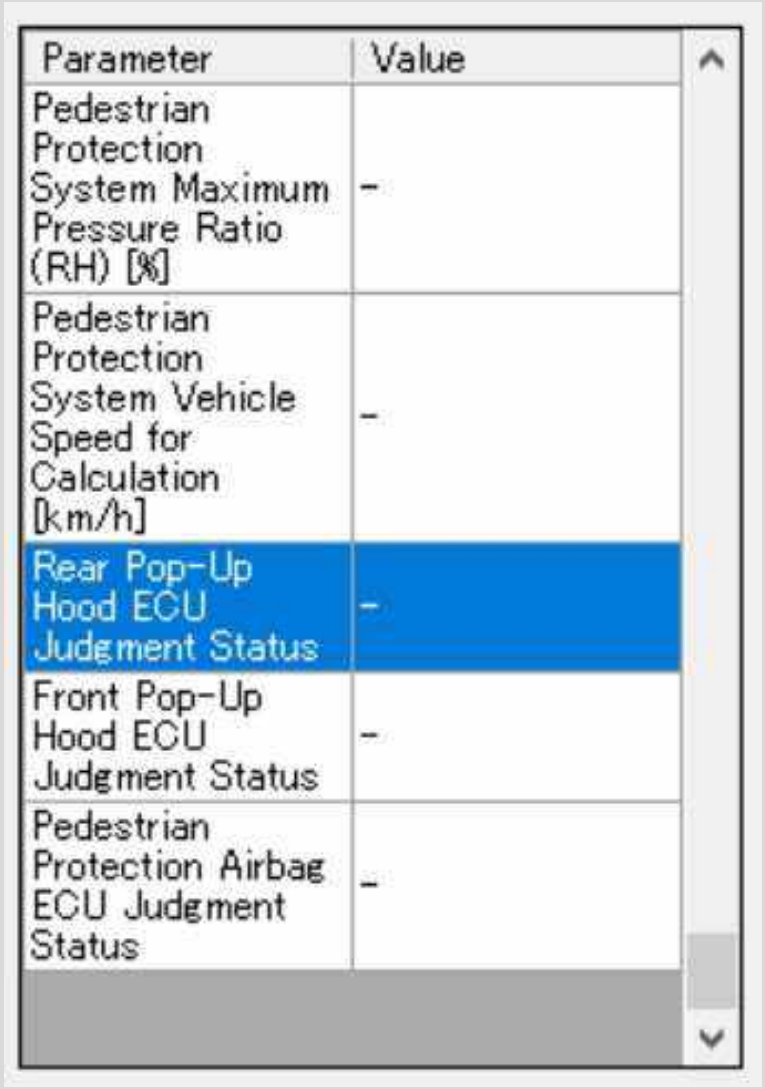
Image Number(1 /22)

Time Stamp : -5.400s

Parameter	Value
Vehicle Speed [km/h]	-
Accelerator Opening Ratio [%]	-
Engine RPM Data [rpm]	-
Cruise Control	-
Shift Position Signal	-
Brake SW	-
Brake Oil Pressure [MPa]	-
Yaw Rate Sensor Signal [deg/s]	-
Longitudinal G Sensor Signal [m/s^2]	-
Lateral G Sensor Signal [m/s^2]	-
Throttle Opening Ratio [%]	-

Parameter	Value
Steering Signal [deg]	-
HV Recognition Flag	-
Diesel Flag	-
Sensor Type Identification (Security)	-
Sensor Type Identification (G Sensor Layout)	-
Sensor Type Identification (G Sensor)	-
Sensor Type Identification (Yaw Rate Sensor)	-
Yaw G Sensor Installation Information	-
PCS Reception	-

Parameter	Value
ALM Request Flag	-
ALM Request Flag 2	-
ABK Request Flag	-
PBA Request Flag	-
FPB Request Flag	-
PB Request Flag	-
PSB Request Flag	-
Prefill Request Flag	-
Stop Lamp Lighting Request Flag	-
PCS Brake Hold Request Flag	-
Collision Avoidance Assist	-



Parameter	Value
Pedestrian Protection System Maximum Pressure Ratio (RH) [%]	-
Pedestrian Protection System Vehicle Speed for Calculation [km/h]	-
Rear Pop-Up Hood ECU Judgment Status	-
Front Pop-Up Hood ECU Judgment Status	-
Pedestrian Protection Airbag ECU Judgment Status	-



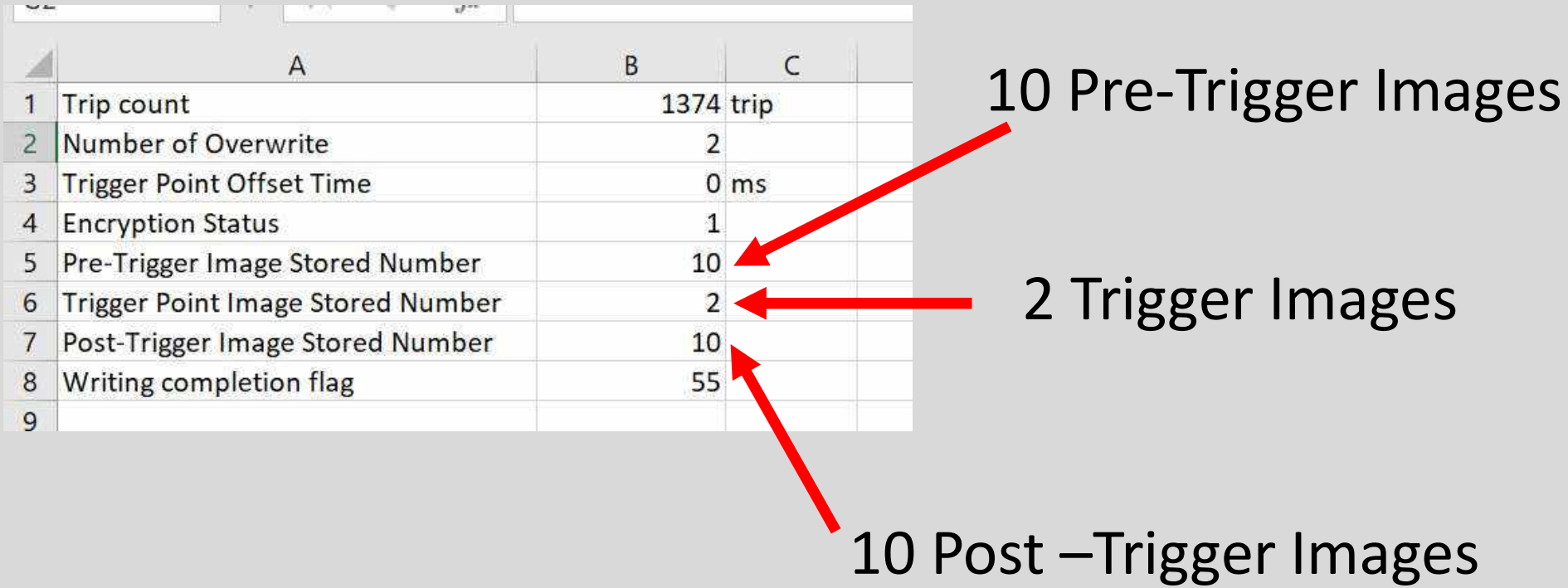
About 22 images with file names and an Excel data file

	A	B	C	D
1	Trip count	1374	trip	
2	Number of Overwrite	2		
3	Trigger Point Offset Time	0	ms	
4	Encryption Status	1		
5	Pre-Trigger Image Stored Number	10		
6	Trigger Point Image Stored Number	2		
7	Post-Trigger Image Stored Number	10		
8	Writing completion flag	55		
9				
10	Trigger ID	Time Stam	File Name	
11	18_3	2022-01-2	010.bmp	
12	18_3	2022-01-2	011.bmp	
13	18_3	2022-01-2	012.bmp	
14	18_3	2022-01-2	013.bmp	
15	18_3	2022-01-2	014.bmp	
16	18_3	2022-01-2	015.bmp	
17	18_3	2022-01-2	016.bmp	
18	18_3	2022-01-2	017.bmp	
19	18_3	2022-01-2	018.bmp	
20	18_3	2022-01-2	019.bmp	

Set2

Excel File

	A	B	C
1	Trip count	1374	trip
2	Number of Overwrite	2	
3	Trigger Point Offset Time	0	ms
4	Encryption Status	1	
5	Pre-Trigger Image Stored Number	10	
6	Trigger Point Image Stored Number	2	
7	Post-Trigger Image Stored Number	10	
8	Writing completion flag	55	
9			

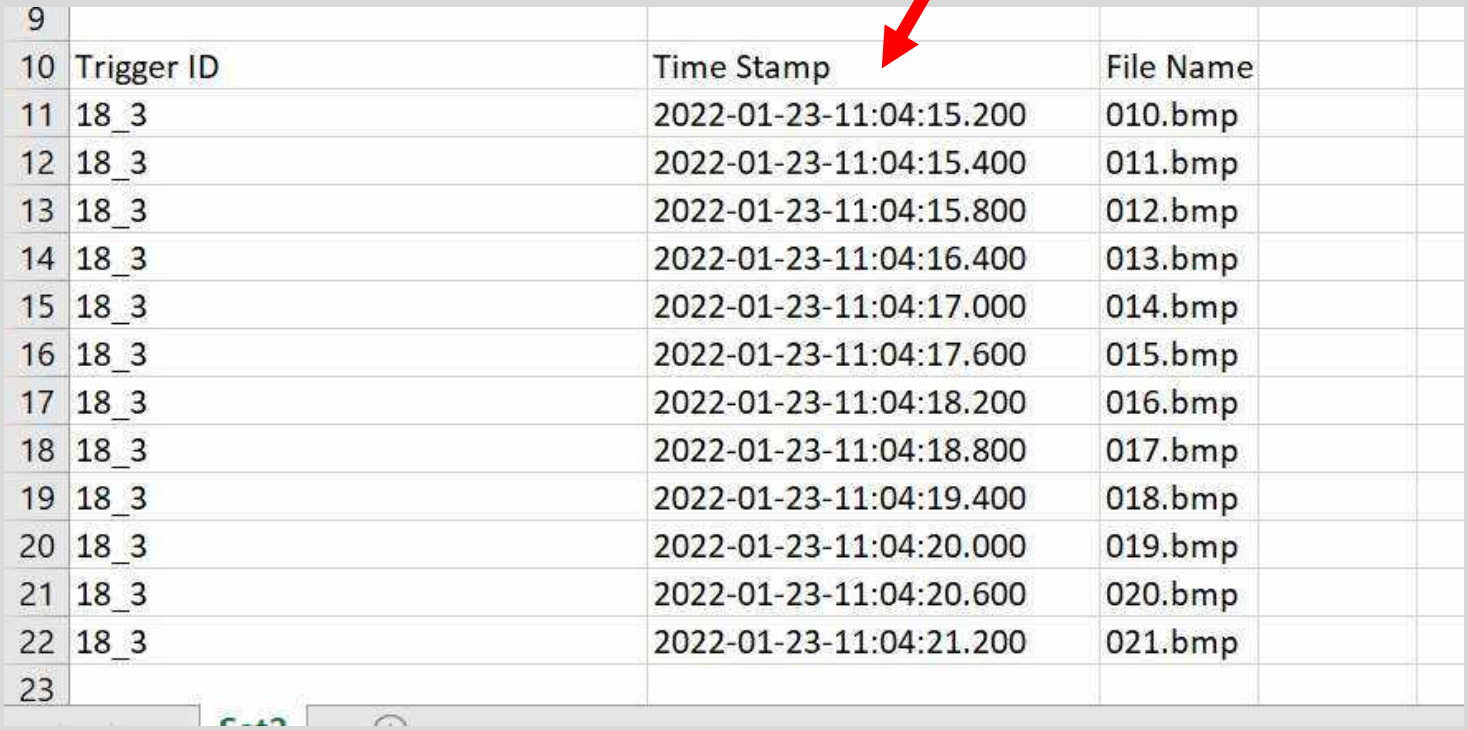


	A	B	C
1	Trip count	1371	trip
2	Number of Overwrite	1	
3	Trigger Point Offset Time	400	ms
4	Encryption Status	1	
5	Pre-Trigger Image Stored Number	10	
6	Trigger Point Image Stored Number	3	
7	Post-Trigger Image Stored Number	9	
8	Writing completion flag	55	
9			



3 Trigger images

Date and Time



	Trigger ID	Time Stamp	File Name
9			
10	Trigger ID	Time Stamp	File Name
11	18_3	2022-01-23-11:04:15.200	010.bmp
12	18_3	2022-01-23-11:04:15.400	011.bmp
13	18_3	2022-01-23-11:04:15.800	012.bmp
14	18_3	2022-01-23-11:04:16.400	013.bmp
15	18_3	2022-01-23-11:04:17.000	014.bmp
16	18_3	2022-01-23-11:04:17.600	015.bmp
17	18_3	2022-01-23-11:04:18.200	016.bmp
18	18_3	2022-01-23-11:04:18.800	017.bmp
19	18_3	2022-01-23-11:04:19.400	018.bmp
20	18_3	2022-01-23-11:04:20.000	019.bmp
21	18_3	2022-01-23-11:04:20.600	020.bmp
22	18_3	2022-01-23-11:04:21.200	021.bmp
23			

Only the Trigger and Post-Trigger Images have a timestamp

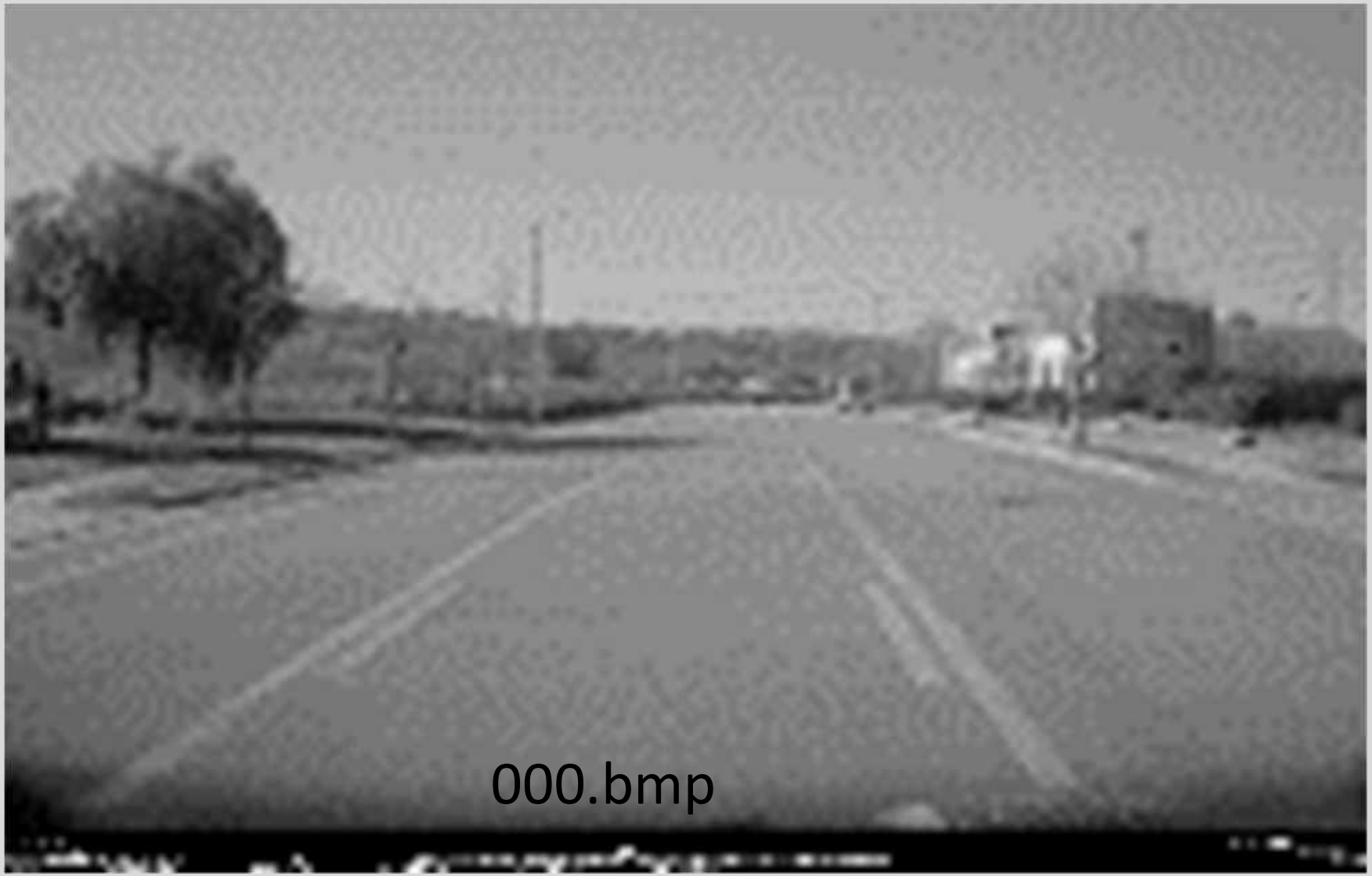
	Trigger ID	Time Stamp	File Name
11	18_3	2022-01-23-11:04:15.200	010.bmp
12	18_3	2022-01-23-11:04:15.400	011.bmp
13	18_3	2022-01-23-11:04:15.800	012.bmp
14	18_3	2022-01-23-11:04:16.400	013.bmp
15	18_3	2022-01-23-11:04:17.000	014.bmp
16	18_3	2022-01-23-11:04:17.600	015.bmp
17	18_3	2022-01-23-11:04:18.200	016.bmp
18	18_3	2022-01-23-11:04:18.800	017.bmp
19	18_3	2022-01-23-11:04:19.400	018.bmp
20	18_3	2022-01-23-11:04:20.000	019.bmp
21	18_3	2022-01-23-11:04:20.600	020.bmp
22	18_3	2022-01-23-11:04:21.200	021.bmp

Timestep 0.200 sec.

Timestep 0.400 sec.

Timestep 0.600 sec.

Usually between the 2nd and 3rd sample or as late as between the 4th and 5th sample the timestep continues at 0.600 seconds


000.bmp


001.bmp


002.bmp


003.bmp


004.bmp


005.bmp

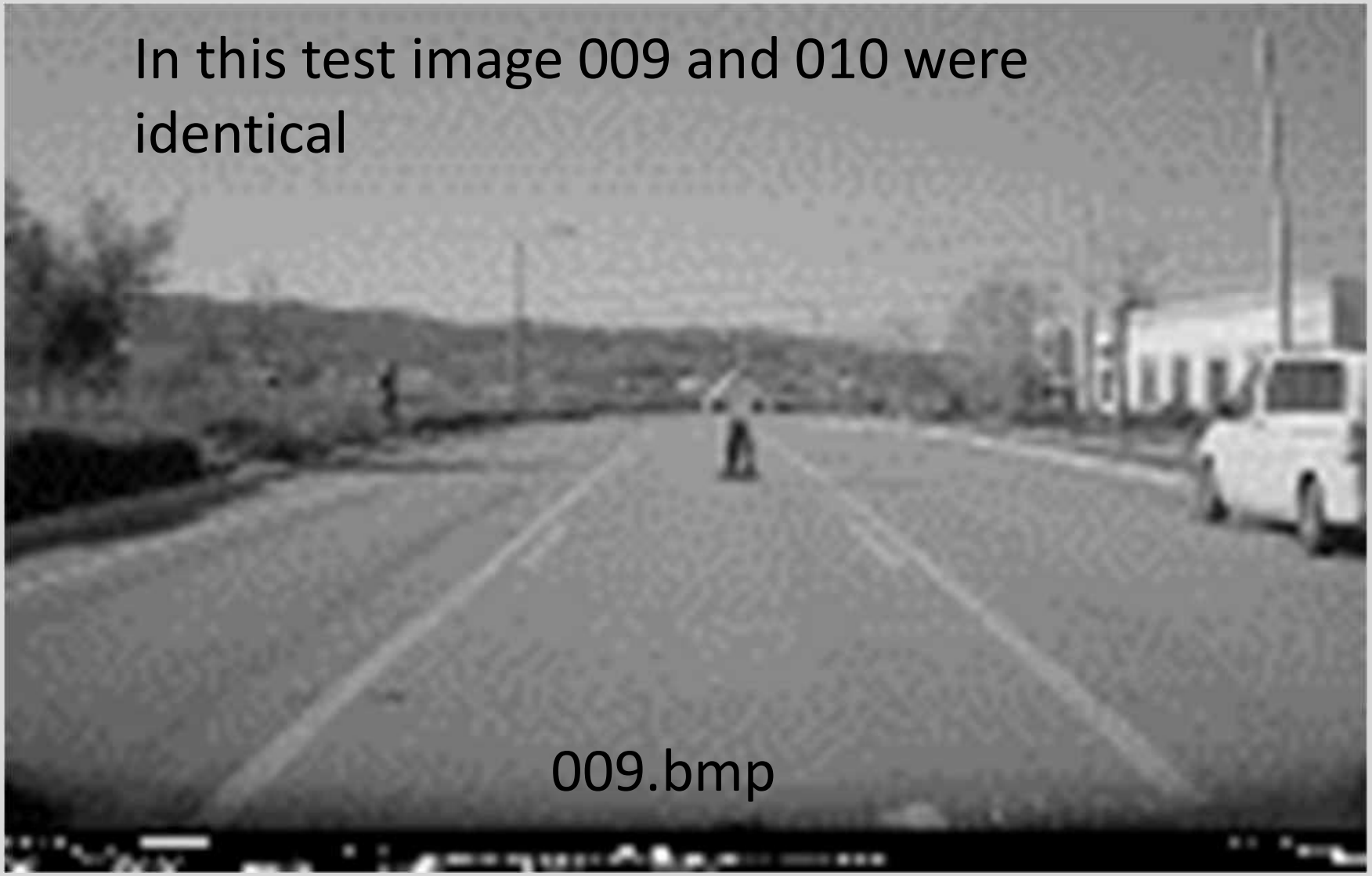

006.bmp

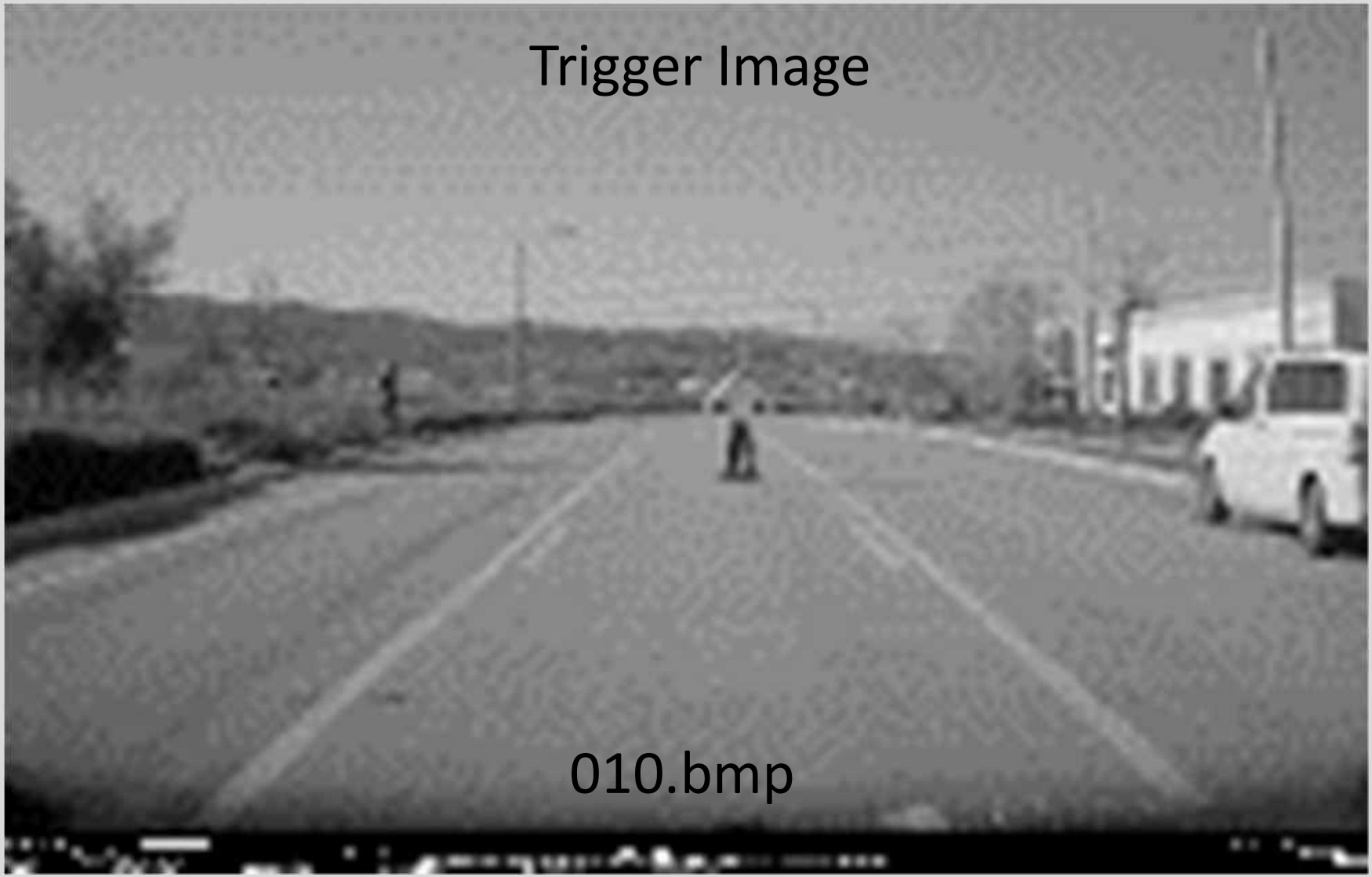


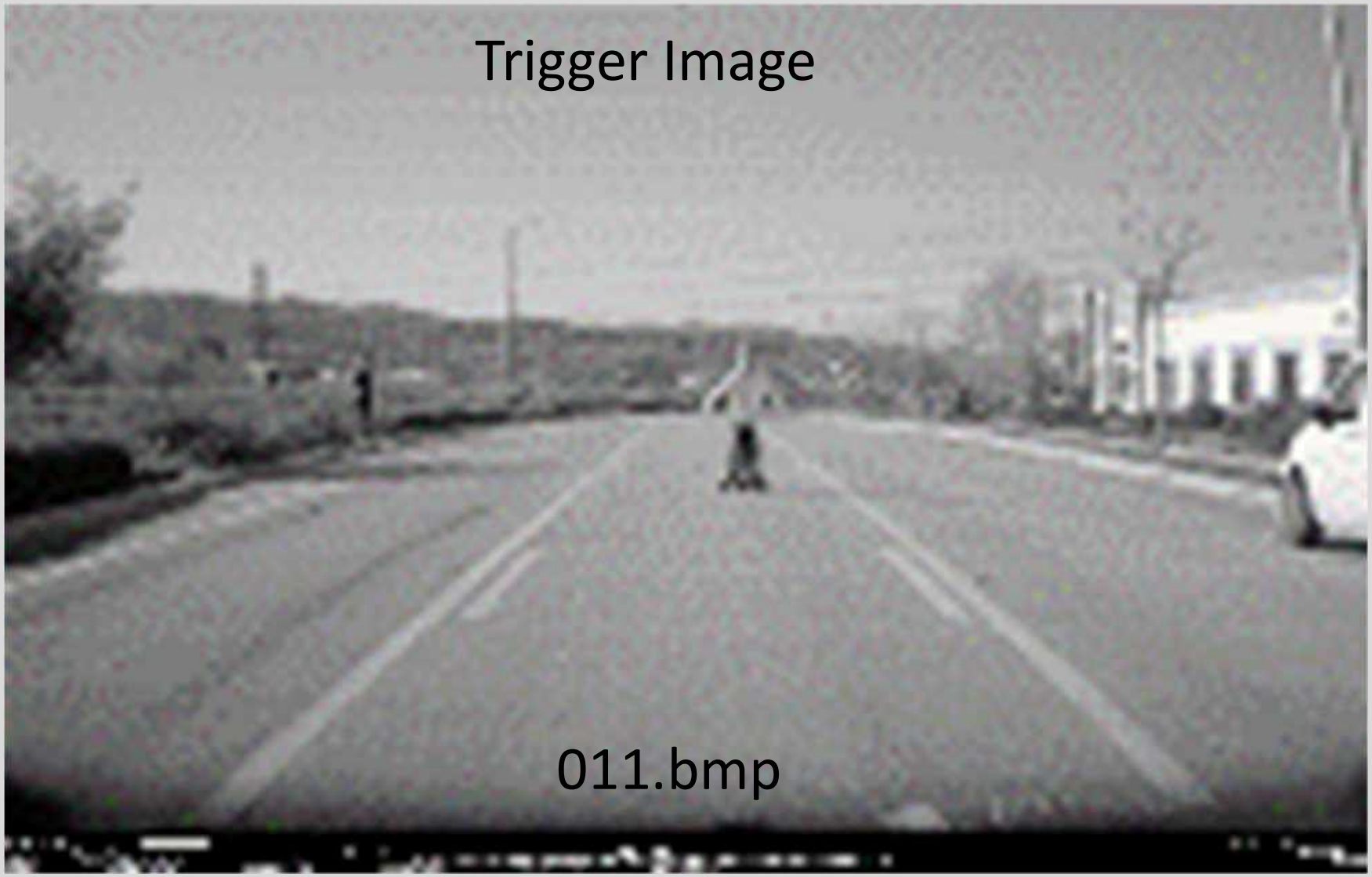
007.bmp



008.bmp


In this test image 009 and 010 were identical
009.bmp


Trigger Image
010.bmp


Trigger Image
011.bmp



012.bmp



013.bmp


014.bmp


015.bmp


016.bmp


017.bmp


018.bmp


019.bmp

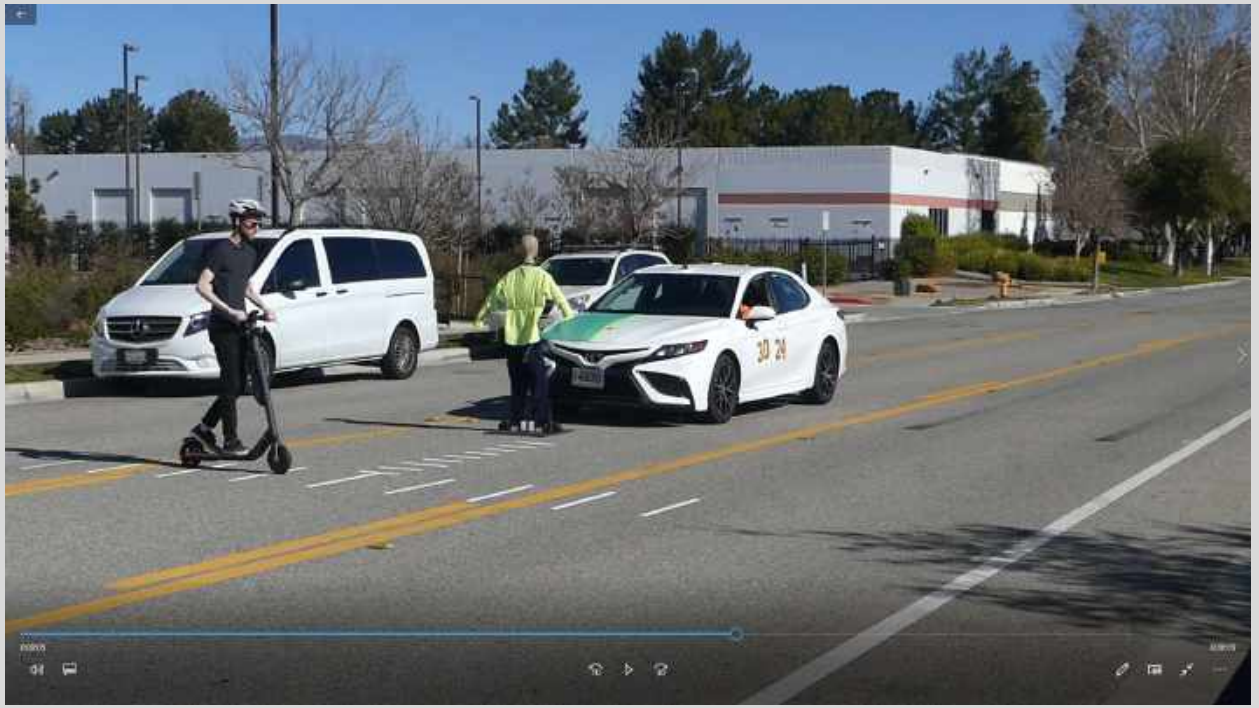

020.bmp


021.bmp

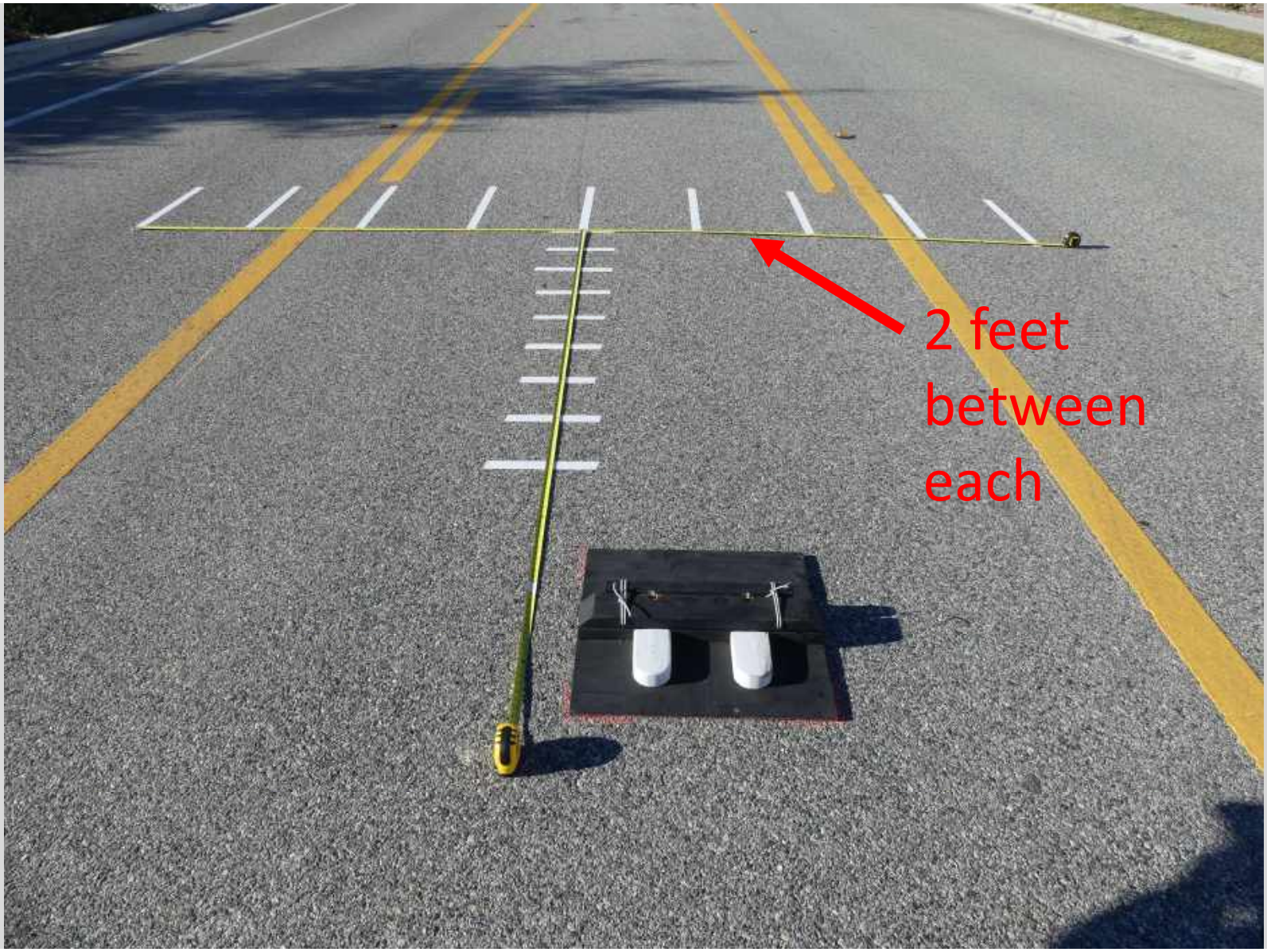
Jim Morgan

Using the images to evaluate the speed of other items of interest

- Bicycle
- Pedestrian
- Scooter
- Cross-traffic




Mechanical Governor
at the throttle of the
scooter to control the
speed


2 feet
between
each

Test 30-24



Image 019



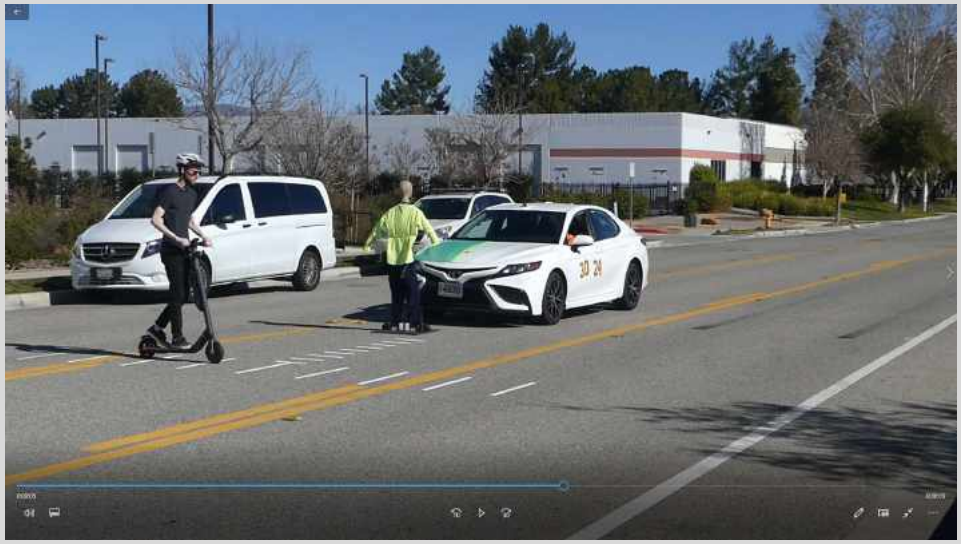
Image 018

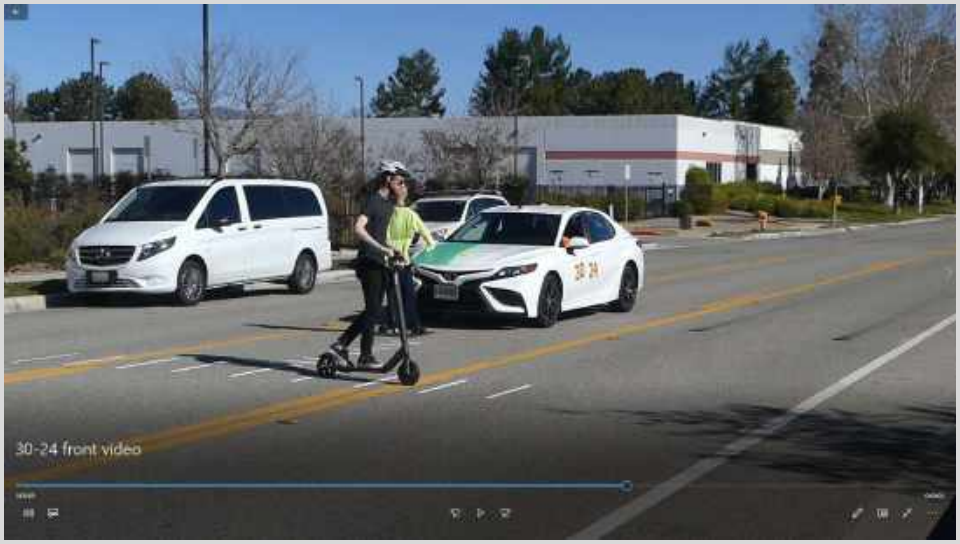
GoPro camera View Test 30-24





Lumix Camera view Test 30-24





9			
10	Trigger ID	Time Stamp	File Name
11	18_4	2022-02-06-10:31:50.400	010.bmp
12	18_3	2022-02-06-10:31:50.500	011.bmp
13	18_5	2022-02-06-10:31:50.600	012.bmp
14	18_4	2022-02-06-10:31:50.700	013.bmp
15	18_4	2022-02-06-10:31:51.300	014.bmp
16	18_4	2022-02-06-10:31:51.900	015.bmp
17	18_4	2022-02-06-10:31:52.500	016.bmp
18	18_4	2022-02-06-10:31:53.100	017.bmp
19	18_4	2022-02-06-10:31:53.700	018.bmp
20	18_4	2022-02-06-10:31:54.300	019.bmp
21	18_4	2022-02-06-10:31:54.900	020.bmp
22	18_4	2022-02-06-10:31:55.500	021.bmp
23	18_4	2022-02-06-10:31:56.100	022.bmp
24			

Test 30-24



Time between 018 and 019
0.600 seconds


7.33 feet

Test 30-24
travel distance 7.33 feet

TSS TIME 0.600 SEC.	GOPRO TIME 0.591 SEC.	LUMIX CAMERA TIME 0.601 SEC.
Calculated Speed	Calculated Speed	Calculated Speed
12.22 fps	12.40 fps	12.20 fps
8.33 mph	8.45 mph	8.32 mph

Test 30-25

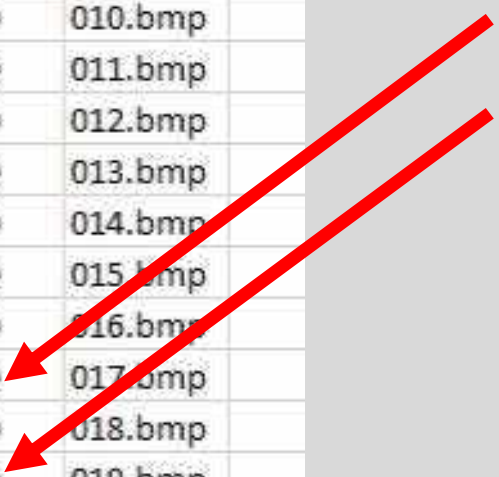


Image 019



Image 017

9			
10	Trigger ID	Time Stamp	File Name
11	18_4	2022-02-06-10:38:17.600	010.bmp
12	18_3	2022-02-06-10:38:17.700	011.bmp
13	18_5	2022-02-06-10:38:17.800	012.bmp
14	18_4	2022-02-06-10:38:18.100	013.bmp
15	18_4	2022-02-06-10:38:18.700	014.bmp
16	18_4	2022-02-06-10:38:19.300	015.bmp
17	18_4	2022-02-06-10:38:19.900	016.bmp
18	18_4	2022-02-06-10:38:20.500	017.bmp
19	18_4	2022-02-06-10:38:21.100	018.bmp
20	18_4	2022-02-06-10:38:21.700	019.bmp
21	18_4	2022-02-06-10:38:22.300	020.bmp
22	18_4	2022-02-06-10:38:22.900	021.bmp
23	18_4	2022-02-06-10:38:23.500	022.bmp
24			



Time between 017 and 019
1.200 seconds


14.5 feet

Test 30-25
travel distance 14.5 feet

TSS TIME 1.200 SEC.	GOPRO TIME 1.166 SEC.	LUMIX CAMERA TIME 1.167 SEC.
Calculated Speed	Calculated Speed	Calculated Speed
12.08 fps	12.44 fps	12.43 fps
8.24 mph	8.48 mph	8.48 mph

Test 30-26



Image 017



Image 016

9			
10	Trigger ID	Time Stamp	File Name
11	18_4	2022-02-06-10:44:09.900	010.bmp
12	18_3	2022-02-06-10:44:10.000	011.bmp
13	18_4	2022-02-06-10:44:10.600	012.bmp
14	18_4	2022-02-06-10:44:11.200	013.bmp
15	18_4	2022-02-06-10:44:11.800	014.bmp
16	18_4	2022-02-06-10:44:12.400	015.bmp
17	18_4	2022-02-06-10:44:13.000	016.bmp
18	18_4	2022-02-06-10:44:13.600	017.bmp
19	18_4	2022-02-06-10:44:14.200	018.bmp
20	18_4	2022-02-06-10:44:14.800	019.bmp
21	18_4	2022-02-06-10:44:15.400	020.bmp
22			



Time between 016 and 017
0.600 seconds


7.5 Feet

Test 30-26
travel distance 7.5 feet

TSS TIME 0.600 SEC.	GOPRO TIME 0.60 SEC.	LUMIX CAMERA TIME 0.583 SEC.
Calculated Speed	Calculated Speed	Calculated Speed
12.50 fps	12.5 fps	12.86 fps
8.52 mph	8.52 mph	8.77 mph

Test 25-27



Image 018



Image 020

10	Trigger ID	Time Stamp	File Name
11	18_4	2022-02-17-16:50:18.100	010.bmp
12	18_3	2022-02-17-16:50:18.300	011.bmp
13	18_4	2022-02-17-16:50:18.500	012.bmp
14	18_4	2022-02-17-16:50:19.100	013.bmp
15	18_4	2022-02-17-16:50:19.700	014.bmp
16	18_4	2022-02-17-16:50:20.300	015.bmp
17	18_4	2022-02-17-16:50:20.900	016.bmp
18	18_4	2022-02-17-16:50:21.500	017.bmp
19	18_4	2022-02-17-16:50:22.100	018.bmp
20	18_4	2022-02-17-16:50:22.700	019.bmp
21	18_4	2022-02-17-16:50:23.300	020.bmp
22	18_4	2022-02-17-16:50:23.900	021.bmp



Time between
018 and 020
1.200 seconds


13.3 Feet

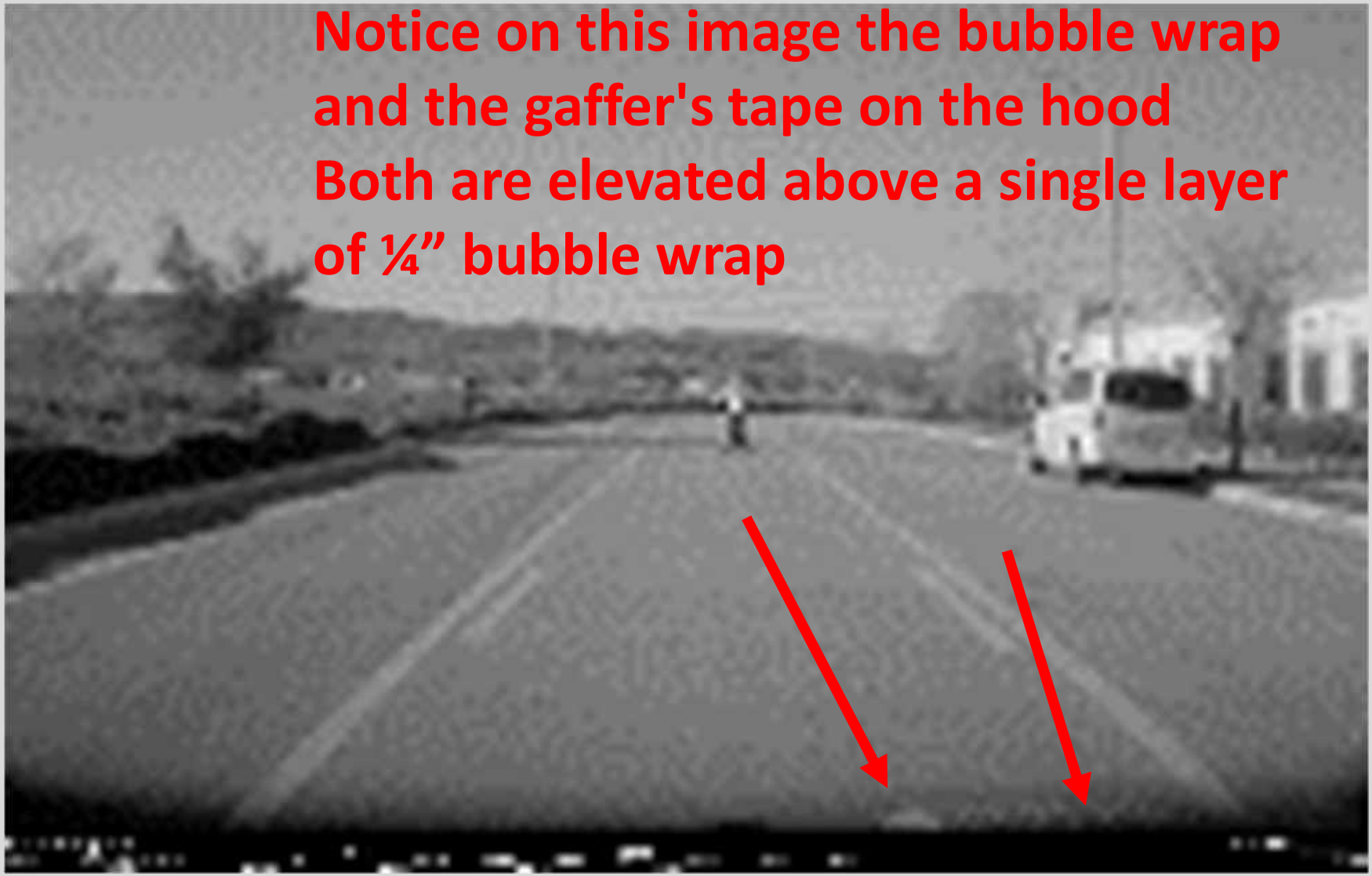
Test 25-27
travel distance 13.5 feet

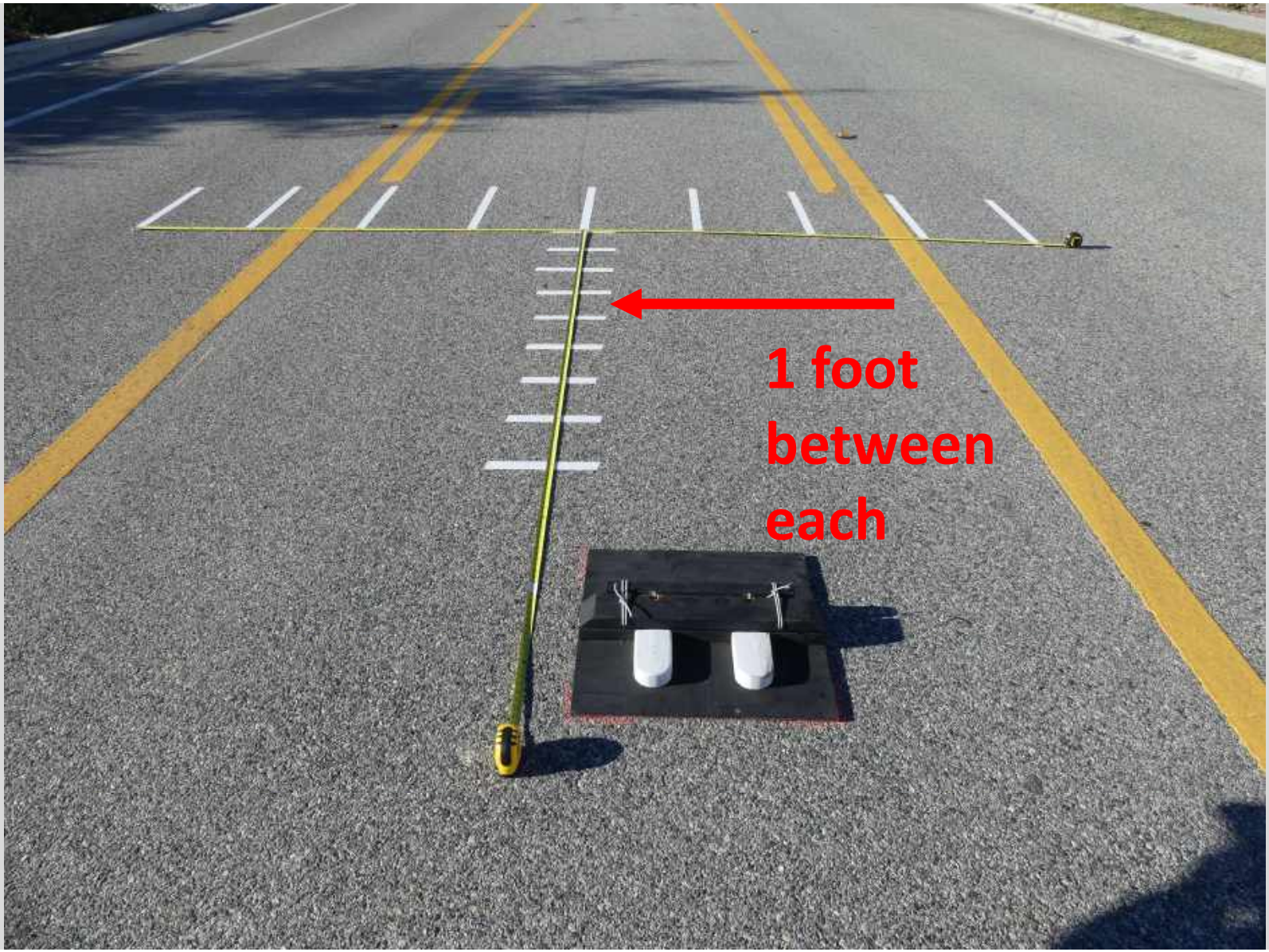
TSS TIME 1.200 SEC.	VBOX SPORT	STALKER RADAR ATS
Calculated Speed	Speed Range	Speed Range
11.08 fps	7.21 to 7.67 mph	7.2 to 7.7 mph
7.56 mph	Average 7.44 mph	Average 7.45 mph

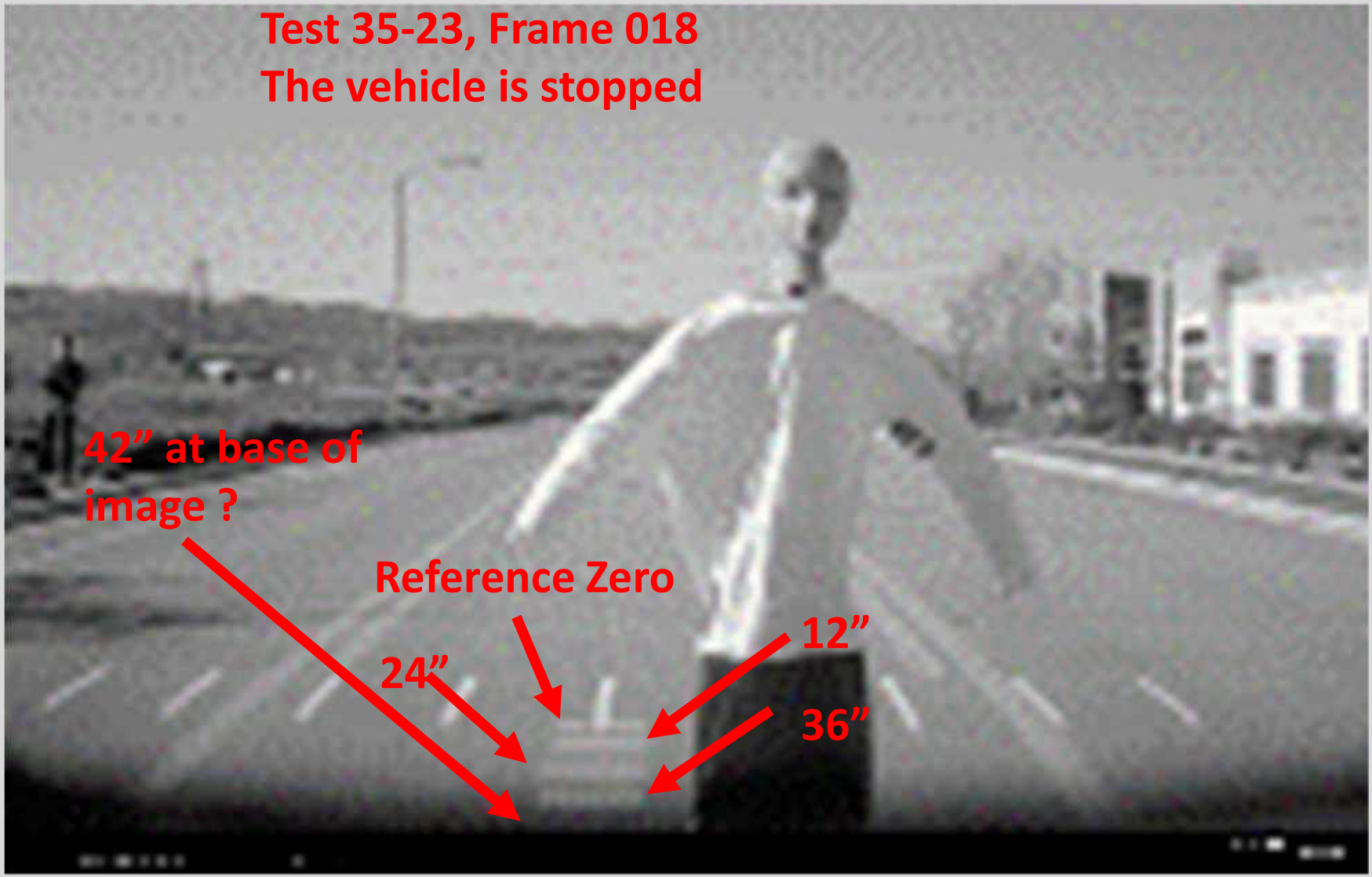
What is the distance from the front Bumper to the bottom of the image at the ground?



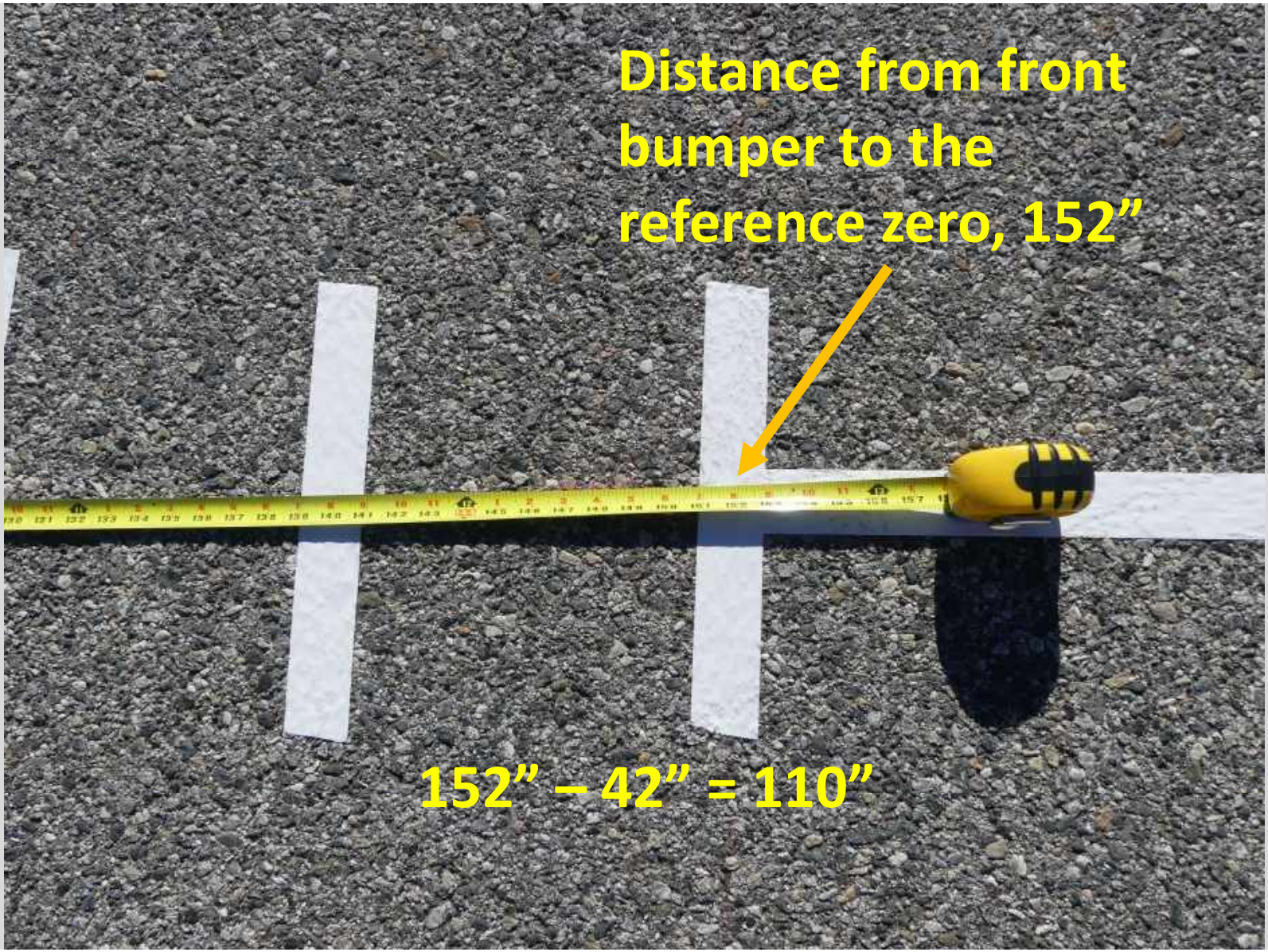


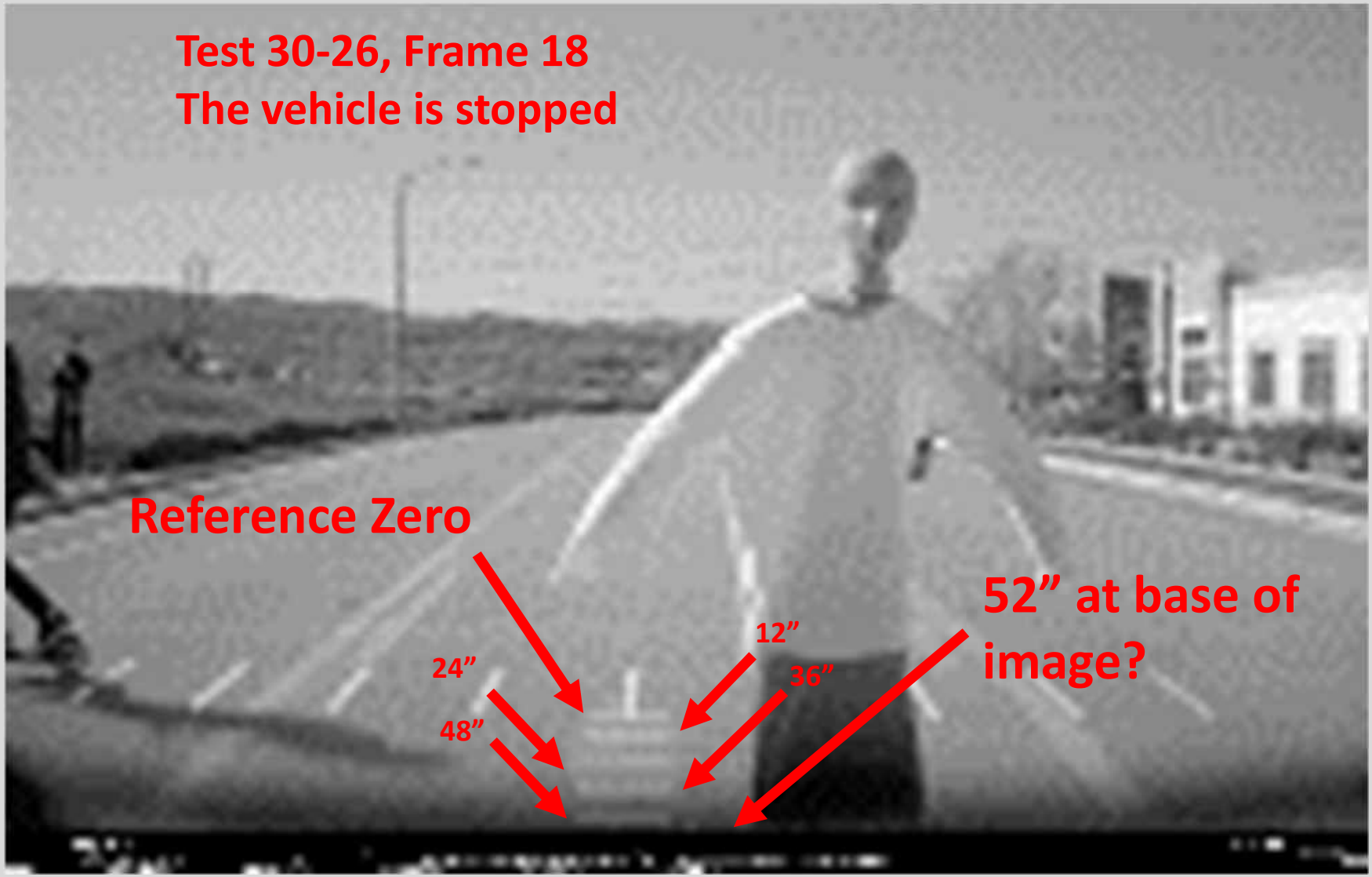

Notice on this image the bubble wrap
and the gaffer's tape on the hood
Both are elevated above a single layer
of ¼” bubble wrap

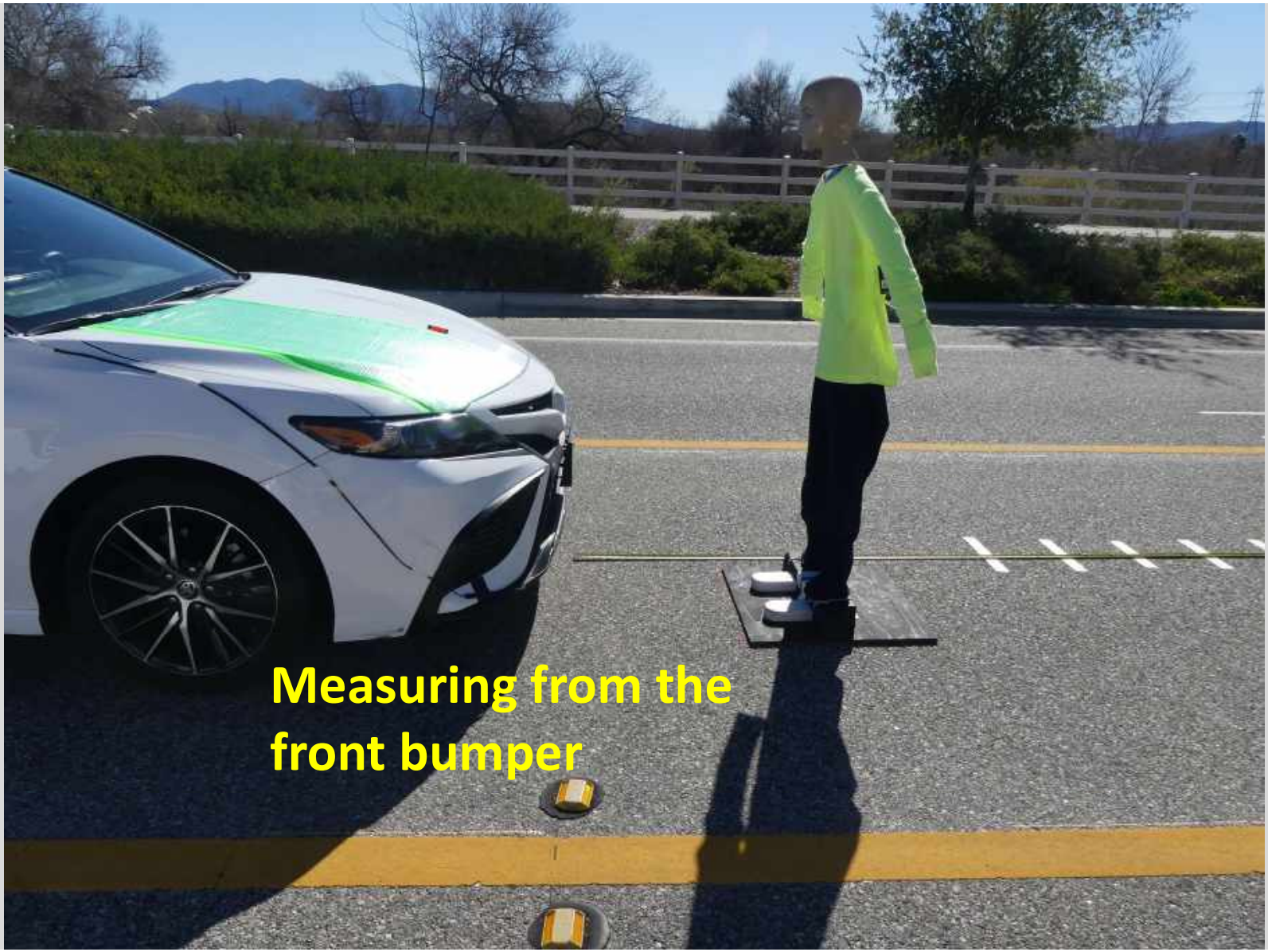

1 foot
between
each


Test 35-23, Frame 018
The vehicle is stopped
42” at base of image ?
Reference Zero
24”
12”
36”

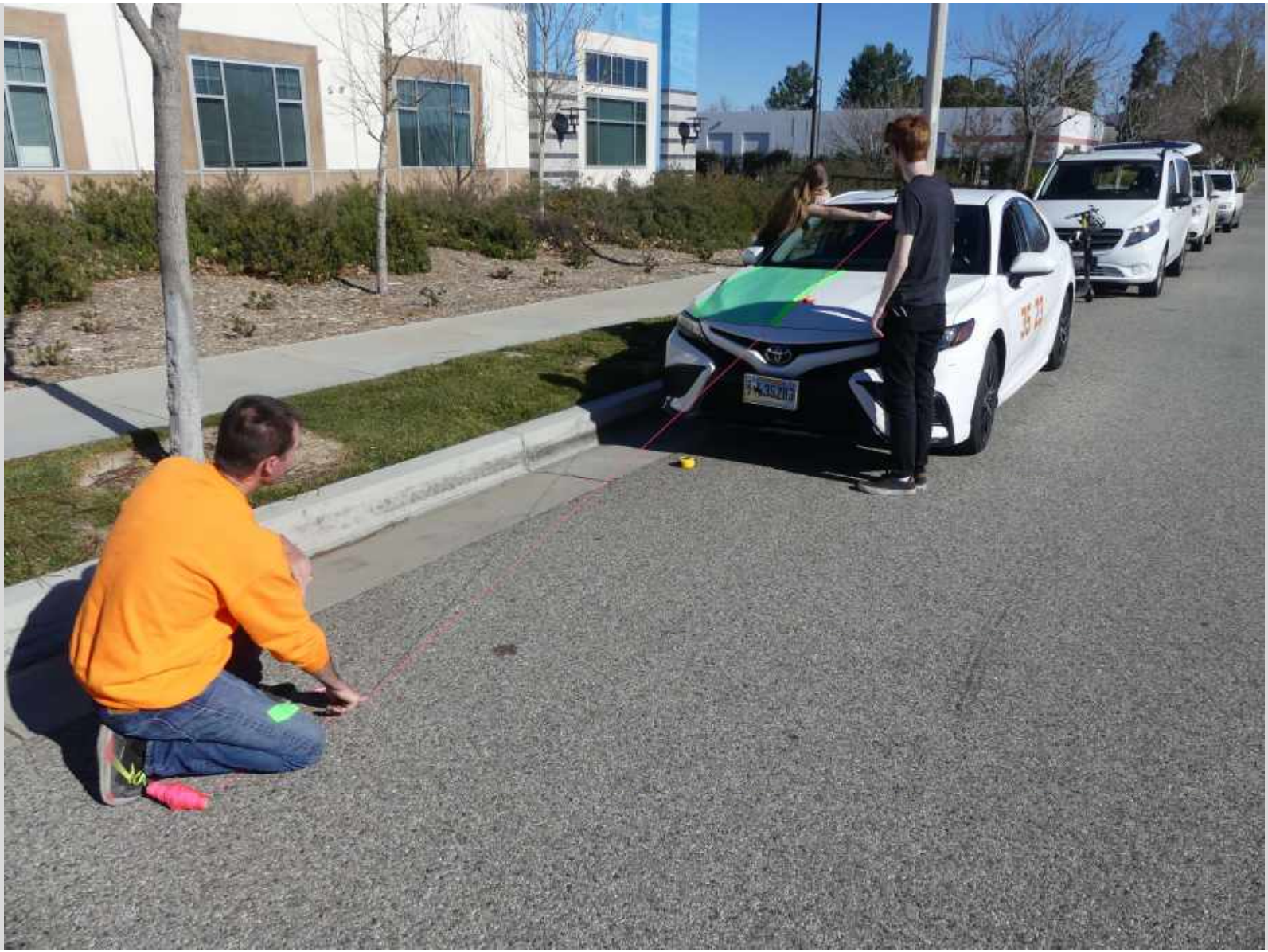

Measuring from
the front bumper

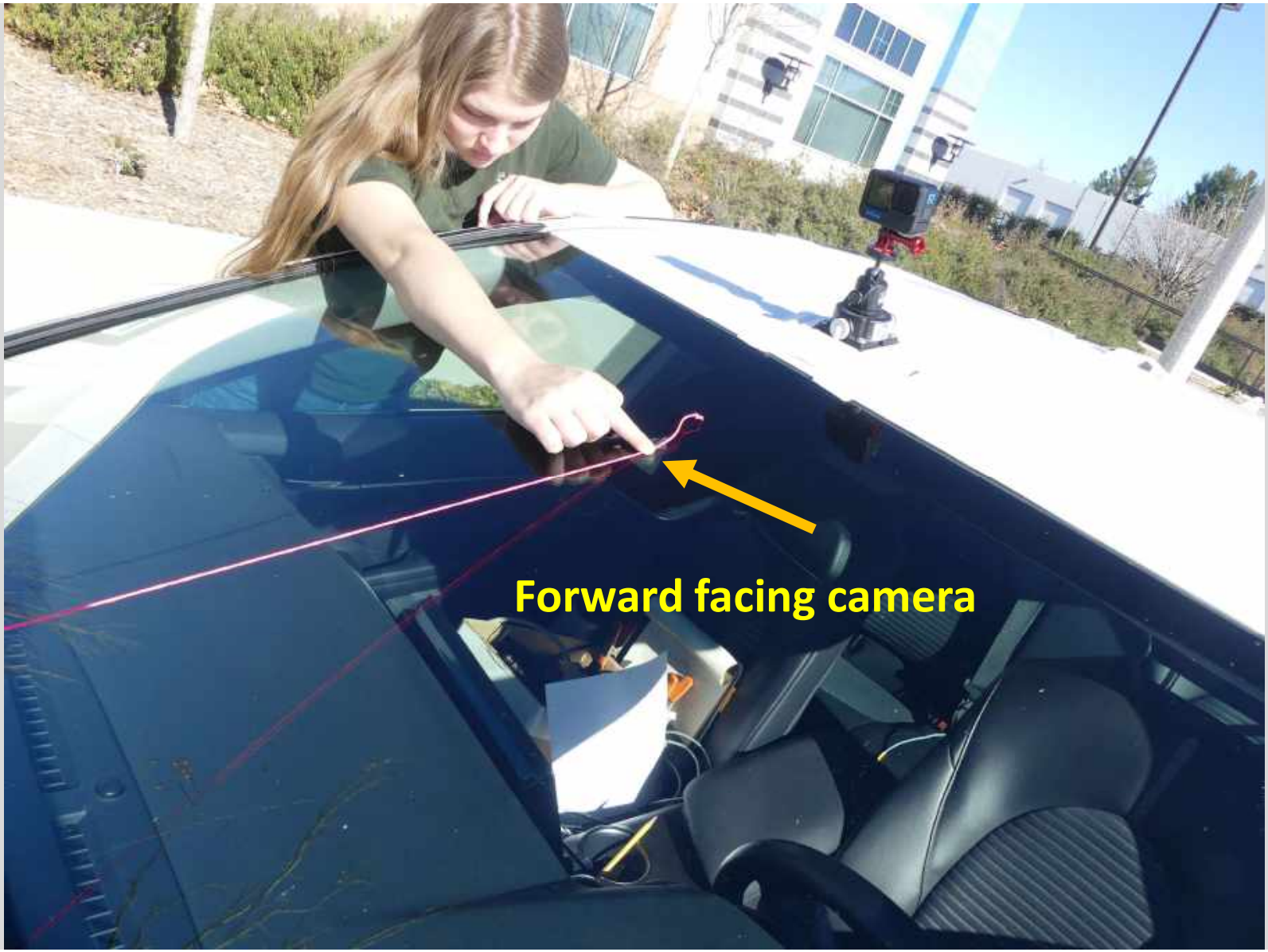

Distance from front bumper to the reference zero, 152”
152” – 42” = 110”


Test 30-26, Frame 18
The vehicle is stopped
Reference Zero
24”
48”
12”
36”
52” at base of image?


Measuring from the
front bumper


Distance form front bumper160.5 “
160.5”-52” = 108.5”

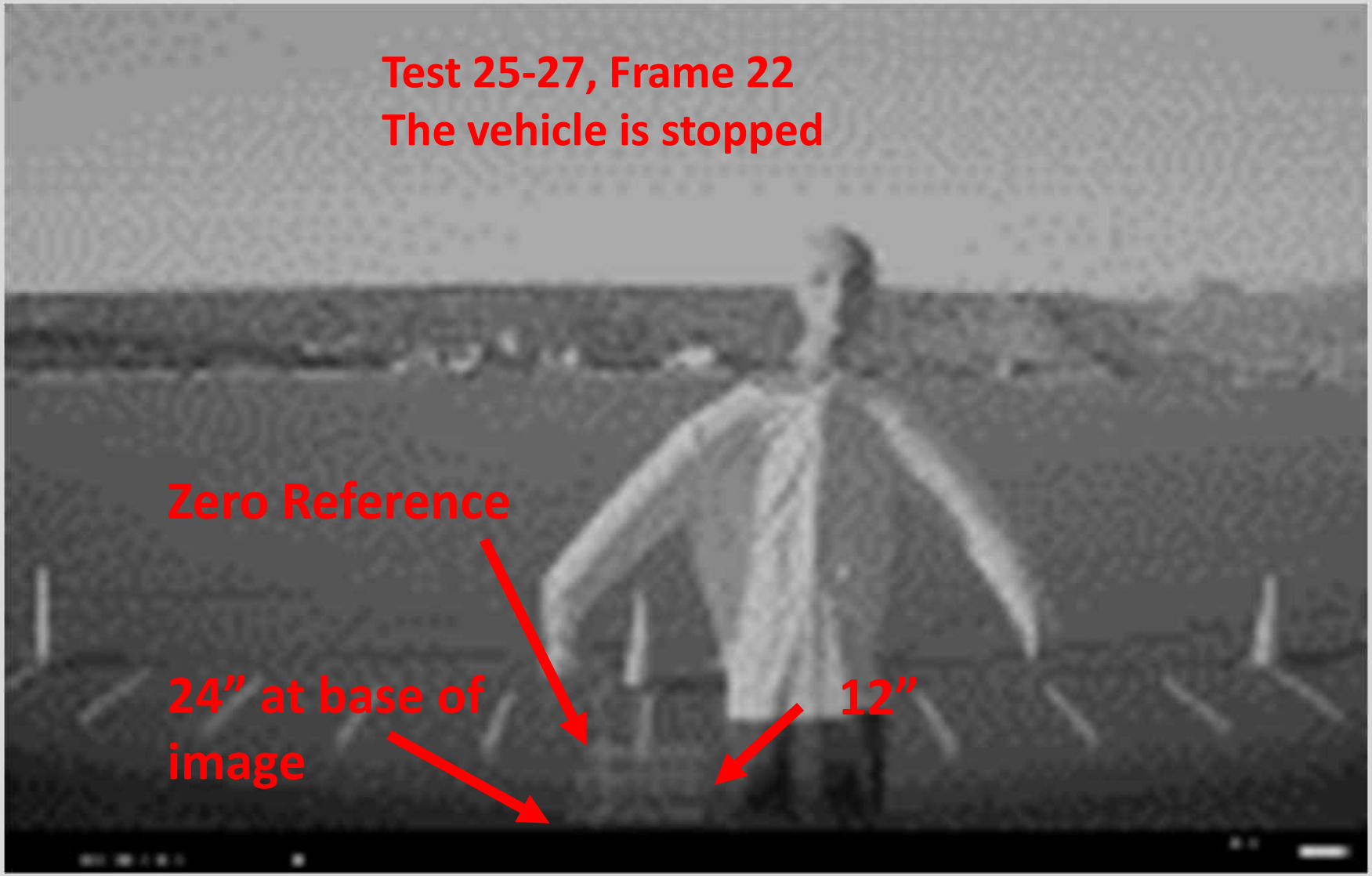


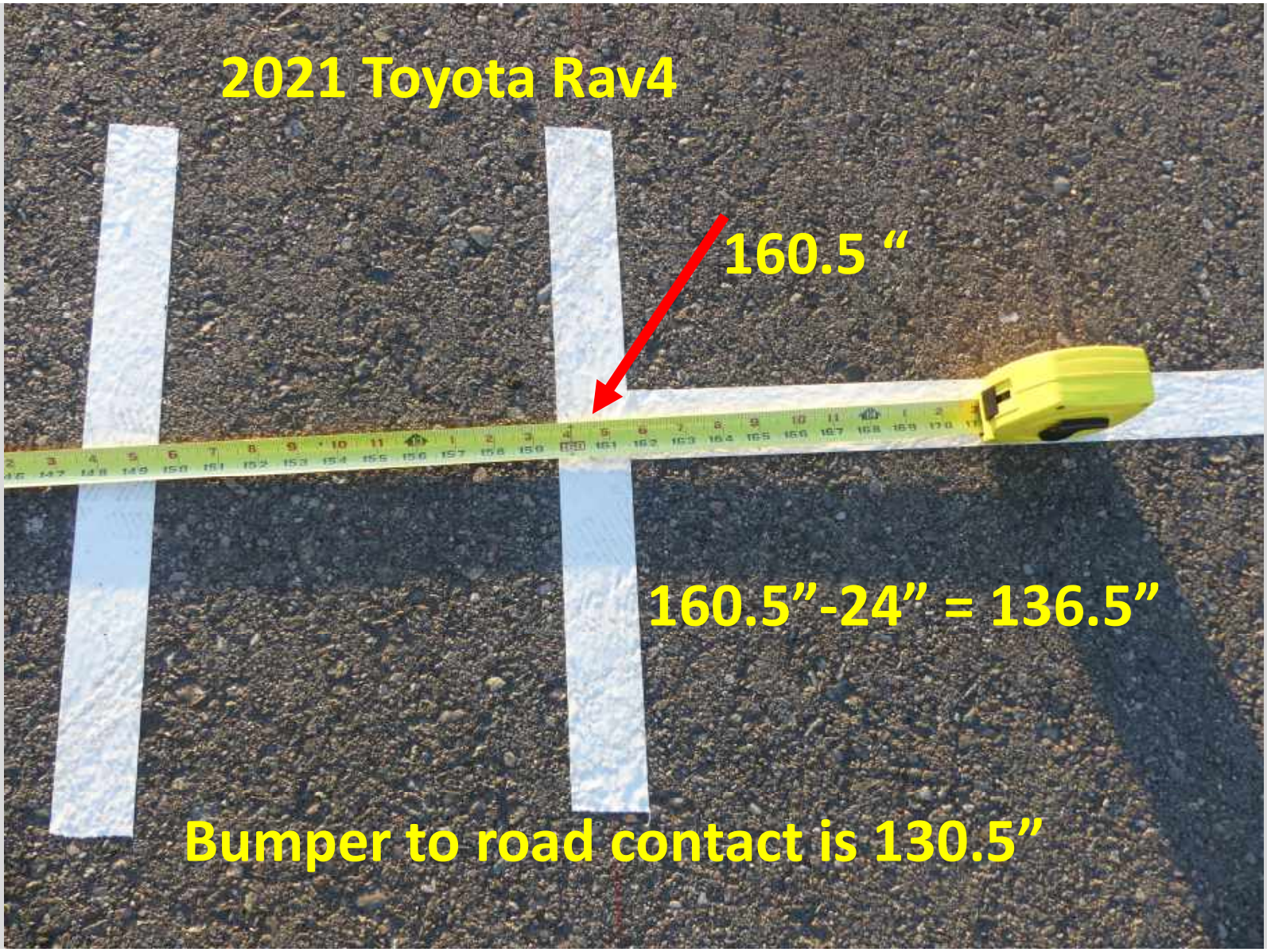

Forward facing camera


String just touching
the hood


Measuring from the front bumper to where the string contacts the road


2021 Toyota Camry
110.5” from bumper to ground contact
Test 35-23, 110”
Test 30-26, 108.5”


Test 25-27, Frame 22
The vehicle is stopped
Zero Reference
24” at base of image
12”


2021 Toyota Rav4
160.5 “
160.5”-24” = 136.5”
Bumper to road contact is 130.5”

PCS events based on tests with the 2021 Toyota Camry Tss 2.5+

PCS

Total Time of Data 9 sec.

Data Time Pre-Trigger 4.95 sec.

Data Time Post-Trigger 4.05 sec.

Sample rate generally ranges between 0.15 to 0.5 sec.

SUDDEN BRAKING HISTORY

Total Time of Data 10.2 sec.

Data Time Pre-Trigger 5 sec.

Data Time Post-Trigger 5.2 sec

Sample rate 0.15 sec. except at trigger time

ABS OPERATION HISTORY

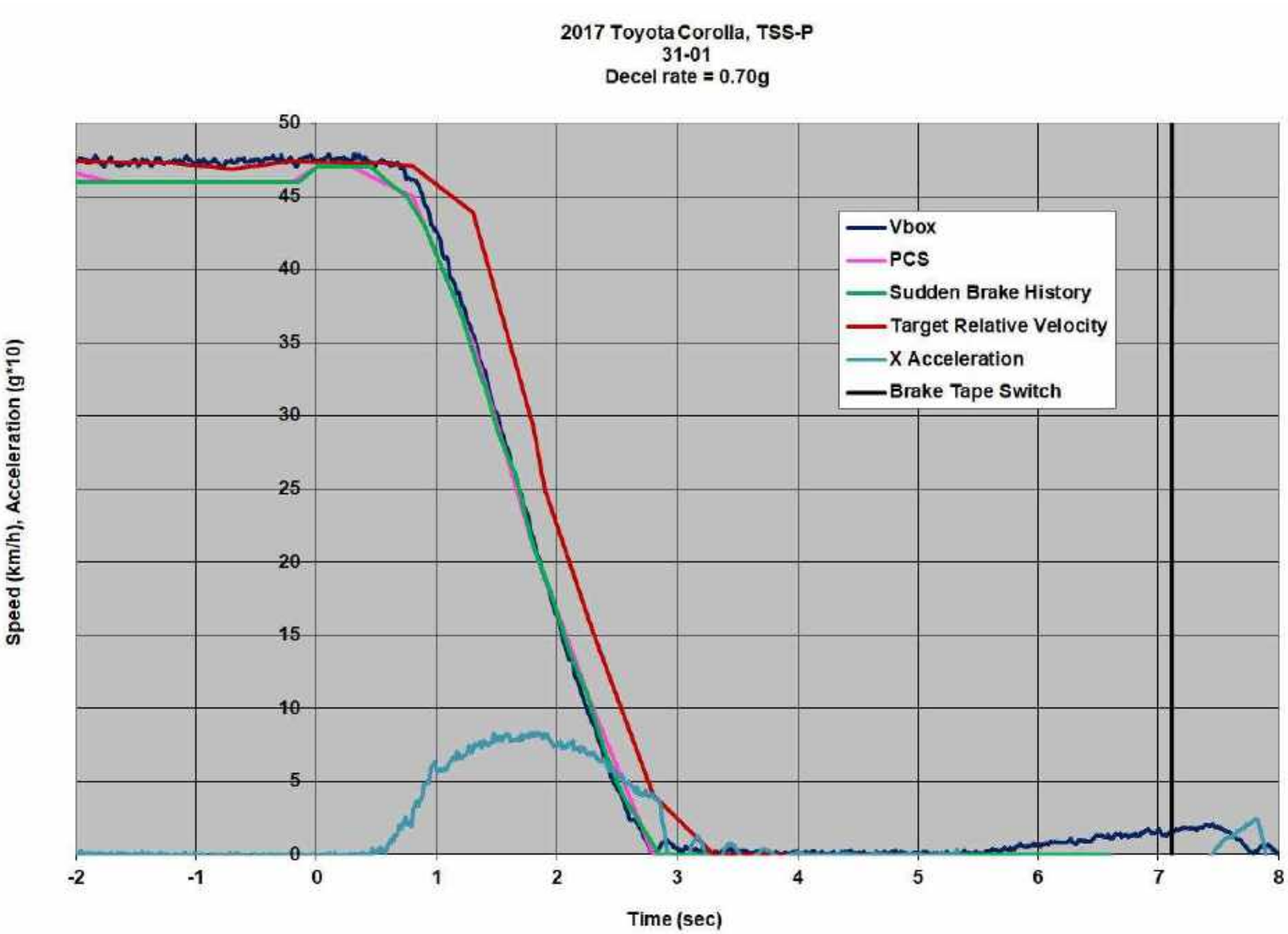
Not always triggered during a PCS event

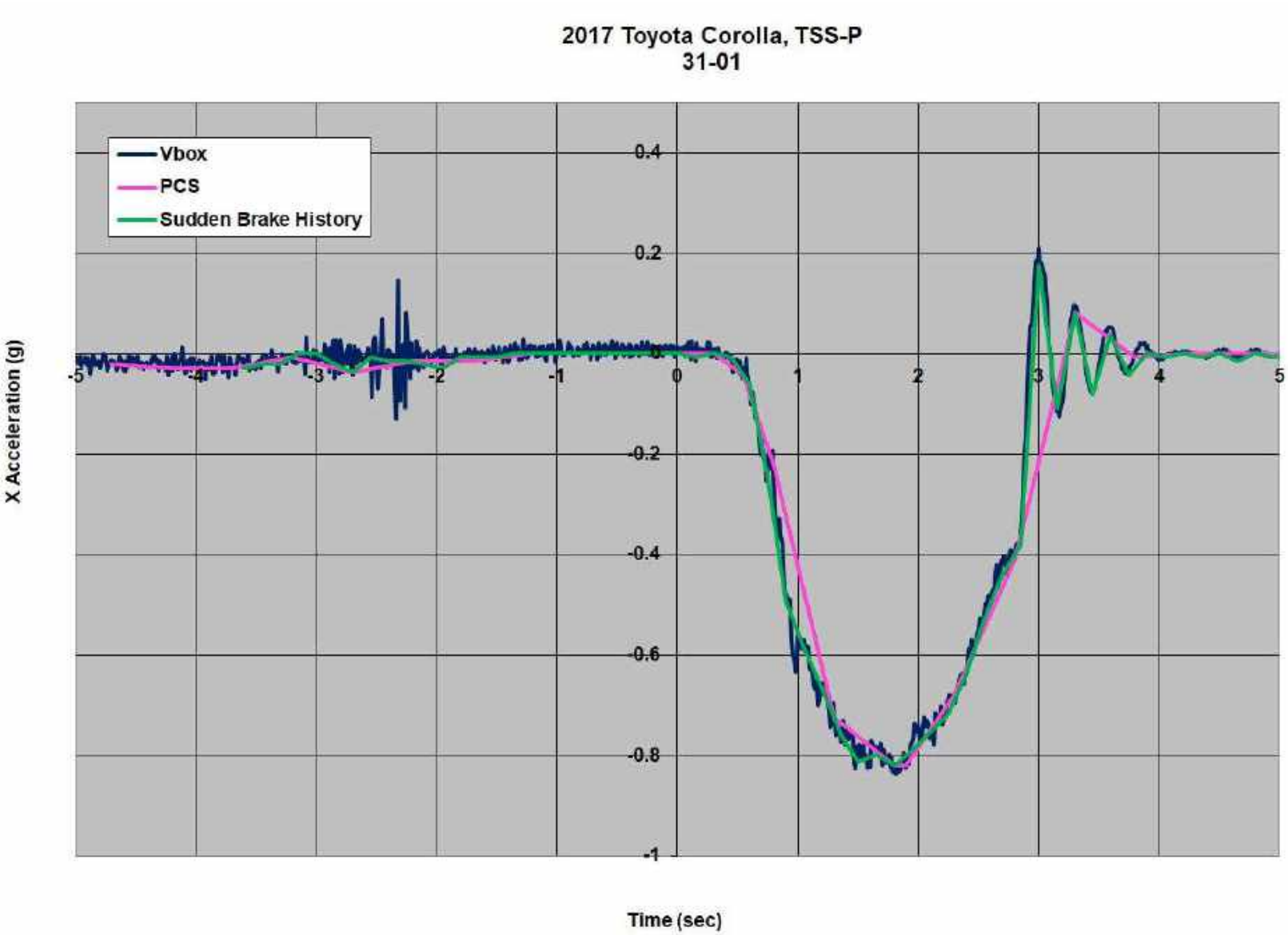
Total Time of Data 10.2 sec.

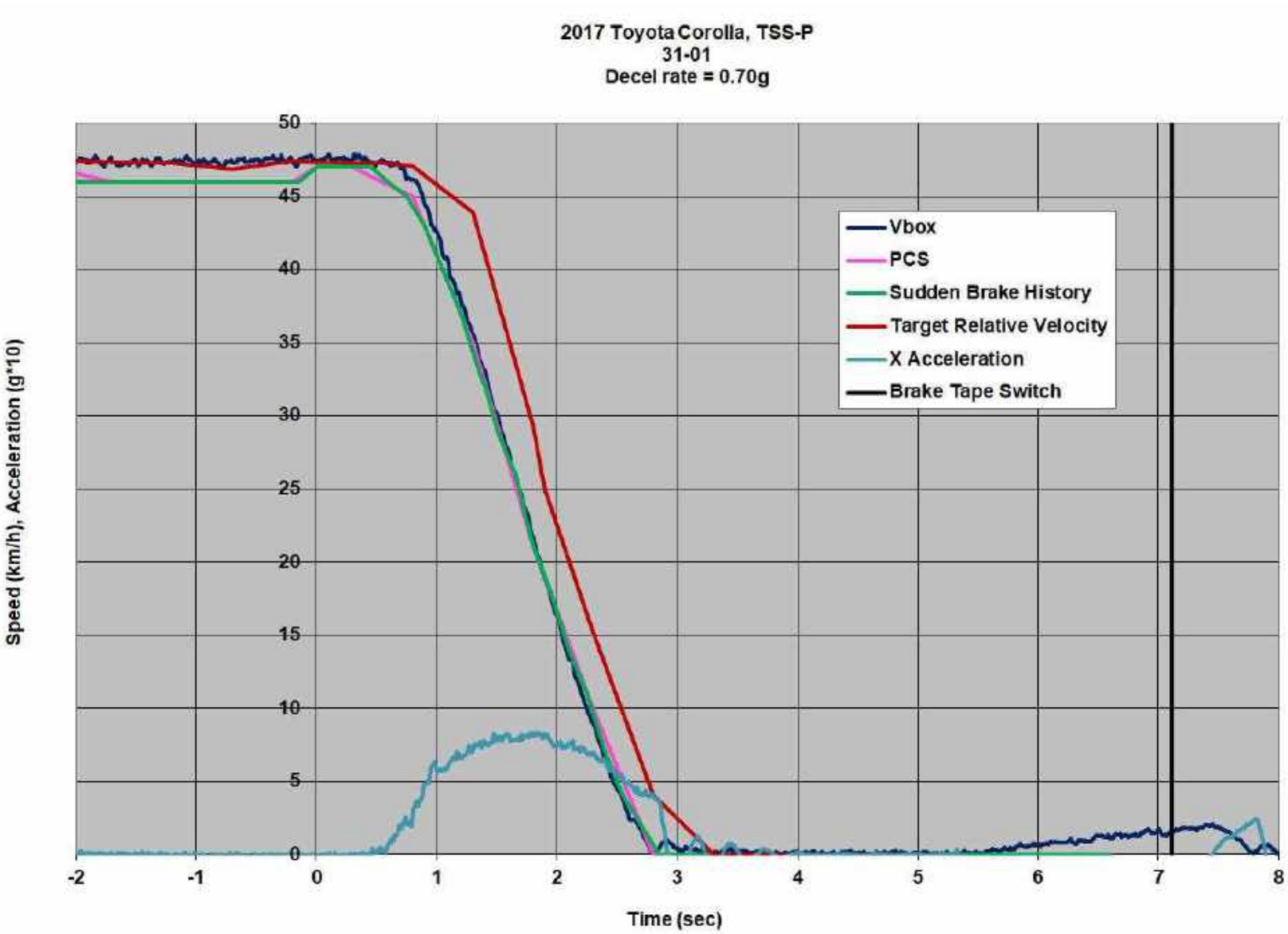
Data Time Pre-Trigger 5.05 sec

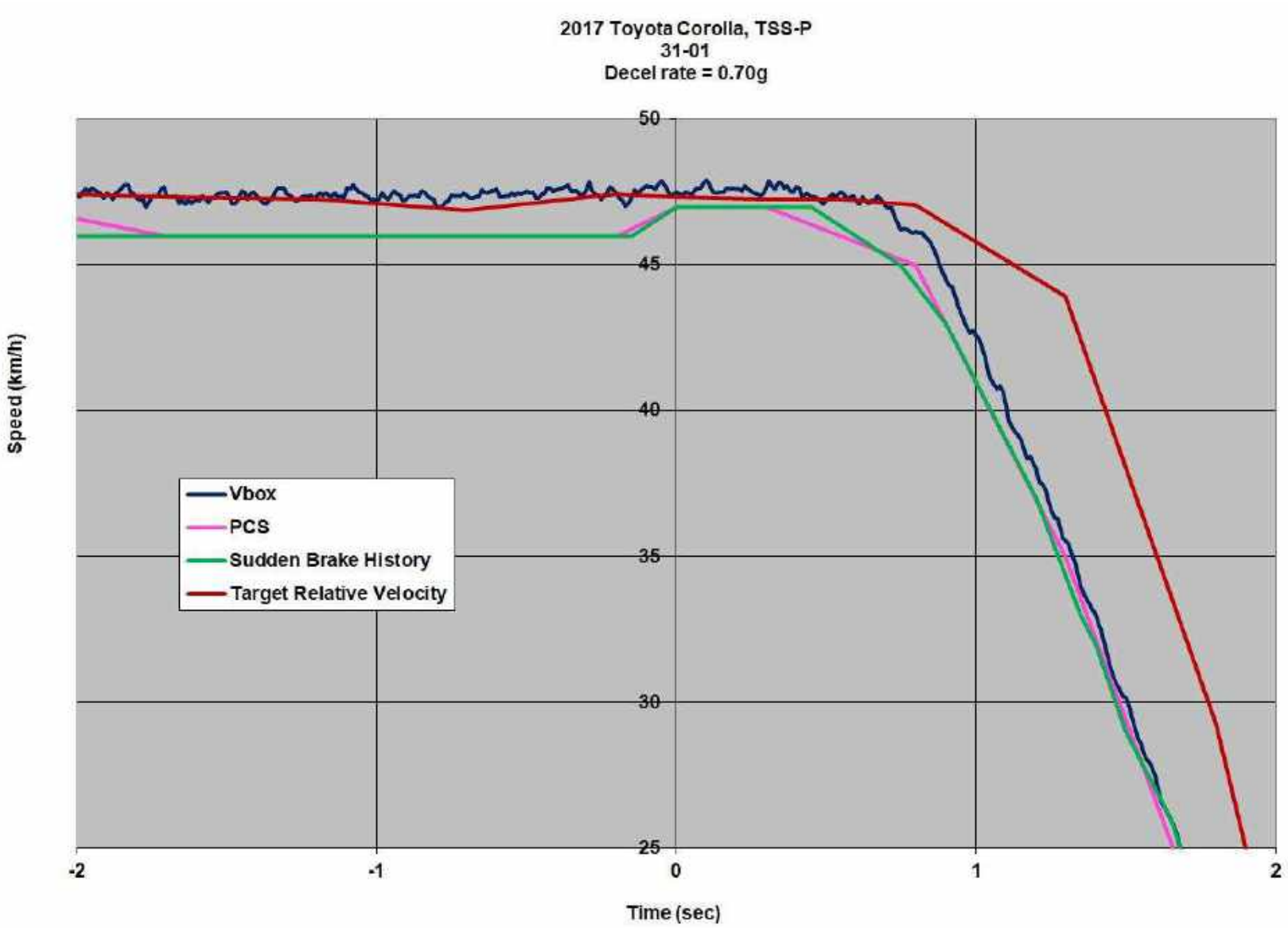
Data Time Pre-Trigger 5.15 sec.

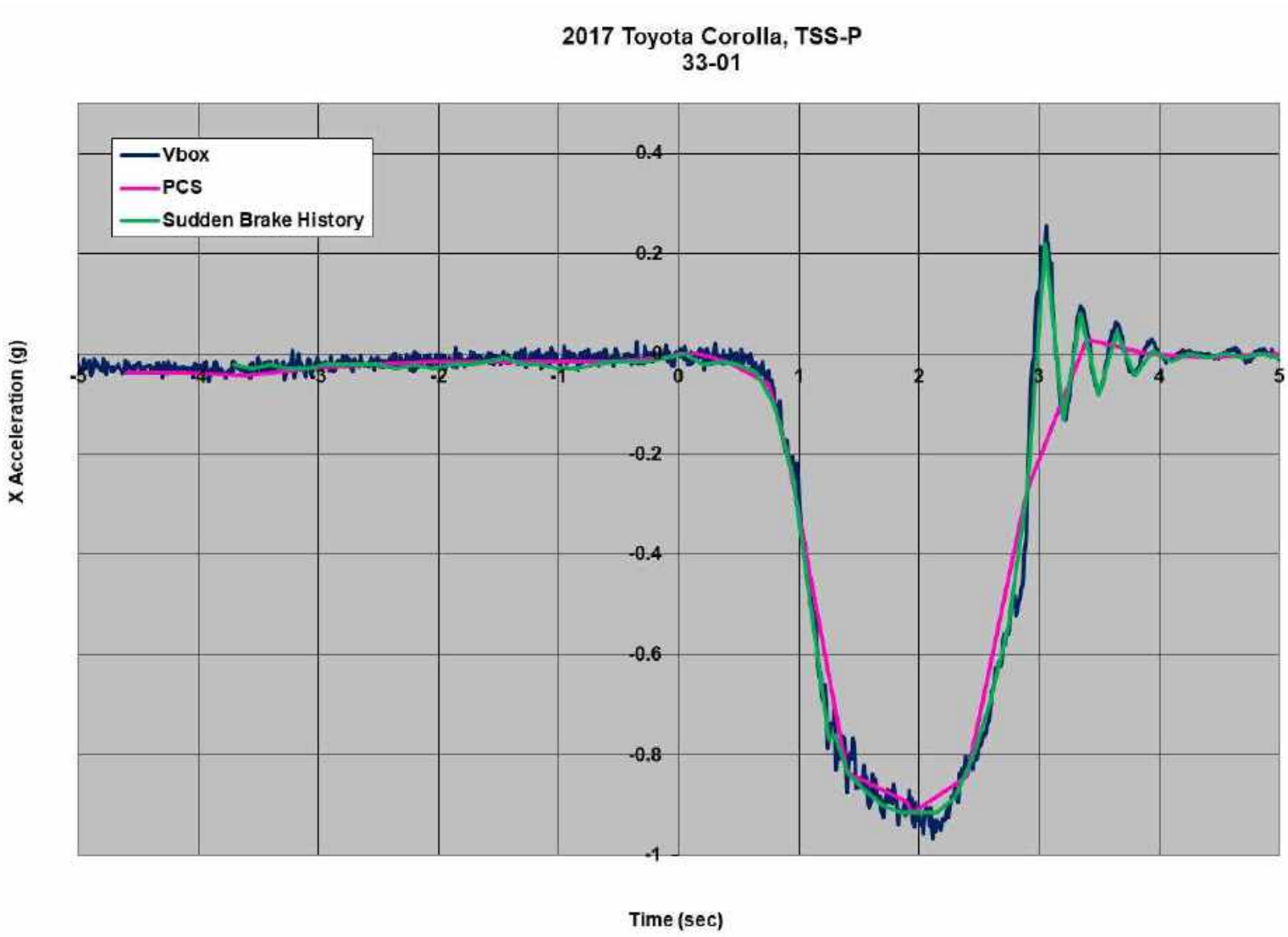
Sample rate 0.15 sec. except at trigger time

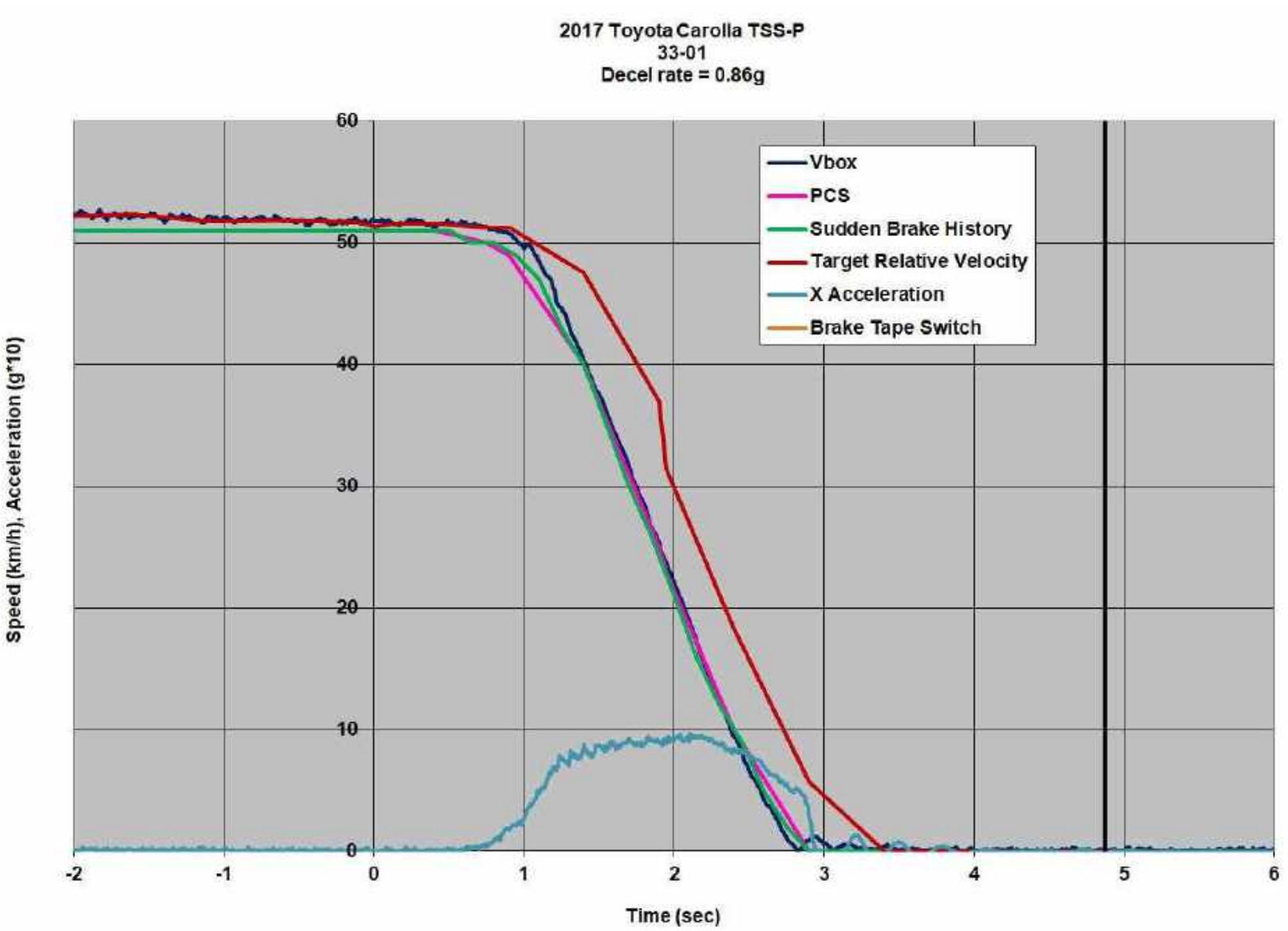

2017 Toyota Corolla, TSS-P
31-01
Decel rate = 0.70g
Speed (km/h), Acceleration (g*10)
Time (sec)
Vbox
PCS
Sudden Brake History
Target Relative Velocity
X Acceleration
Brake Tape Switch
50
45
40
35
30
25
20
15
10
5
0
-2
-1
0
1
2
3
4
5
6
7
8

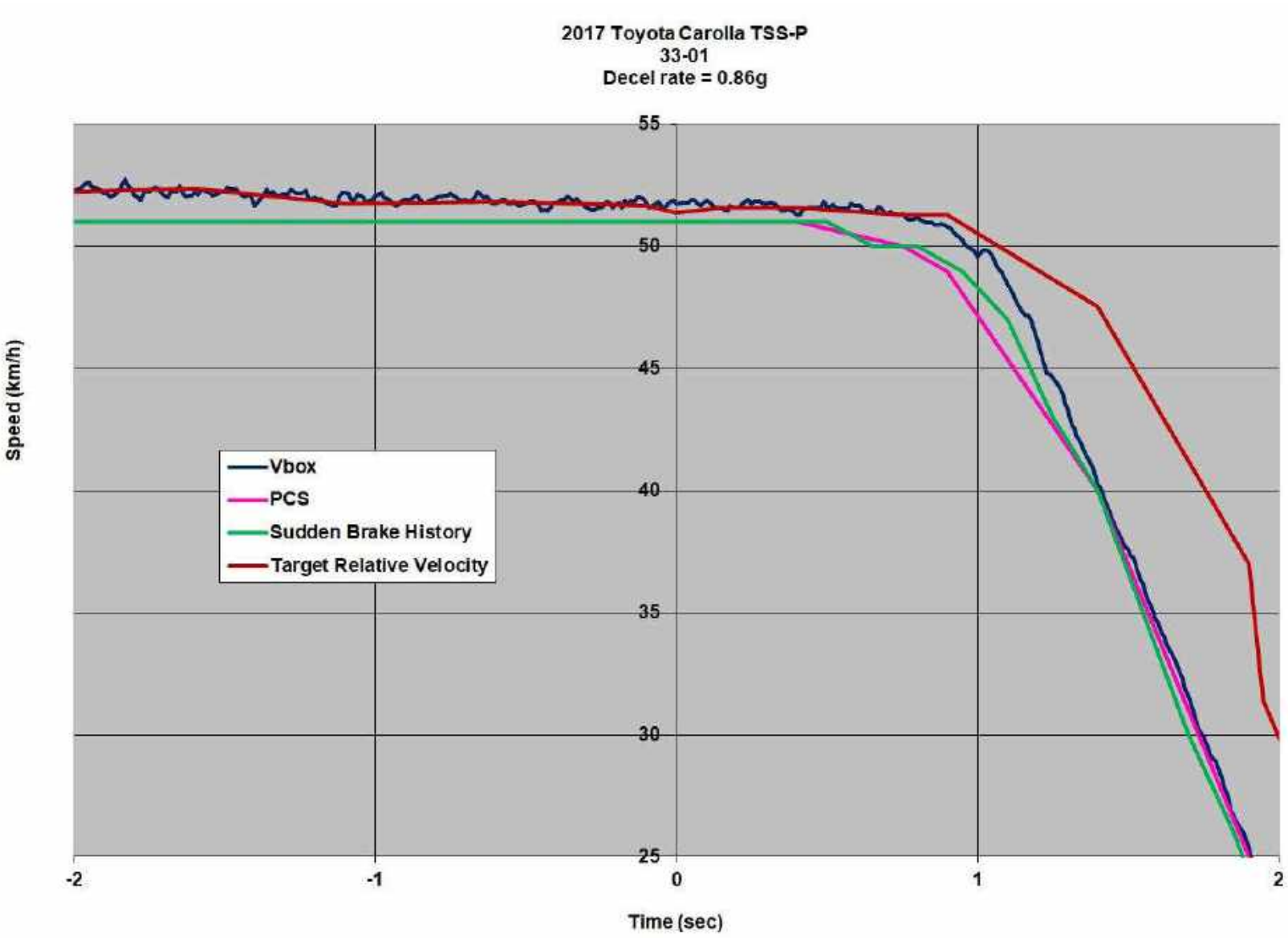

2017 Toyota Corolla, TSS-P
31-01
Vbox
PCS
Sudden Brake History
X Acceleration (g)
0.4
0.2
-0.2
-0.4
-0.6
-0.8
-1
-5
-4
-3
-2
-1
0
1
2
3
4
5
Time (sec)

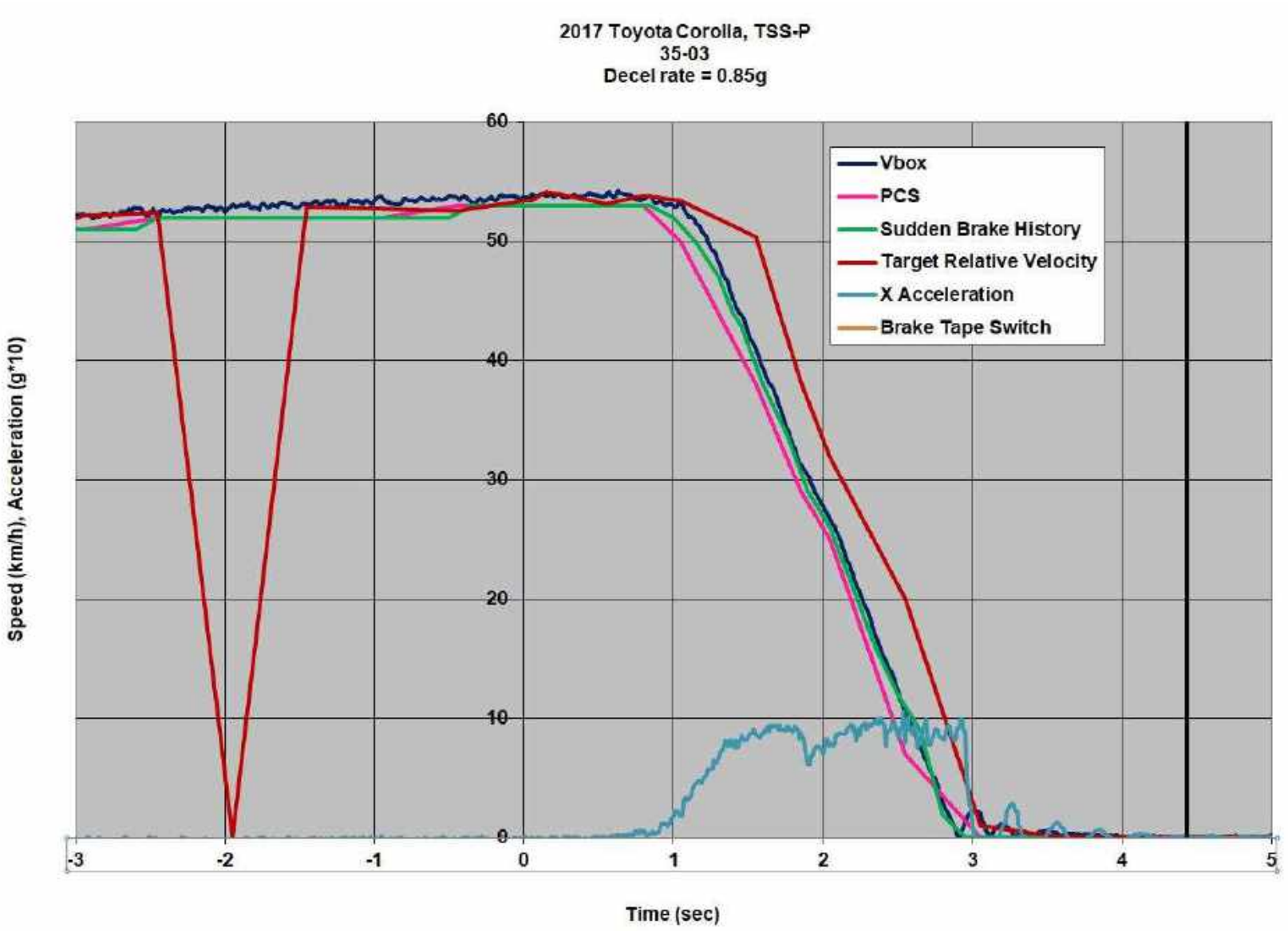

2017 Toyota Corolla, TSS-P
31-01
Decel rate = 0.70g
Speed (km/h), Acceleration (g*10)
Time (sec)
Vbox
PCS
Sudden Brake History
Target Relative Velocity
X Acceleration
Brake Tape Switch
50
45
40
35
30
25
20
15
10
5
0
-2
-1
0
1
2
3
4
5
6
7
8

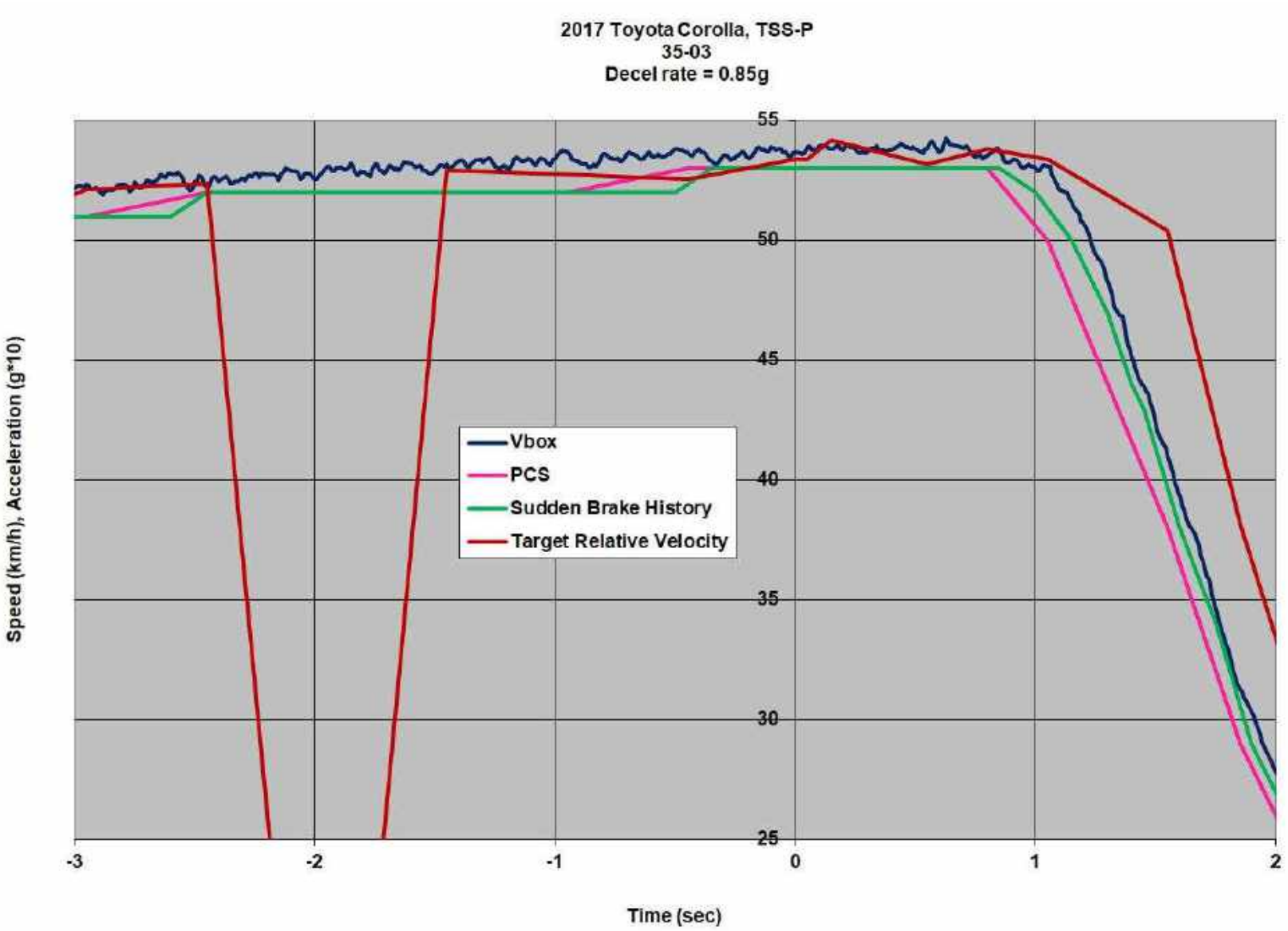

2017 Toyota Corolla, TSS-P
31-01
Decel rate = 0.70g
Speed (km/h)
50
45
40
35
30
25
-2
-1
0
1
2
Time (sec)
Vbox
PCS
Sudden Brake History
Target Relative Velocity

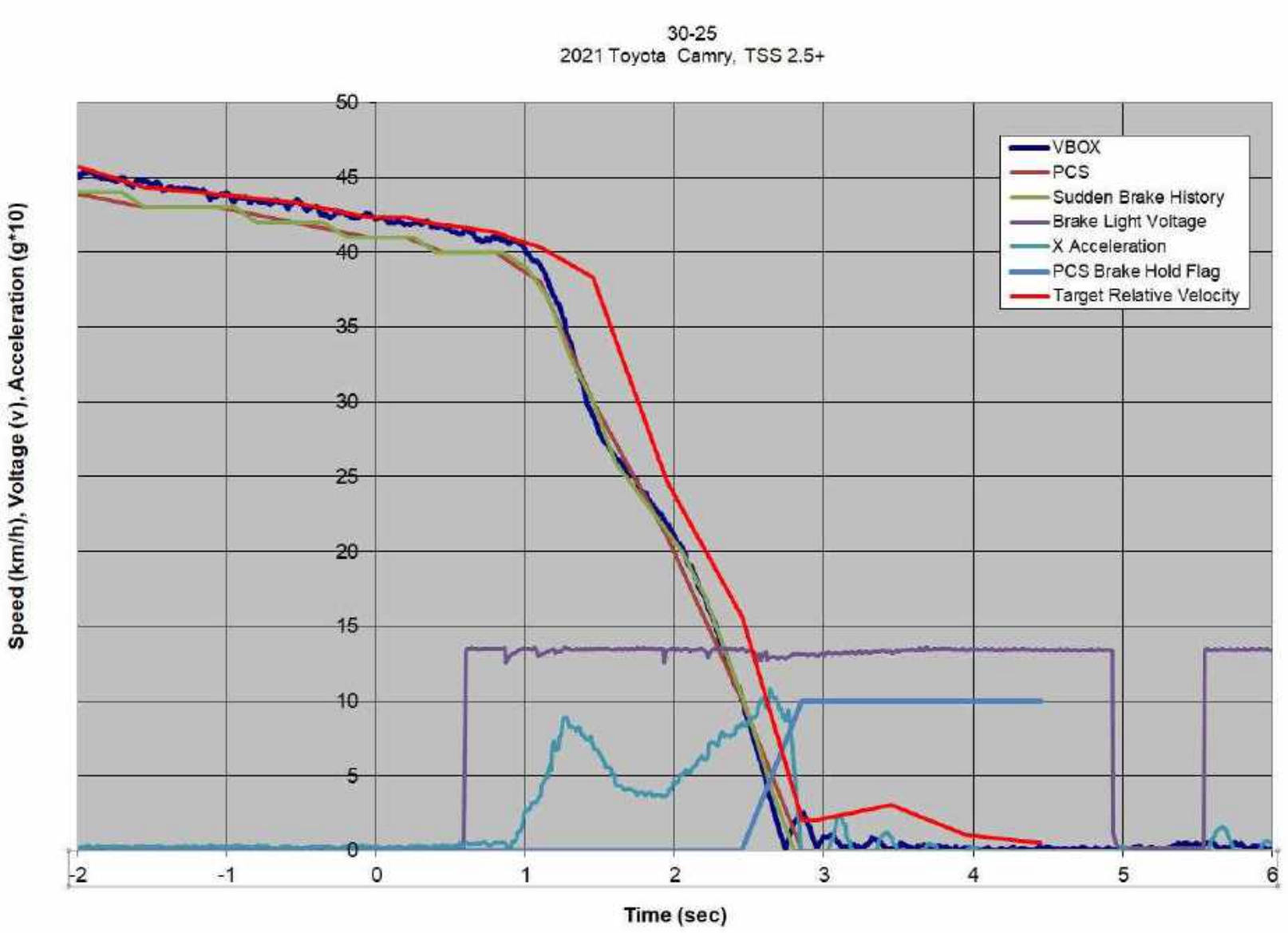

2017 Toyota Corolla, TSS-P
33-01
Vbox
PCS
Sudden Brake History
X Acceleration (g)
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
-5
-4
-3
-2
-1
0
1
2
3
4
5
Time (sec)

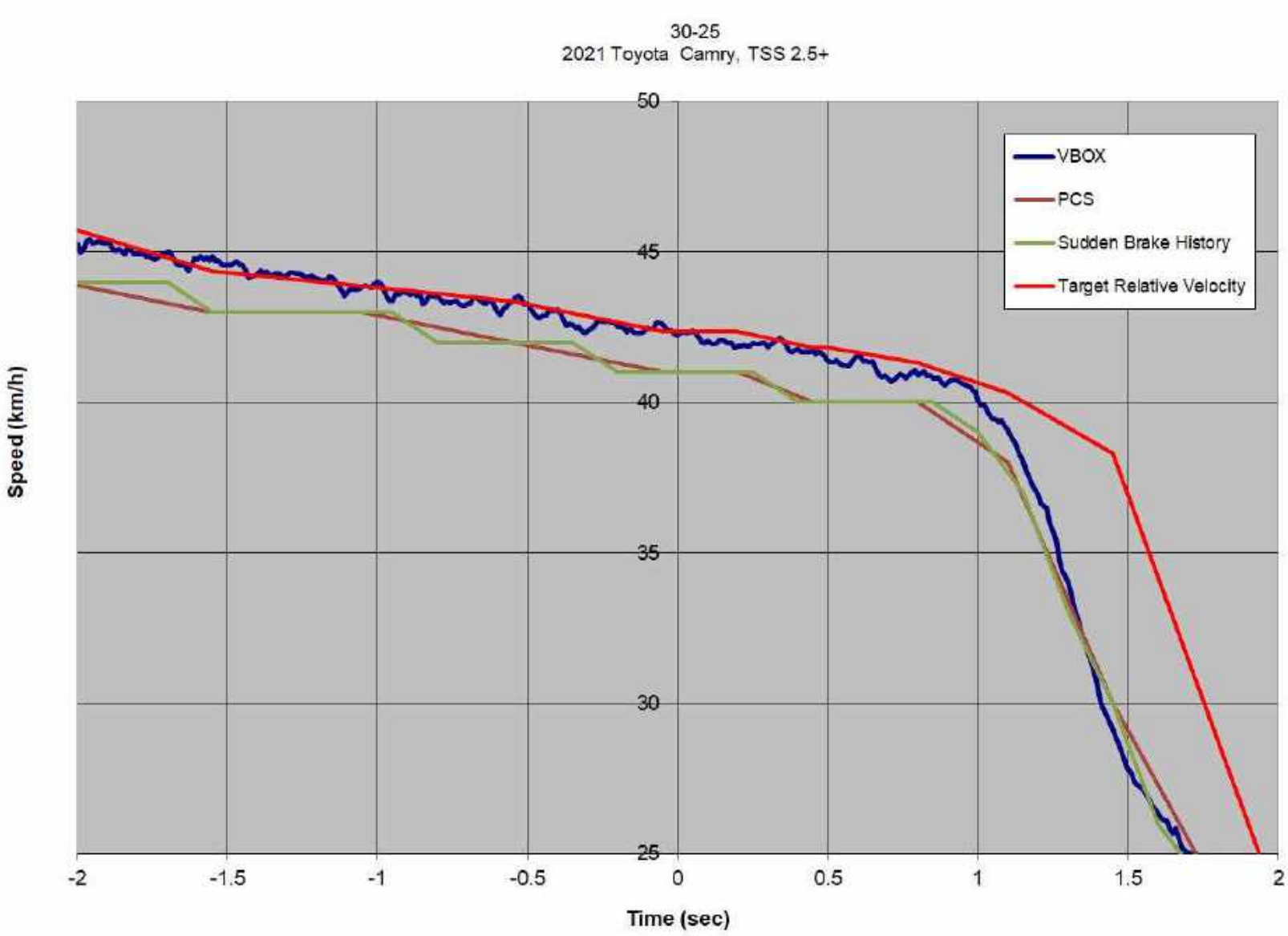

2017 Toyota Carolla TSS-P
33-01
Decel rate = 0.86g
Vbox
PCS
Sudden Brake History
Target Relative Velocity
X Acceleration
Brake Tape Switch
Speed (km/h), Acceleration (g*10)
Time (sec)
60
50
40
30
20
10
0
-2
-1
0
1
2
3
4
5
6

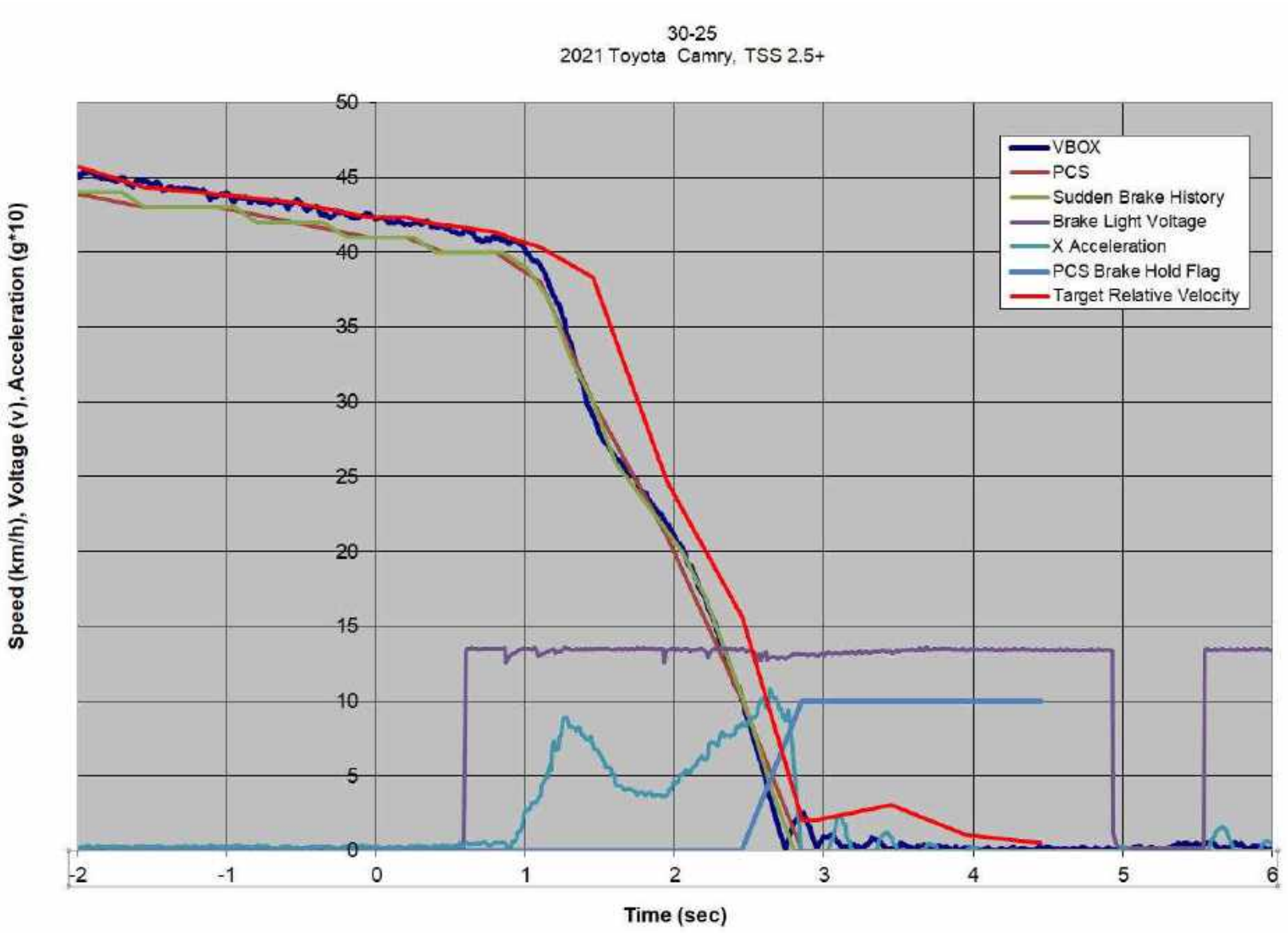

2017 Toyota Carolla TSS-P
33-01
Decel rate = 0.86g
Speed (km/h)
55
50
45
40
35
30
25
-2
-1
0
1
2
Time (sec)
Vbox
PCS
Sudden Brake History
Target Relative Velocity

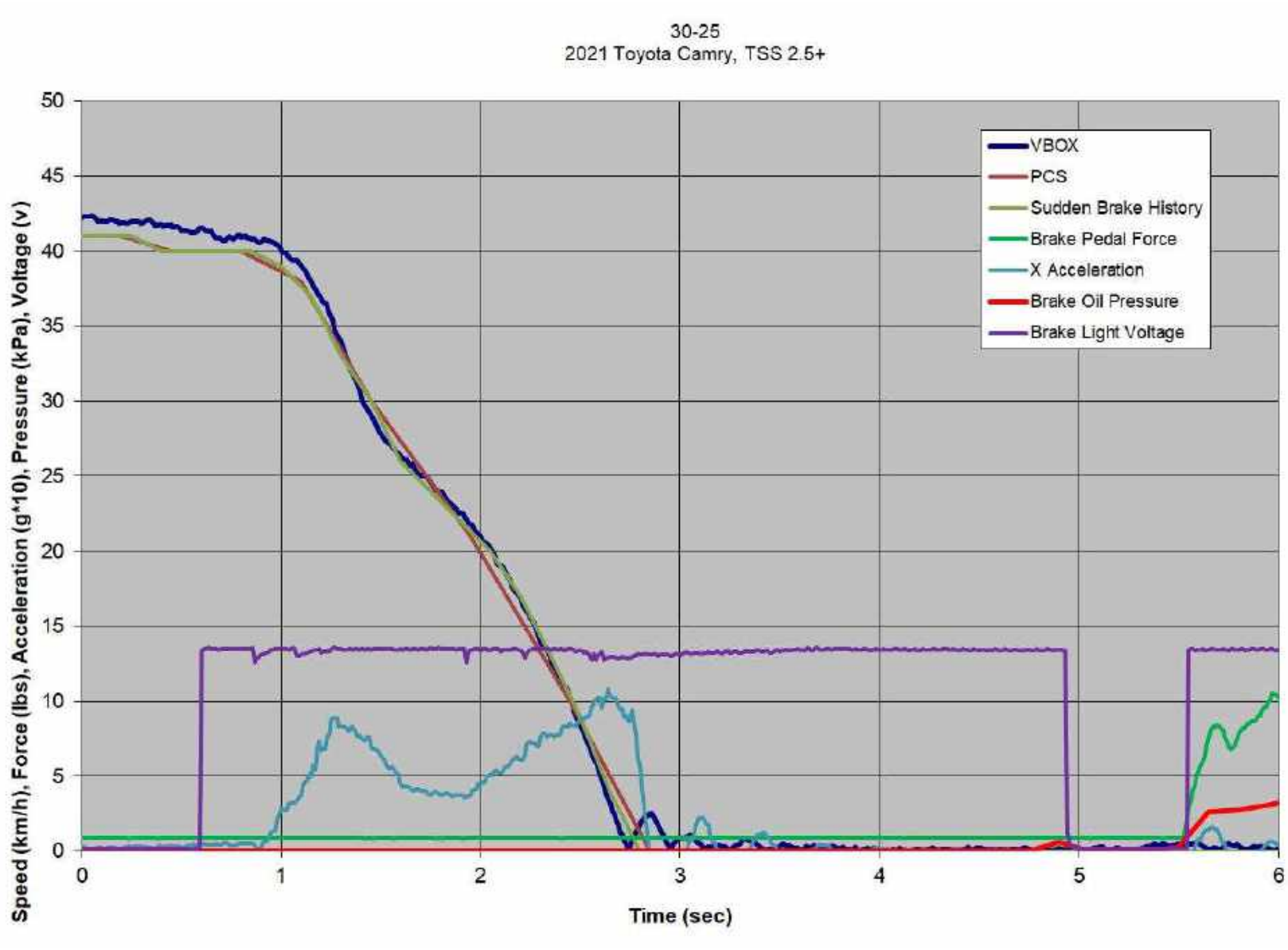

2017 Toyota Corolla, TSS-P
35-03
Decel rate = 0.85g
Speed (km/h), Acceleration (g*10)
Time (sec)
Vbox
PCS
Sudden Brake History
Target Relative Velocity
X Acceleration
Brake Tape Switch
60
50
40
30
20
10
0
-3
-2
-1
0
1
2
3
4
5

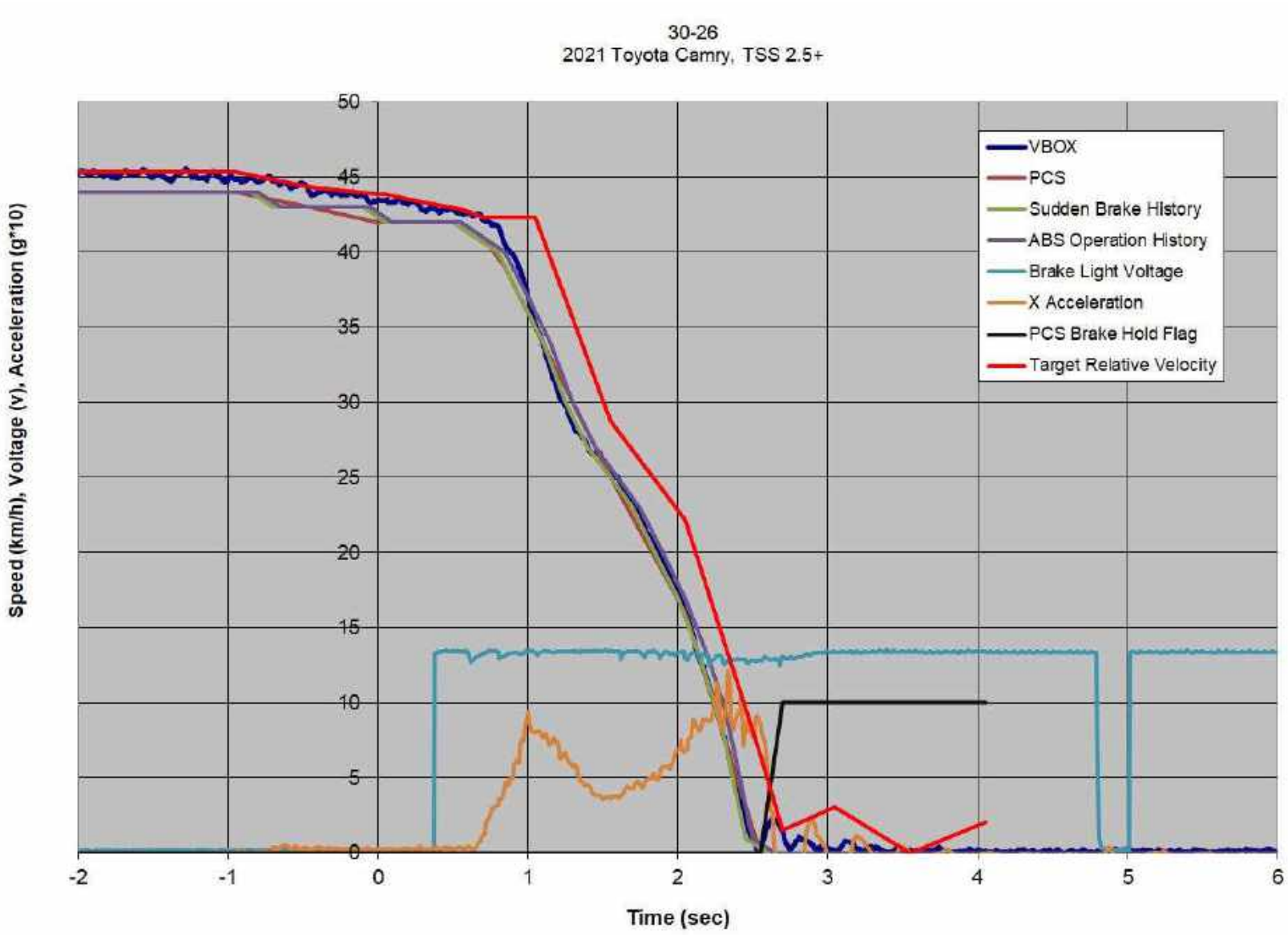

2017 Toyota Corolla, TSS-P
35-03
Decel rate = 0.85g
Speed (km/h), Acceleration (g*10)
55
50
45
40
35
30
25
-3
-2
-1
0
1
2
Time (sec)
Vbox
PCS
Sudden Brake History
Target Relative Velocity

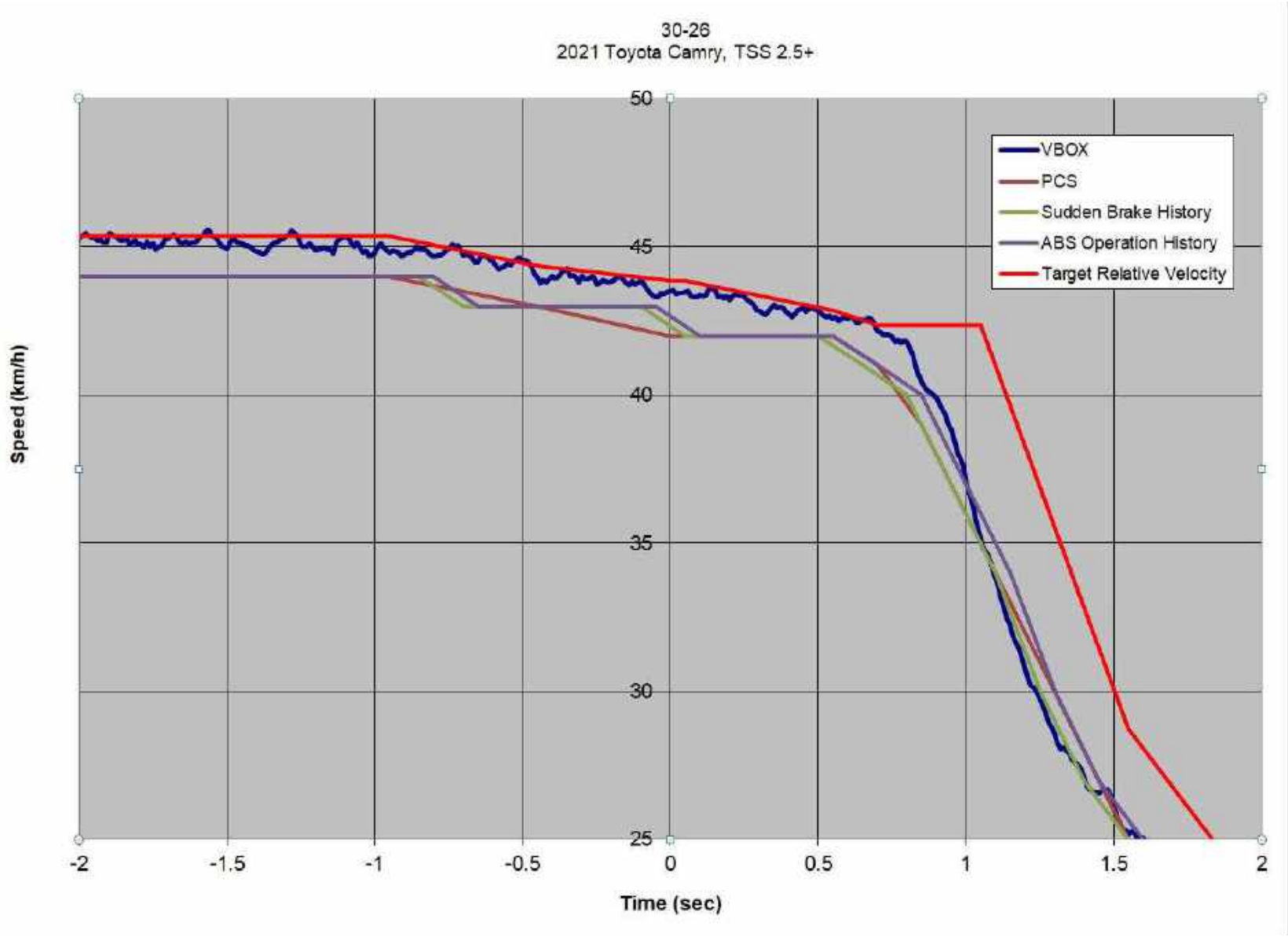

30-25
2021 Toyota Camry, TSS 2.5+
Speed (km/h), Voltage (v), Acceleration (g*10)
Time (sec)
VBOX
PCS
Sudden Brake History
Brake Light Voltage
X Acceleration
PCS Brake Hold Flag
Target Relative Velocity
50
45
40
35
30
25
20
15
10
5
0
-2
-1
0
1
2
3
4
5
6

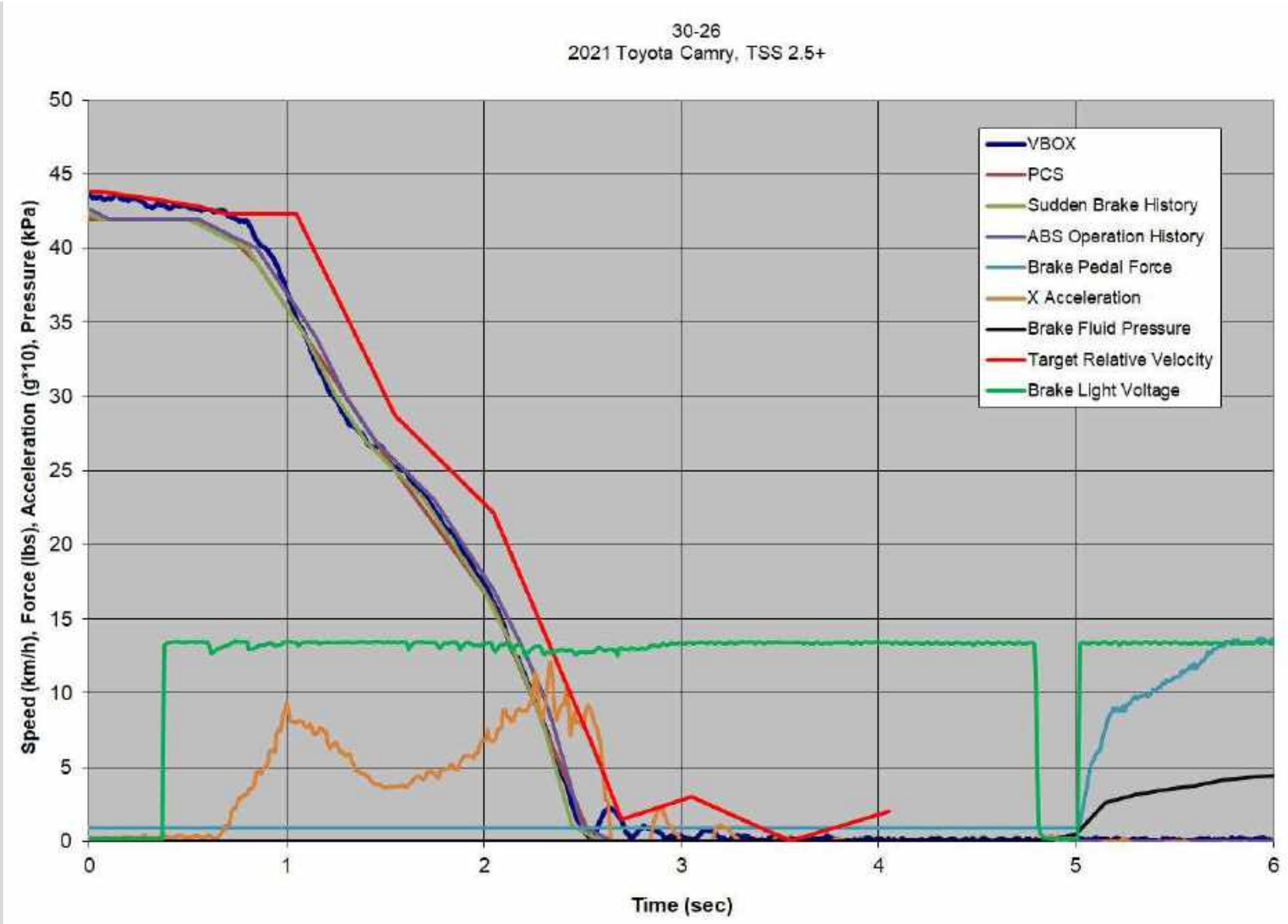

30-25
2021 Toyota Camry, TSS 2.5+
Speed (km/h)
50
45
40
35
30
25
-2
-1.5
-1
-0.5
0
0.5
1
1.5
2
Time (sec)
VBOX
PCS
Sudden Brake History
Target Relative Velocity

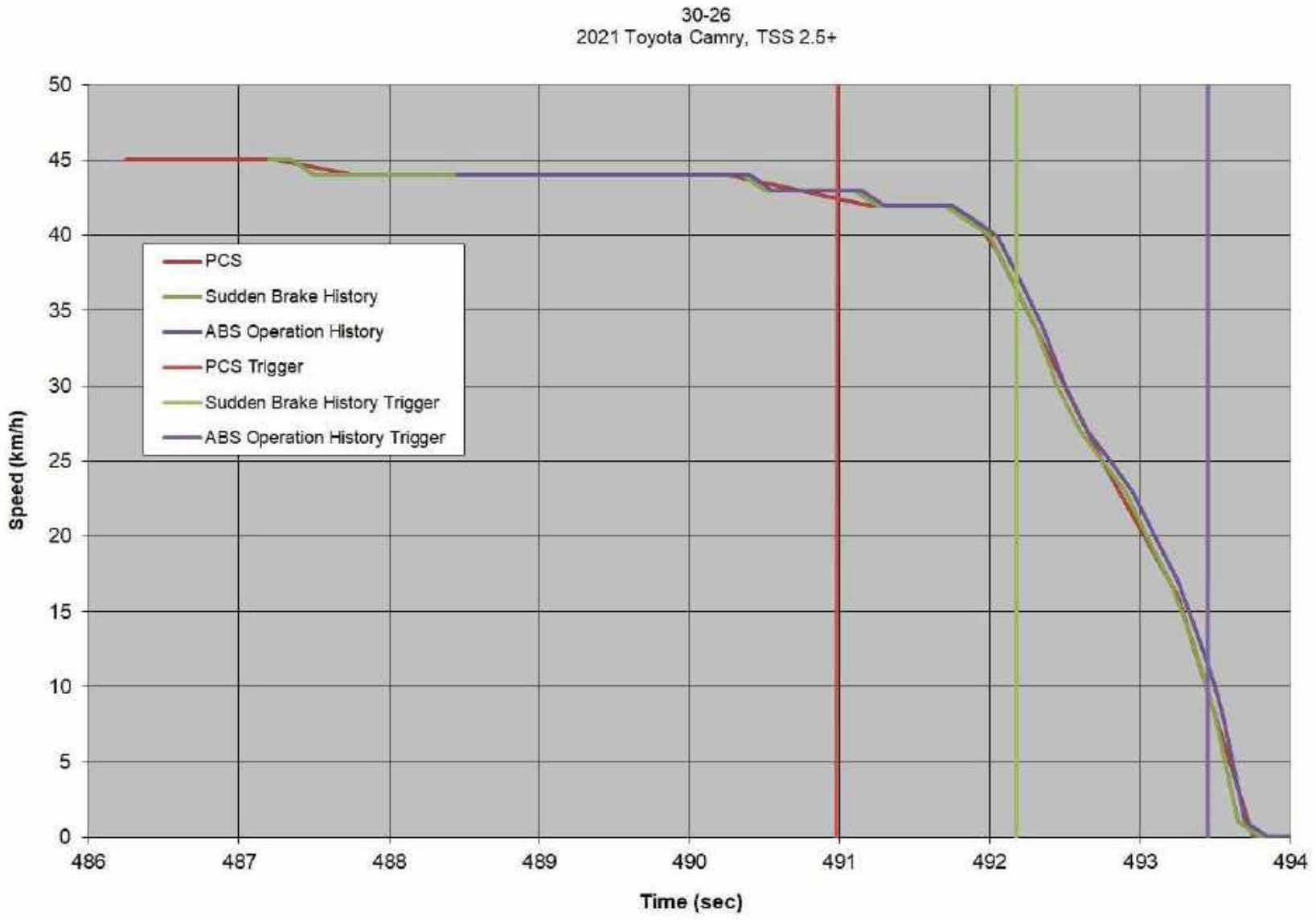

30-25
2021 Toyota Camry, TSS 2.5+
Speed (km/h), Voltage (v), Acceleration (g*10)
Time (sec)
VBOX
PCS
Sudden Brake History
Brake Light Voltage
X Acceleration
PCS Brake Hold Flag
Target Relative Velocity
50
45
40
35
30
25
20
15
10
5
0
-2
-1
0
1
2
3
4
5
6


30-25
2021 Toyota Camry, TSS 2.5+
Speed (km/h), Force (lbs), Acceleration (g*10), Pressure (kPa), Voltage (v)
Time (sec)
VBOX
PCS
Sudden Brake History
Brake Pedal Force
X Acceleration
Brake Oil Pressure
Brake Light Voltage
0
5
10
15
20
25
30
35
40
45
50
0
1
2
3
4
5
6


30-26
2021 Toyota Camry, TSS 2.5+
Speed (km/h), Voltage (v), Acceleration (g*10)
Time (sec)
VBOX
PCS
Sudden Brake History
ABS Operation History
Brake Light Voltage
X Acceleration
PCS Brake Hold Flag
Target Relative Velocity
50
45
40
35
30
25
20
15
10
5
0
-2
-1
0
1
2
3
4
5
6

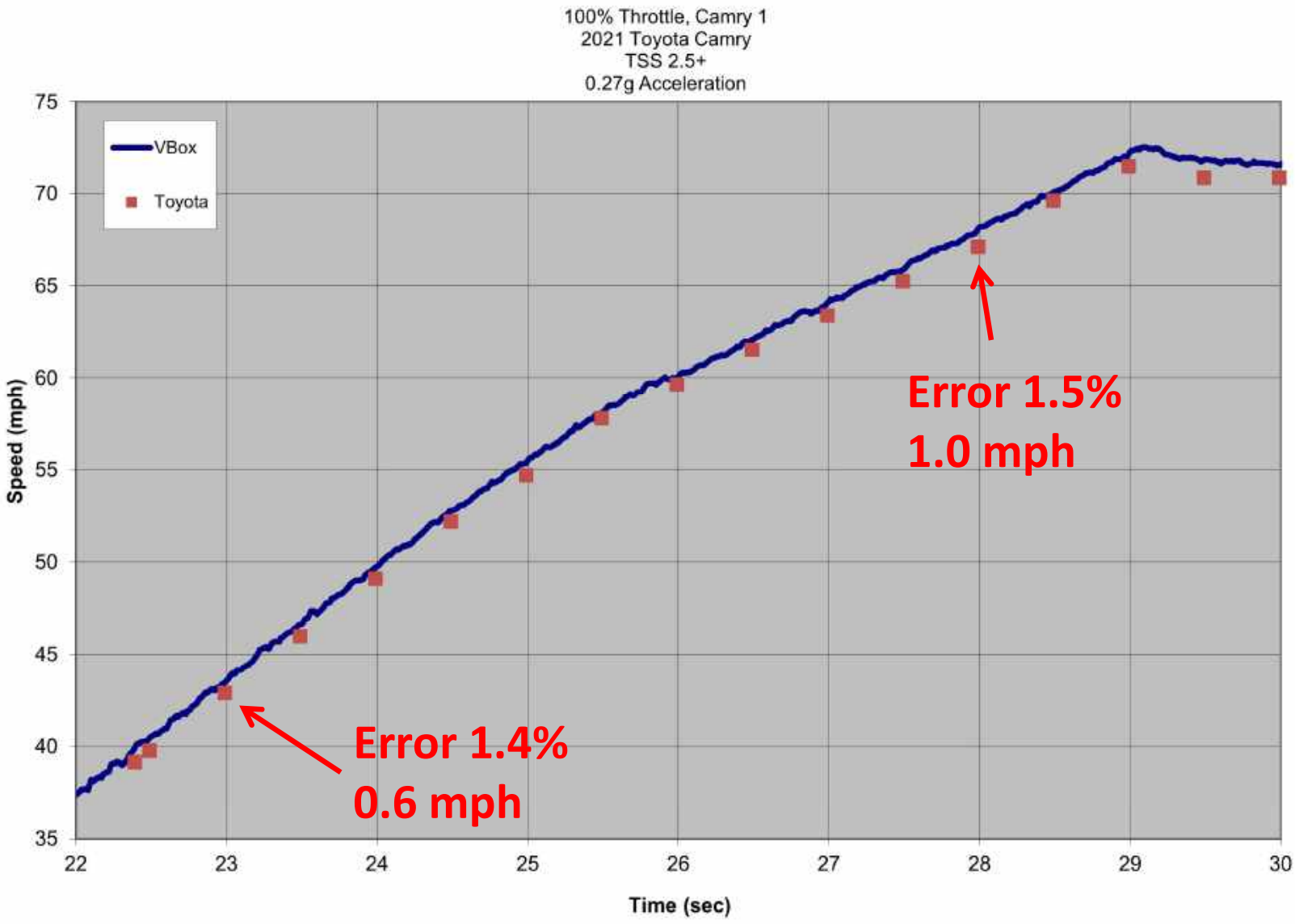

30-26
2021 Toyota Camry, TSS 2.5+
Speed (km/h)
50
45
40
35
30
25
-2
-1.5
-1
-0.5
0
0.5
1
1.5
2
Time (sec)
VBOX
PCS
Sudden Brake History
ABS Operation History
Target Relative Velocity

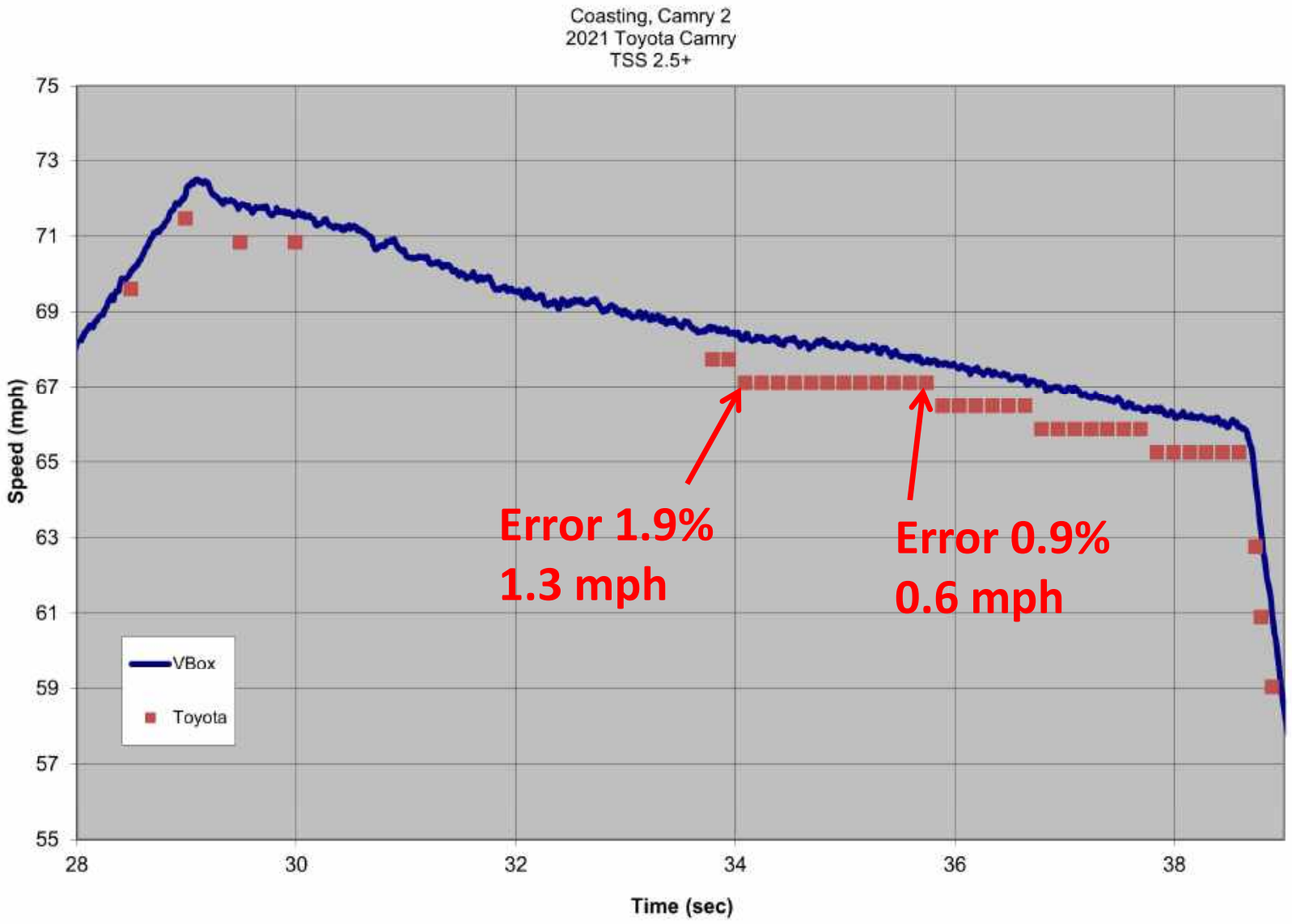

30-26
2021 Toyota Camry, TSS 2.5+
Speed (km/h), Force (lbs), Acceleration (g*10), Pressure (kPa)
Time (sec)
VBOX
PCS
Sudden Brake History
ABS Operation History
Brake Pedal Force
X Acceleration
Brake Fluid Pressure
Target Relative Velocity
Brake Light Voltage
0
5
10
15
20
25
30
35
40
45
50
0
1
2
3
4
5
6

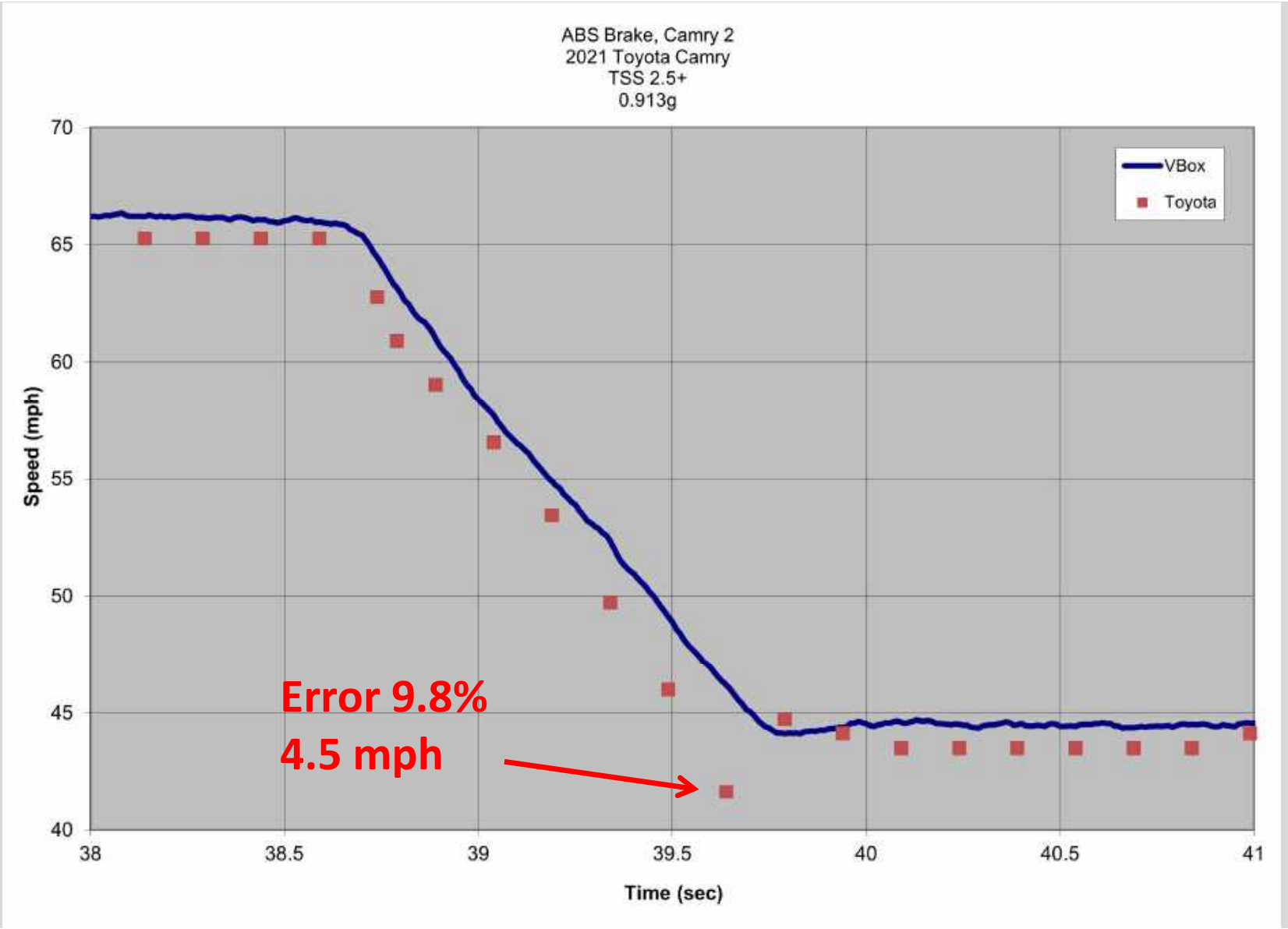

30-26
2021 Toyota Camry, TSS 2.5+
PCS
Sudden Brake History
ABS Operation History
PCS Trigger
Sudden Brake History Trigger
ABS Operation History Trigger
Speed (km/h)
Time (sec)
0
5
10
15
20
25
30
35
40
45
50
486
487
488
489
490
491
492
493
494

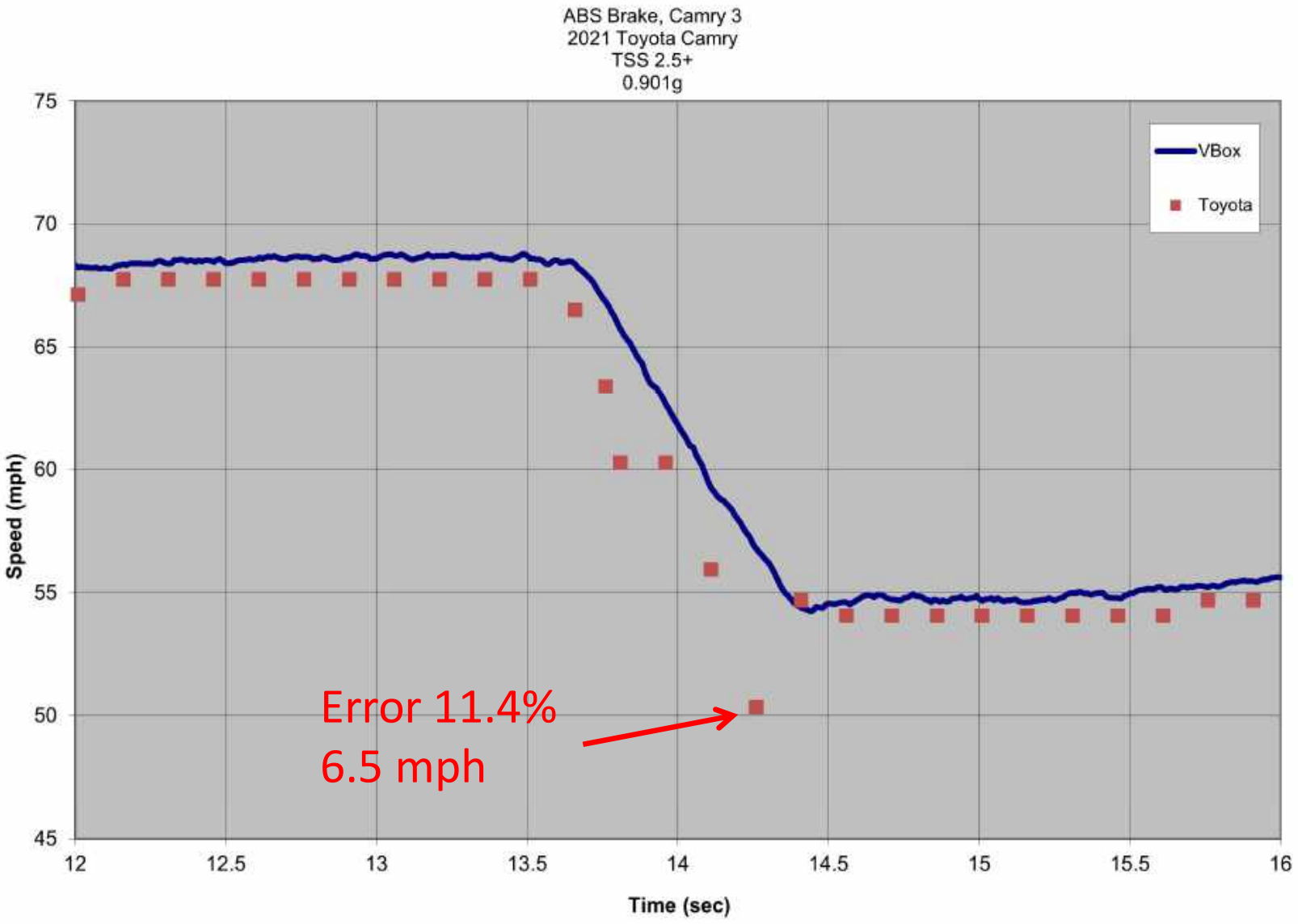
The following slides showing speed accuracy are based on tests with a 2021 Toyota Camry SE

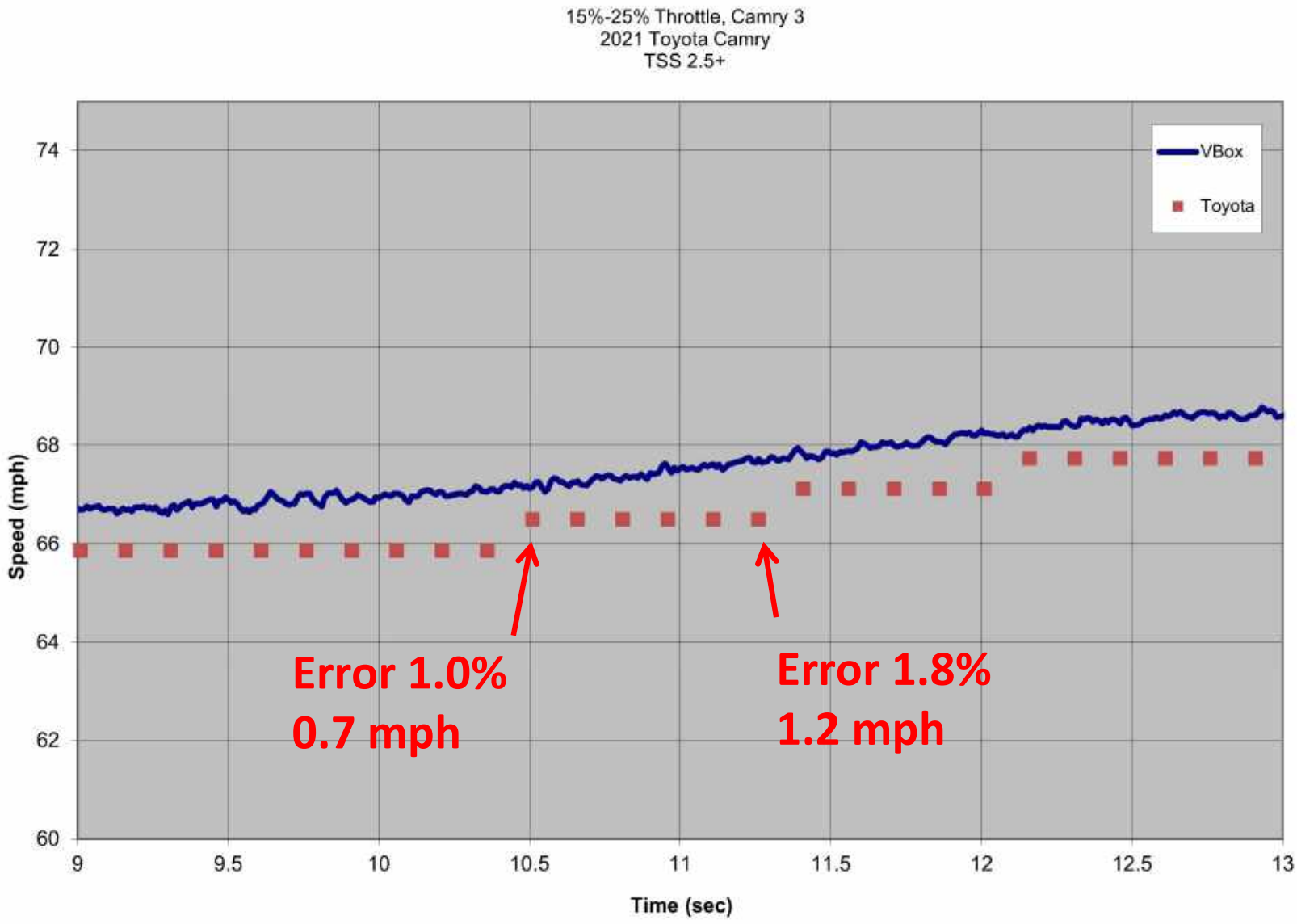
- OEM tire size 235/45R18
- Installed tires Hankook Kinergy GT 235/45R18
- Original tread depth 10/32
- Measured tread depth 7/32
- Tread wear 3/32
- The tires were inflated to the recommended pressure

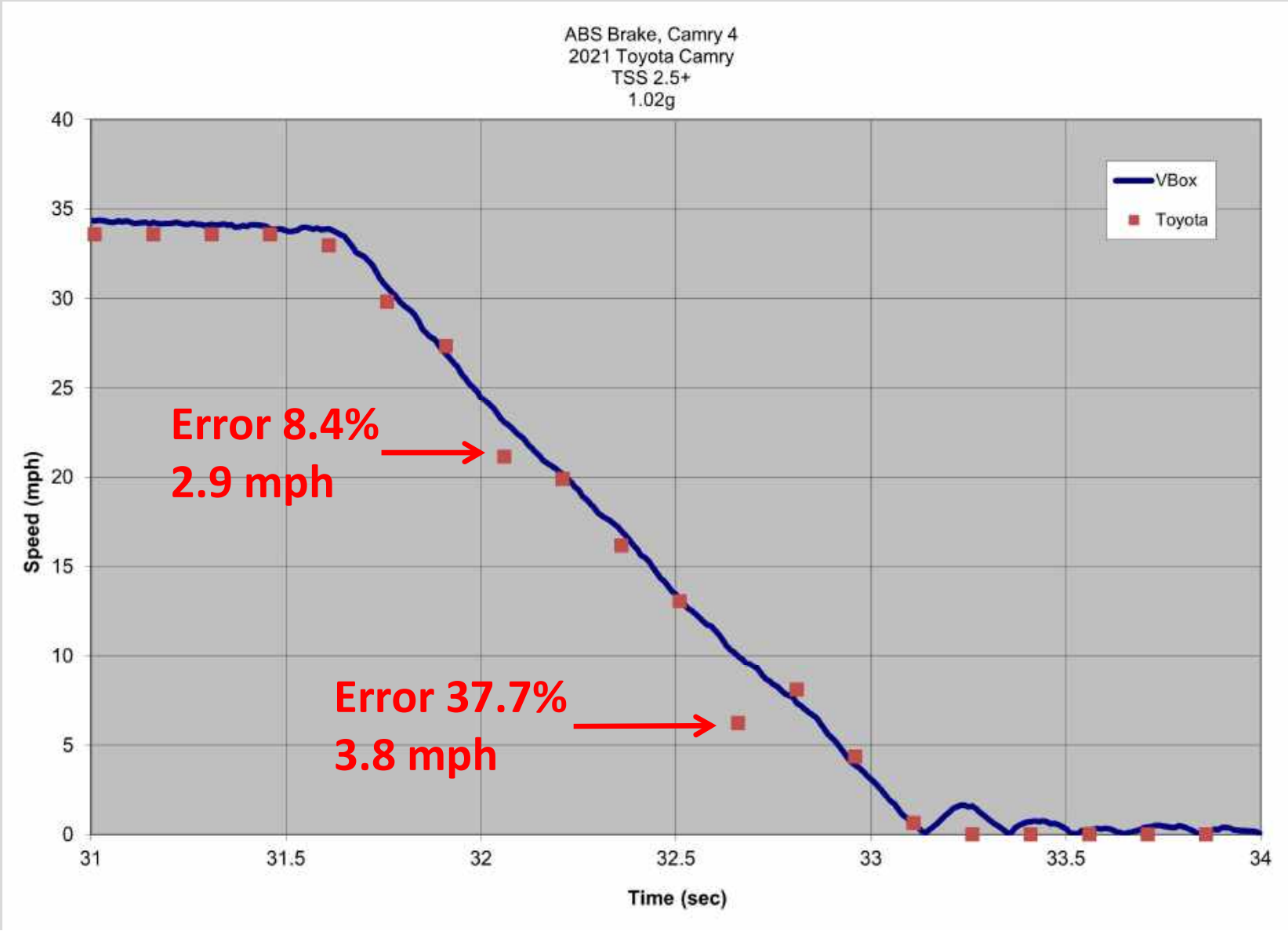

100% Throttle, Camry 1
2021 Toyota Camry
TSS 2.5+
0.27g Acceleration
VBox
Toyota
Speed (mph)
Time (sec)
75
70
65
60
55
50
45
40
35
22
23
24
25
26
27
28
29
30
Error 1.5%
1.0 mph
Error 1.4%
0.6 mph

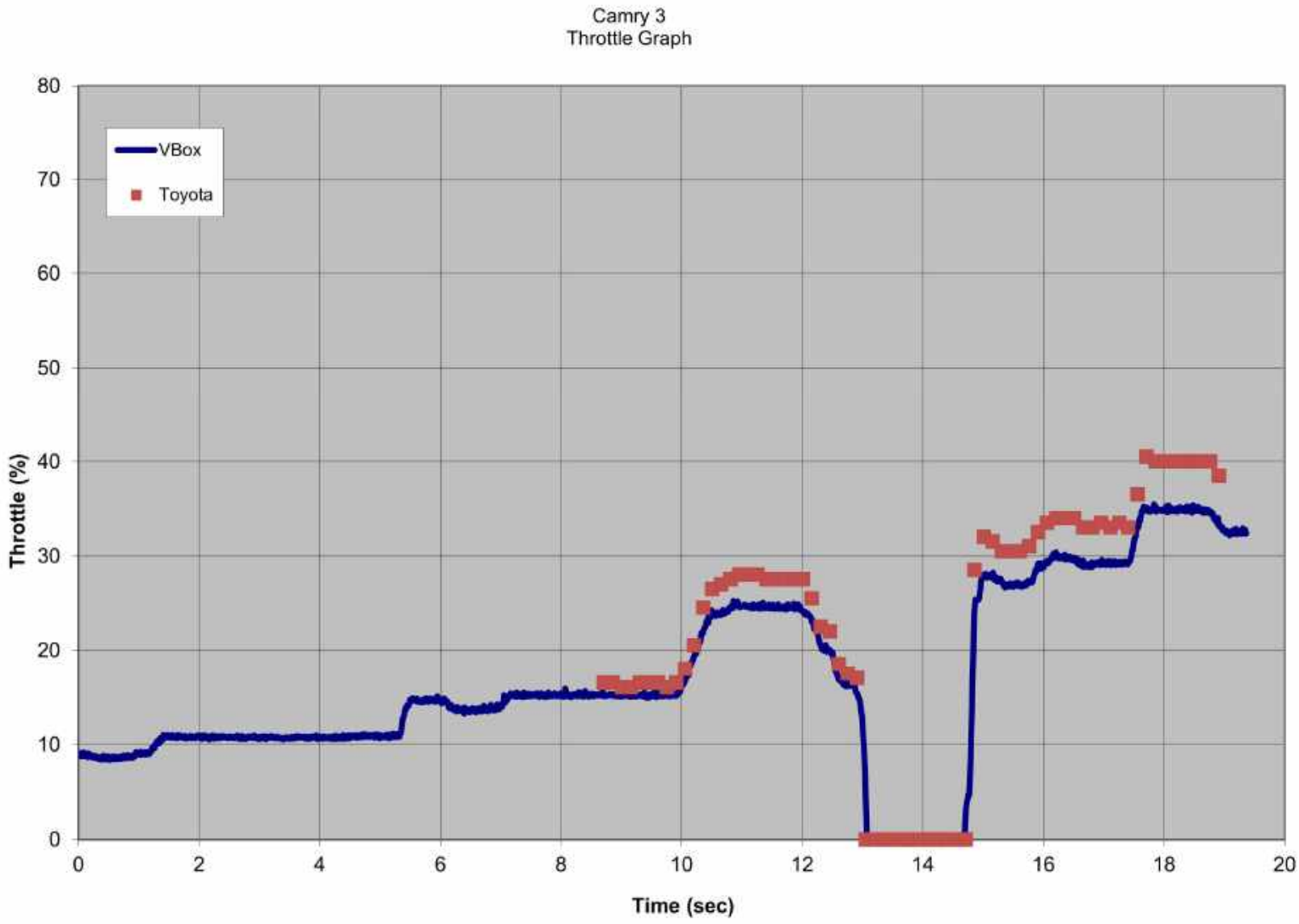

Coasting, Camry 2
2021 Toyota Camry
TSS 2.5+
Speed (mph)
75
73
71
69
67
65
63
61
59
57
55
28
30
32
34
36
38
Time (sec)
VBox
Toyota
Error 1.9%
1.3 mph
Error 0.9%
0.6 mph

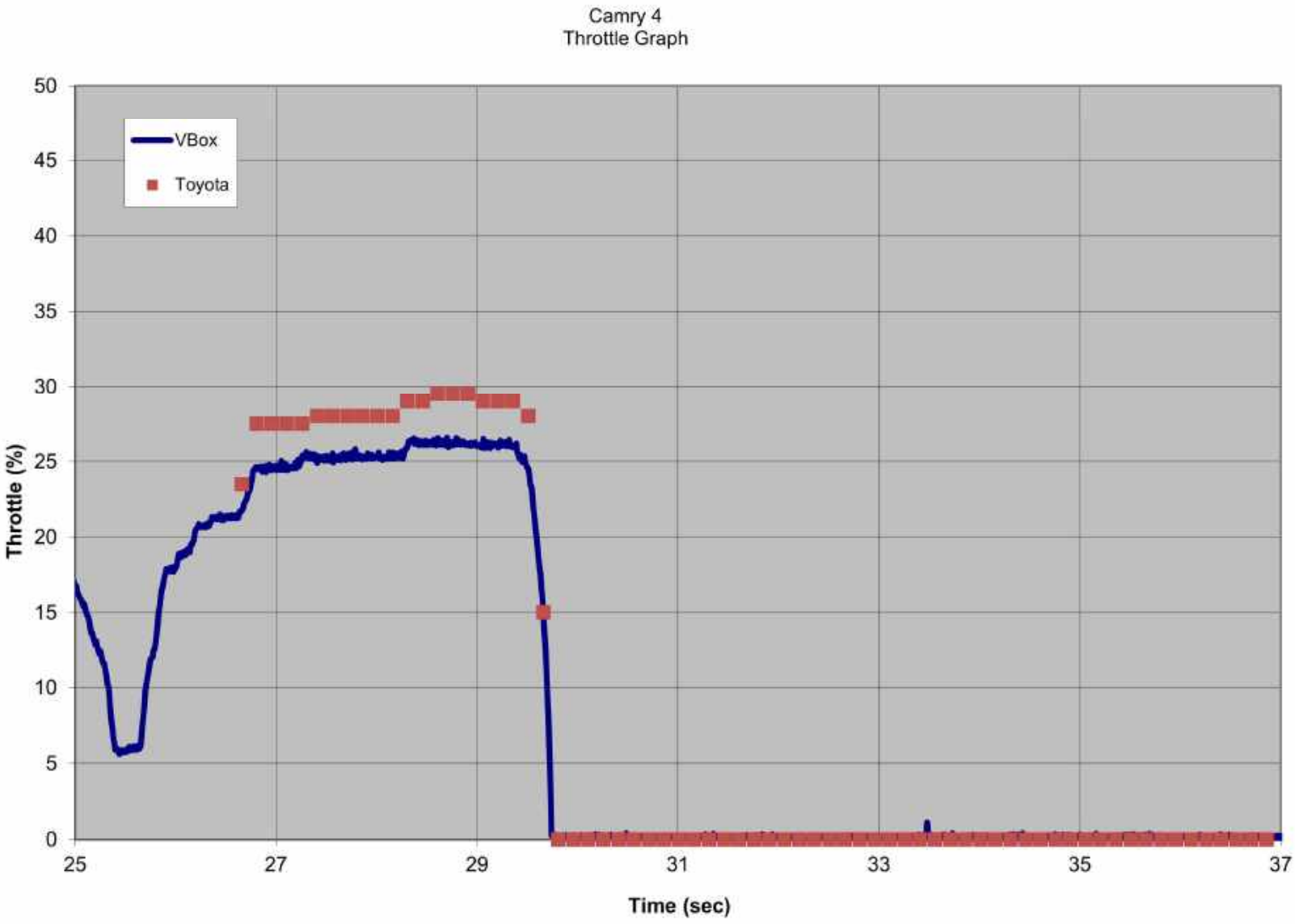

ABS Brake, Camry 2
2021 Toyota Camry
TSS 2.5+
0.913g
VBox
Toyota
Speed (mph)
70
65
60
55
50
45
40
38
38.5
39
39.5
40
40.5
41
Time (sec)
Error 9.8%
4.5 mph


ABS Brake, Camry 3
2021 Toyota Camry
TSS 2.5+
0.901g
VBox
Toyota
Speed (mph)
75
70
65
60
55
50
45
12
12.5
13
13.5
14
14.5
15
15.5
16
Time (sec)
Error 11.4%
6.5 mph


15%-25% Throttle, Camry 3
2021 Toyota Camry
TSS 2.5+
VBox
Toyota
Speed (mph)
74
72
70
68
66
64
62
60
9
9.5
10
10.5
11
11.5
12
12.5
13
Time (sec)
Error 1.0%
0.7 mph
Error 1.8%
1.2 mph


ABS Brake, Camry 4
2021 Toyota Camry
TSS 2.5+
1.02g
VBox
Toyota
Error 8.4%
2.9 mph
Error 37.7%
3.8 mph
Speed (mph)
Time (sec)
40
35
30
25
20
15
10
5
0
31
31.5
32
32.5
33
33.5
34

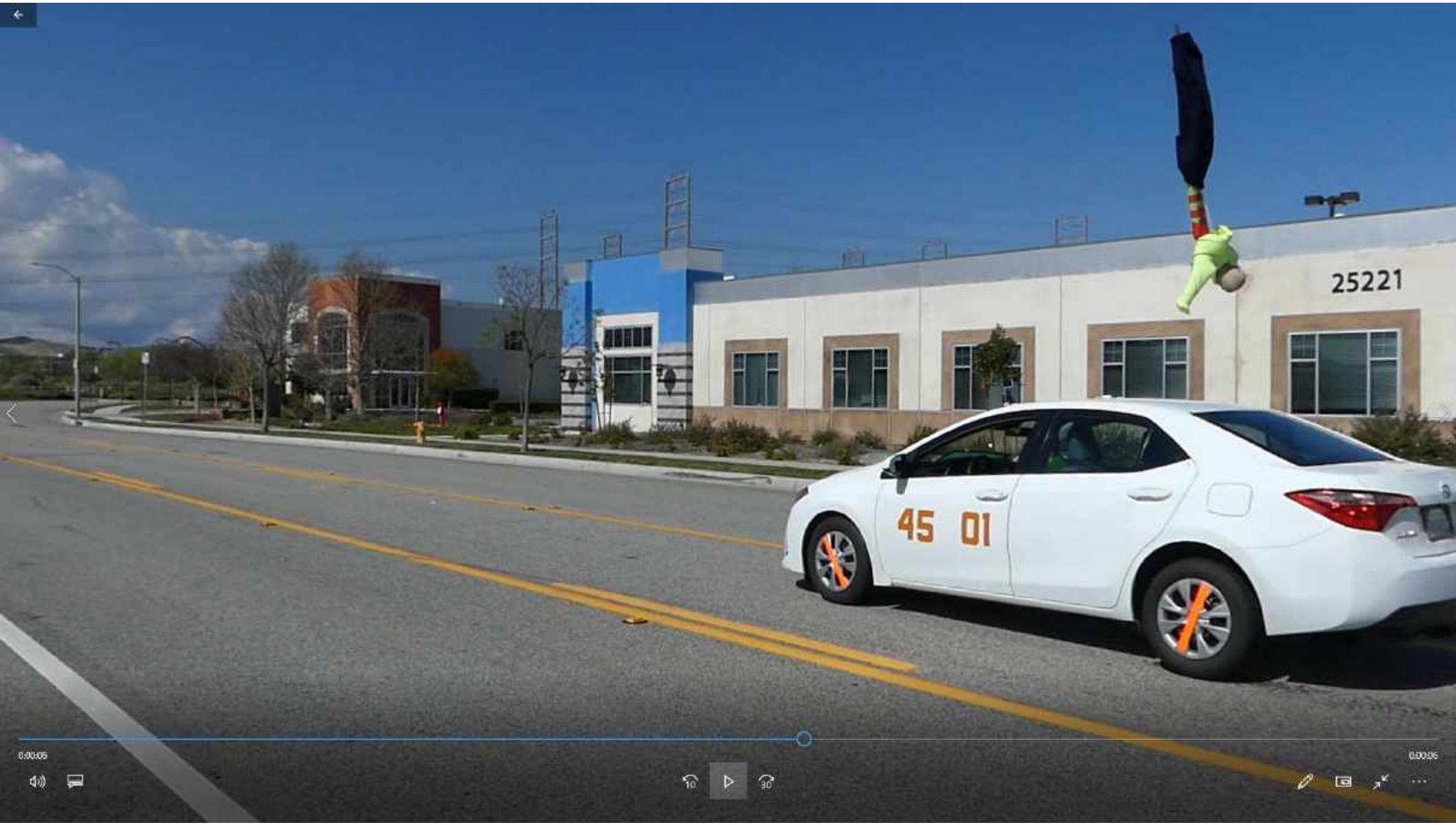

Camry 3
Throttle Graph
VBox
Toyota
Throttle (%)
Time (sec)
0
10
20
30
40
50
60
70
80
0
2
4
6
8
10
12
14
16
18
20


Camry 4
Throttle Graph
VBox
Toyota
Throttle (%)
50
45
40
35
30
25
20
15
10
5
0
25
27
29
31
33
35
37
Time (sec)

Back to Tim


What if you break the windshield?


25221
0:00:51
0:00:37


25221
45 01
0:00:05
0:00:06

Camera is mounted to the windshield



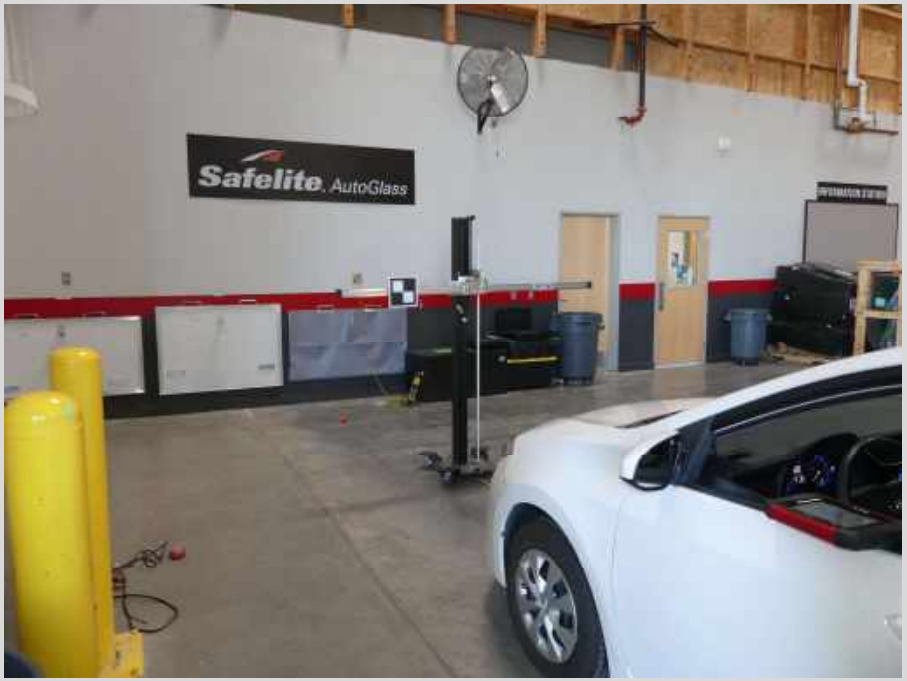


The camera unit





Calibration with targets at left, center and right





Step by step calibration



Back-powering the vehicle if ignition key or fob is not available

Can you still do a Techstream download?


Little Trees



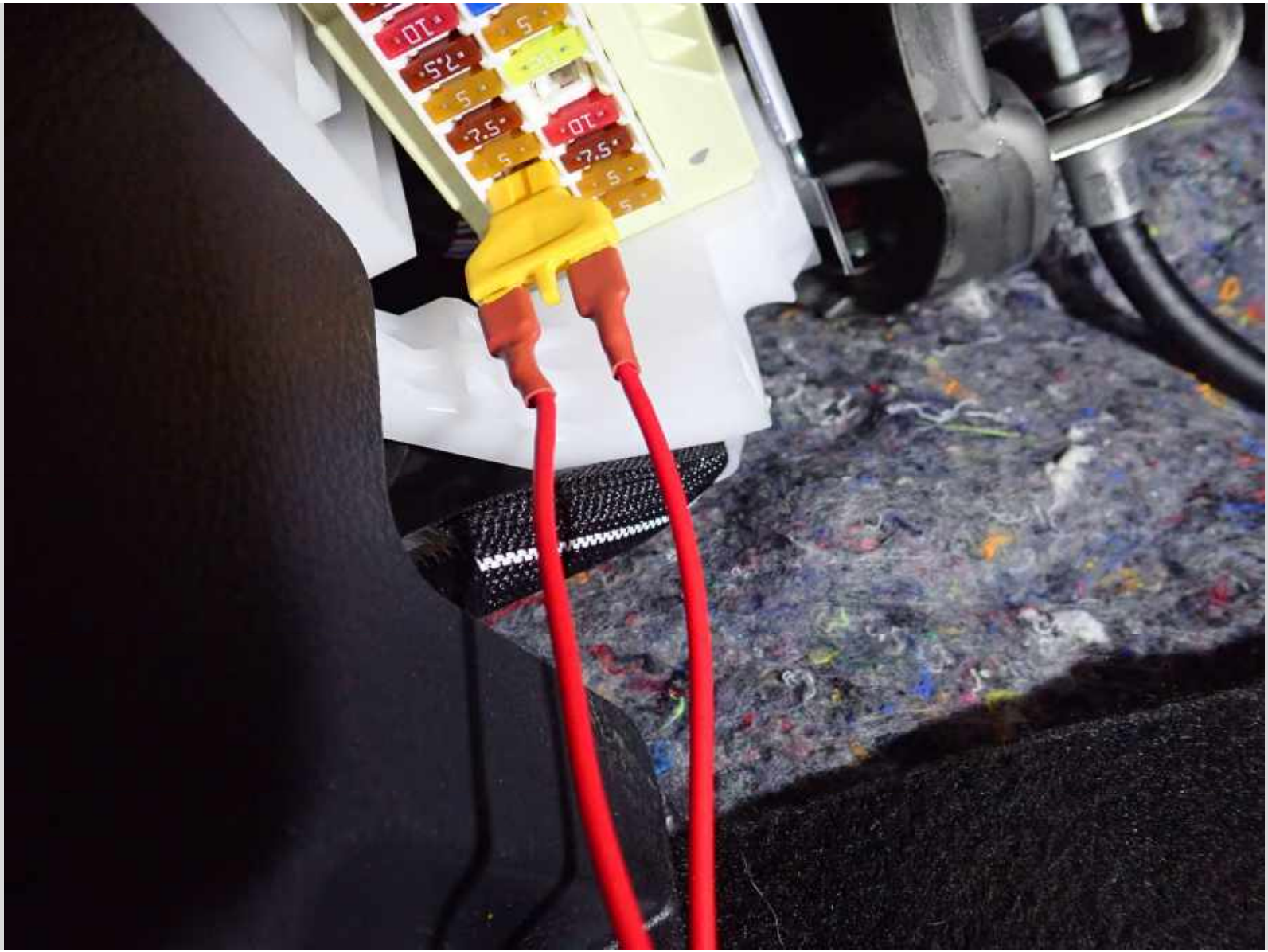
Airbag fuse


7.5A
DCM
5A
SFT LOCK-ACC
5A
ECU-ACC
20A
WASHER
7.5A
ECU-IG1 NO.3
5A
EPS-IG1
10A
ECU-IG1 NO.4
10A
A/BAG-IG2
7.5A
BKUP LP
5A
METER-IG2
5A
ECU-IG1 NO.2
5A
ECU-IG2 NO.3
5A
ECU-IG1 NO.1
USE THE DESIGNATED FUSES ONLY.
.4L

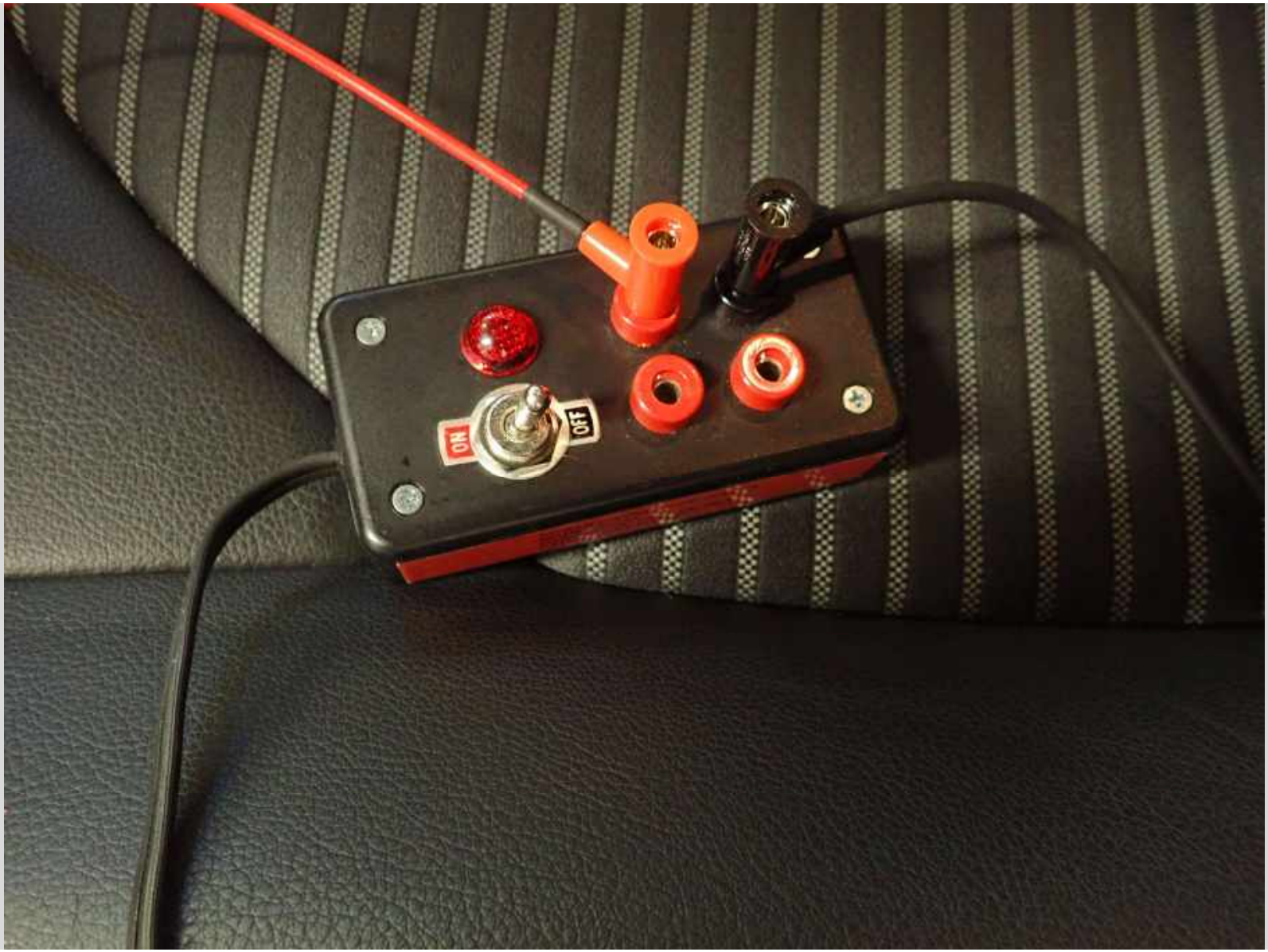
TOYOTA


GTC063
FUSE SOCKET CONNECTORS
micro-3
micro-2
LPM
mini
ATO/ATC
MAXI




10
5
15
5
7.5
5
10
7.5
5
5




ON
OFF


ON
OFF



The download will still work but no images are available

Current thought is that the system needs to confirm that the ignition key or key fob is present before it will provide the images

Questions?